

Project Data Page:





Table of Contents

Cover Page	1
Vessel Design Summary:	2
Nozzle Summary:	5
Nozzle Schedule:	6
Bill of Material:	7
MDMT Summary:	9
Warnings and Errors:	11
Input Echo:	12
XY Coordinate Calculations:	25
Flg Calc [Int P]: New Flange	26
Flg Calc [Int P]: New Flange	29
Flg Calc [Int P]: New Flange	37
Flg Calc [Int P]: New Flange	45
Flg Calc [Int P]: New Flange	53
Flg Calc [Int P]: New Flange	61
Internal Pressure Calculations:	69
Element: LHS Ch Shell	70
Element: Shell-1	72
Element: Shell 2	74
Element: Shell 3	76
Element: Rhs ch shell	78
Element: Dish End	80
Shop Pressure Test Results:	82
Field Pressure Test Results:	84
ASME TS Calc: Pressure Test	86
ASME Fl-TS Calc: Pressure Test	96
Floating Head: Operating	104
External Pressure Calculations:	109
Element and Detail Weights:	111
Nozzle Flange MAWP:	114
Wind Load Calculation:	116
Earthquake Load Calculation:	119
Center of Gravity Calculation:	121
Saddle Calcs: Bundle Pull New:	122
Saddle Calcs: Bundle Pull Cor:	130
Saddle Calcs: Road Transport:	138
Saddle Calcs: Operating Case:	152
Saddle Calcs: Test Case:	168
Nozzle Calcs.: N1-Tube Inlet	183
Nozzle Calcs.: N2-Tube outlet	197
Nozzle Calcs.: N3-Shell inlet	211
Nozzle Calcs.: N4-Shell outlet	217
Nozzle Calcs.: Drain	229
Nozzle Calcs.: Vent	233
ASME TS Calc:	237
ASME Fl-TS Calc:	248
Floating Head: Operating	257
Problems/Failures Summary:	262



DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\197-CChannel MOC= CS.pvdb

Date of Analysis : Sep 9, 2026 10:15am

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 2 of 262

Vessel Design Summary:

Step: 39 10:16am Sep 9,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2700.000 mm ID
Vessel Design Length, Tangent to Tangent 10146.85 mm
Specified Datum Line Distance 0.00 mm
Shell Side Design Temperature 240 °C
Channel Side Design Temperature 82 °C
Shell Side Design Pressure 1000.000 KPa.
Channel Side Design Pressure 1000.000 KPa.
Shop Shell Side Test Pressure 1596.923 KPa.
Field Shell Side Test Pressure 1300.000 KPa.
Shop Channel Side Test Pressure 1472.169 KPa.
Field Channel Side Test Pressure 1300.000 KPa.
Wind Design Code ASCE/SEI 7-16
Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-266 2	...	...	K03506	
Nozzle	SA-105	...	...	K03504	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-266 2	...	...	K03506	
Tubes	SA-789	...	...	S32205	
Tubesheet	SA-182 F60	...	...	S32205	
Flo. Head	SA-240	...	...	S32205	
Bk. Bolting	SA-193 B7	...	<= 64	G41400	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-307 A	...	6 <= t <= 100		
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Ch Cover	1000.000	103.42	1045.000	3.0000	N/A	No
LHS Ch Flg-1	1026.465	103.42	1292.000	3.0000	N/A	No
LHS Ch Shell	1026.495	103.42	1094.000	3.0000	N/A	No
LHS Ch Flg-2	1026.465	103.42	1292.000	3.0000	N/A	No
LHS Shell Flg-1	1026.465	0.00	1359.000	0.8000	N/A	No
Shell-1	1026.473	0.00	1156.000	0.8000	N/A	No
Shell 2	1026.473	0.00	1840.000	0.8000	N/A	No
Shell 3	1026.473	0.00	1156.000	0.8000	N/A	No



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 3 of 262

Vessel Design Summary:

Step: 39 10:16am Sep 9,2026

Rhs Shell Flg-1	1026.465	0.00	1359.000	0.8000	N/A	No
Rhs Shell Flg-2	1026.465	0.00	1359.000	0.8000	N/A	No
Rhs ch shell	1026.473	0.00	1156.000	0.8000	N/A	No
Dish End	1026.473	0.00	1085.000	0.8000	No	No

Liquid Level: 2700.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	191.3	185.0	193.0	185.0	180.3	138.2	1.00	1.00
Body Flg	506.4	315.0	268.0	260.0	254.4	254.4	1.00	1.00
Cylinder	1706.3	1200.0	16.0	16.0	14.9	9.2	0.85	0.85
Body Flg	2022.7	310.0	263.0	255.0	251.6	251.6	1.00	1.00
Body Flg	2469.7	307.0	260.0	252.0	250.2	250.2	1.00	1.00
Cylinder	4369.7	1900.0	16.0	14.5	12.7	...	0.85	0.85
Cylinder	6869.7	2500.0	24.0	22.5	12.7	...	0.85	0.85
Cylinder	8769.7	1900.0	16.0	14.5	12.7	...	0.85	0.85
Body Flg	9086.1	310.0	263.0	255.0	250.2	250.2	1.00	1.00
Body Flg	9396.1	310.0	263.0	255.0	250.2	250.2	1.00	1.00
Cylinder	10096.0	700.0	16.0	14.5	12.7	...	0.85	0.85
Ellipse	10146.8	50.8	16.0	13.6	12.6	...	0.85	0.85

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1-Tube Inlet	WRC-297	0.995	Shell	Passed
N2-Tube outlet	WRC-297	0.986	Shell	Passed
N4-Shell outle	WRC-297	0.986	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width 585.000 mm
 Saddle Bearing Angle 140.000 deg.
 Centerline Dimension 1981.200 mm
 Wear Pad Width 875.000 mm
 Wear Pad Thickness 20.000 mm
 Wear Pad Bearing Angle 152.000 deg.
 Distance from Saddle to Tangent 1260.000 mm

Baseplate Length 2349.500 mm
 Baseplate Thickness 50.000 mm
 Baseplate Width 610.000 mm
 Number of Ribs (including outside ribs) 5
 Rib Thickness 20.000 mm
 Web Thickness 20.000 mm
 Height of Center Web 650.000 mm
 Number of Bolts in Baseplate 8

Baseplate Sketch

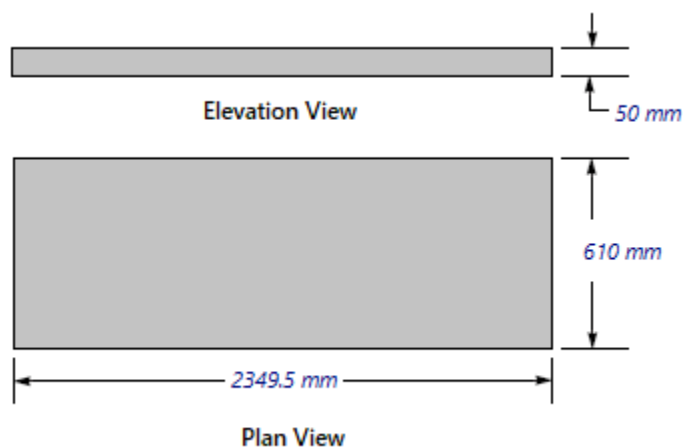
PV Elite 28 Licensee: SL User

FileName : 197-C Channel MOC= CS

Page 4 of 262

Vessel Design Summary:

Step: 39 10:16am Sep 9, 2026



Maximum Tensile Bolt Load 6070. Kgf

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	117747.80	Kgf
Maximum Transverse Saddle Shear Load	5297.66	Kgf
Maximum Longitudinal Saddle Shear Load	8817.99	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	104016.84	Kgf
Maximum Transverse Saddle Shear Load	359.34	Kgf
Maximum Longitudinal Saddle Shear Load	361.90	Kgf

Weights:

Fabricated	- Bare w/o removable internals	95968.0	kg.
Shop Test	- Fabricated + water (full)	151830.7	kg.
Shipping	- Fab. + rem. intls.+ shipping app.	95968.0	kg.
Erected	- Fab. + rem. intls.+ insul. (etc)	95968.0	kg.
Empty	- Fab. + intls. + details + wghts.	98065.0	kg.
Operating	- Empty + operating liquid (no ca)	154172.3	kg.
Field Test	- Empty weight + test medium (full)	153091.2	kg.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 5 of 262

Nozzle Summary:

Step: 37 10:16am Sep 9,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1-Tube Inlet	1094	OK	...	OK	11.33	OK	Passed
N2-Tube outlet	1094	OK	...	OK	11.33	OK	Passed
N3-Shell inlet	1840	...	...	OK	9.13	OK	Passed
N4-Shell outlet	1840	...	...	OK	10.89	OK	Passed
Drain	1840	...	...	OK	6.42	OK	No Calc[*]
Vent	1840	...	...	OK	6.42	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 1840 Nozzle : N3-Shell inlet [Shellside]
 Minimum MAWP Nozzles : 1094 Nozzle : N1-Tube Inlet [Tubeside]

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N1-Tube Inlet	1106.350	0.000	739.151
30	N2-Tube outlet	1106.350	180.000	739.151
70	N3-Shell inlet	5869.701	0.000	1832.000
70	N4-Shell outlet	5869.701	180.000	1840.800
70	Drain	4869.701	180.000	103.725
70	Vent	4869.701	0.000	103.725

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 6 of 262

Nozzle Schedule:

Step: 36 10:16am Sep 9,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
Drain	2.000 in	XXS	LWN	2.375	11.074	162.00	24.00	422.84	2500
Vent	2.000 in	XXS	WNF	2.375	11.074	162.00	24.00	422.84	150
N1-Tube Inlet	16.000 in	80	WNF	16.000	21.412	906.40	16.00	381.38	150
N2-Tube outlet	16.000 in	80	WNF	16.000	21.412	926.40	16.00	381.38	150
N3-Shell inlet	36.000 in Actual		WNF	37.575	20.000	...	...	598.90	300
N4-Shell outlet	36.000 in Actual		WNF	37.575	20.000	...	...	598.90	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
Drain	SA-105	22.500	10.000	18.000	24.000	...
Vent	SA-105	22.500	10.000	18.000	24.000	...
N1-Tube Inl	SA-106 B	16.000	10.000	12.000	16.000	...
N2-Tube out	SA-106 B	16.000	10.000	12.000	16.000	...
N3-Shell in	SA-266 2	14.000	10.000	...	...	...
N4-Shell ou	SA-266 2	14.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
Drain	4869.701	180.0	400.00	0.00	Shell 2
Vent	4869.701	0.0	400.00	0.00	Shell 2
N1-Tube Inlet	1106.350	0.0	350.00	0.00	LHS Ch Shell
N2-Tube outlet	1106.350	180.0	350.00	0.00	LHS Ch Shell
N3-Shell inlet	5869.701	0.0	400.00	0.00	Shell 2
N4-Shell outlet	5869.701	180.0	400.00	0.00	Shell 2

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 7 of 262

Bill of Material:

Step: 35 10:16am Sep 9,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 193.0mm THK X 3000.0mm OD	SA-266 2
1	WELD NECK FLANGE: 268.0mm THK X 3000.0mm OD	SA-266 2
1	CYLINDER: 16.0mm THK X 2700.0mm ID X 1200.0mm	SA-516 70
3	WELD NECK FLANGE: 263.0mm THK X 3000.0mm OD	SA-266 2
1	WELD NECK FLANGE: 260.0mm THK X 3000.0mm OD	SA-266 2
1	CYLINDER: 24.0mm THK X 2700.0mm ID X 2500.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 16.0mm THK X 2700.0mm ID X 50.8mm	SA-516 70
1	TUBESHEET: 2806.0mm X 140.0mm THK	SA-182 F60
4924	TUBES: 6550.0mm X 25.4mm DIA X 1.7mm THK	SA-789
1	FLOATING TUBESHEET: 2650.0mm X 140.0mm THK	SA-182 F60
40	FLOATING HEAD FLANGE BOLTS: 47.6mm DIA	SA-193 B7
80	NUTS FOR FLOATING HEAD FLANGE BOLTS: 47.6mm DIA	...
1	FLOATING HEAD FLANGE GASKET: 2483.0mm OD X 2457.0mm ID	...
2	INSULATION: 272mm X 100mm THK	Hot Insulation
3	LINING: 272.000mm X 6.000mm THK	...
2	SADDLE: 585.0mm X 140 DEG	SA-516 70
2	INSULATION: 1899mm X 100mm THK	Hot
2	CLADDING: 1900.000mm X 3.000mm THK	...
1	INSULATION: 2500mm X 100mm THK	Hot Insulation
1	CLADDING: 2500.000mm X 3.000mm THK	...
1	LINING: 700.000mm X 3.000mm THK	...
1	LINING: 725.800mm X 3.000mm THK	...
1	GASKET: 2776.0mm OD X 2732.0mm ID	...
168	BODY FLANGE BOLTS: 47.6mm DIA	SA-193 B7
336	NUTS FOR BODY FLANGE BOLTS: 47.6mm DIA	...
2	GASKET: 2780.0mm OD X 2736.0mm ID	...
4	CYLINDRICAL SEGMENT 16.0mm THK X 3048.0mm X 1900.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 16.0mm THK X 2386.3mm X 1900.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm THK X 3039.4mm X 700.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm THK X 3063.0mm X 700.0mm	SA-516 70



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 8 of 262

Bill of Material:

Step: 35 10:16am Sep 9,2026

1	CYLINDRICAL SEGMENT 16.0mm THK X 2379.7mm X 700.0mm	SA-516 70
1	NAMEPLATE	...

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 9 of 262

MDMT Summary:

Step: 38 10:16am Sep 9,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
LHS Shell Flg[11]	B	-17	-31	-29	0.749	14.500	1.00	No
Rhs Shell Flg[11]	B	-17	-31	-29	0.749	14.500	1.00	No
Bolting [21]		-48						
Rhs Shell Flg[11]	B	-17	-31	-29	0.749	14.500	1.00	No
Shell-1 [8]	B	-17	-27	-29	0.831	14.500	0.85	No
Shell 2 [8]	B	-4	-33	-29	0.525	22.500	0.85	No
Shell 3 [8]	B	-17	-27	-29	0.831	14.500	0.85	No
Left Saddle f[24]	B	-17	-27	-29	0.831	14.500	0.85	No
RH fixed sadd[24]	B	-17	-27	-29	0.831	14.500	0.85	No
N3-Shell inlet[1]	B	-4	-43	-29	0.453	22.500	1.00	No
Nozzle Flg [5]	B	-18	-104					
N4-Shell outlet[1]	B	-4	-41	-29	0.465	22.500	1.00	No
Nozzle Flg [5]	B	-18	-104					
Drain [1]	A	-4	-41	-29	0.465	22.500	1.00	No
Nozzle Flg [5]	A	-18	-104					
Vent [1]	A	-4	-43	-29	0.453	22.500	1.00	No
Nozzle Flg [5]	A	-18	-48					
Warmest MDMT:		-4	-27					
LHS Ch Cover [11]	B	15	10		0.905	46.250	1.00	No
LHS Ch Flg-1 [11]	B	-14	-34	-29	0.644	16.000	1.00	No
LHS Ch Flg-2 [11]	B	-14	-34	-29	0.644	16.000	1.00	No
LHS Ch Shell [8]	B	-14	-26	-29	0.793	16.000	0.85	No
Rhs ch shell [8]	B	-17	-31	-29	0.751	14.500	0.85	No
Dish End [10]	B	-19	-30	-29	0.799	13.600	0.85	No
Dish End [7]	B	-14	-32	-29	0.677	16.000	0.85	No
N1-Tube Inlet [1]	B	-14	-29	-29	0.758	16.000	1.00	No
Nozzle Flg [5]	A	-18	-48					
N2-Tube outlet[1]	B	-14	-29	-29	0.778	16.000	1.00	No
Nozzle Flg [5]	A	-18	-47					
Bolting [21]		-48						
Bolting [21]		-48						
Warmest MDMT:		15	10					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-29.0	-12.0	Pass
Channel/Tube	10.0	-12.0	Fail

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 10 of 262

MDMT Summary:

Step: 38 10:16am Sep 9, 2026

- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 11 of 262

Warnings and Errors:

Step: 0 10:15am Sep 9, 2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Warning:

There is a flange in this model connected to an element whose temperature is different. The bolt allowable stress is based on it's parent design temperature. You may need to manually adjust the operating bolt allowable stress to a value based upon a representative maximum temperature the bolts will be subject to.

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 12 of 262

Input Echo:

Step: 1 10:15am Sep 9,2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	999.99	KPa.
Channel Side Design Pressure	999.99	KPa.
Shell Side Design Temperature	240.0	°C
Channel Side Design Temperature	82.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 13 of 262

Input Echo:

Step: 1 10:15am Sep 9,2026

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	9000 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 14 of 262

Input Echo:

Step: 1 10:15am Sep 9,2026

Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes

Do not use Nozzle MDMT Interpretation VIII-1 01-37 No

Use Table G instead of exact equation for "A" Yes

Shell Head Joints are Tapered Yes

Compute "K" in corroded condition Yes

Use Code Case 2286 No

Use the MAWP to compute the MDMT Yes

For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes

For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

Using Metric Material Databases, ASME II D Yes

Calculate B31.3 type stress for Nozzles with Loads Yes

Reduce the MDMT due to lower membrane stress Yes

Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/12
Element From Node	10
Element To Node	20
Element Type	Flange
Description	LHS Ch Cover
Distance "FROM" to "TO"	185 mm
Flange Outside Diameter	3000 mm
Element Thickness	185 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	193 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	230.08 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	0 mm
Liquid Density	0.0009984 kg./cm ³

Element#	2/12
Element From Node	20



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 15 of 262

Input Echo:

Step: 1 10:15am Sep 9,2026

Element To Node	30
Element Type	Flange
Description	LHS Ch Flg-1
Distance "FROM" to "TO"	315 mm
Flange Inside Diameter	2700 mm
Element Thickness	260 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	268 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2700 mm
Liquid Density	0.0009984 kg./cm³

Element#	3/12
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	LHS Ch Shell
Distance "FROM" to "TO"	1200 mm
Inside Diameter	2700 mm
Element Thickness	16 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm²
Allowable Stress, Operating	138 N./mm²
Allowable Stress, Hydrotest	234 N./mm²
Material Density	0.00775 kg./cm³
P Number Thickness	32.004 mm
Yield Stress, Operating	242.6 N./mm²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	Yes

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2700 mm
Liquid Density	0.0009984 kg./cm³



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 16 of 262

Input Echo:

Step: 1 10:15am Sep 9,2026

Element From Node	30
Detail Type	Nozzle
Detail ID	N1-Tube Inlet
Dist. from "FROM" Node / Offset dist	600 mm
Nozzle Diameter	16 in.
Nozzle Schedule	80
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	204.09 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	30
Detail Type	Nozzle
Detail ID	N2-Tube outlet
Dist. from "FROM" Node / Offset dist	600 mm
Nozzle Diameter	16 in.
Nozzle Schedule	80
Nozzle Class	150
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	207.66 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	30
Detail Type	Weight
Detail ID	Pass partition
Dist. from "FROM" Node / Offset dist	600 mm
Miscellaneous Weight	400 Kg
Offset from Element Centerline	0 mm

Element#	4/12
Element From Node	40
Element To Node	50
Element Type	Flange
Description	LHS Ch Flg-2
Distance "FROM" to "TO"	310 mm
Flange Inside Diameter	2700 mm
Element Thickness	255 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	263 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	230.08 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 17 of 262

Input Echo:

Step: 1 10:15am Sep 9,2026

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2700 mm
Liquid Density	0.0009984 kg./cm ³

Element#	5/12
Element From Node	50
Element To Node	60
Element Type	Flange
Description	LHS Shell Flg-1
Distance "FROM" to "TO"	307 mm
Flange Inside Diameter	2710 mm
Element Thickness	252 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	260 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	240 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2700 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	50
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	272 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element From Node	50
Detail Type	Lining
Detail ID	cladd
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	272 mm
Thickness of Lining	6 mm
Density	0.008 kg./cm ³

Element#	6/12
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	Shell-1
Distance "FROM" to "TO"	1900 mm
Inside Diameter	2700 mm
Element Thickness	14.5 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 18 of 262

Input Echo:

Step: 1 10:15am Sep 9, 2026

Design Internal Pressure	1000	KPa.
Design Temperature Internal Pressure	240	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	218	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	60	
Detail Type	Saddle	
Detail ID	Left Saddle flo	
Dist. from "FROM" Node / Offset dist	1260	mm
Width of Saddle	585	mm
Height of Saddle at Bottom	1981.2	mm
Saddle Contact Angle	140.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	875	mm
Thickness of Wear Plate	20	mm
Contact Angle, Wear Plate (degrees)	152.0	
Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2700	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	60	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1900	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	60	
Detail Type	Lining	
Detail ID	Cladding	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	1900	mm
Thickness of Lining	3	mm
Density	0.008	kg./cm ³

Element From Node	60	
Detail Type	Weight	
Detail ID	Fireproofing we	
Dist. from "FROM" Node / Offset dist	1260	mm



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 19 of 262

Input Echo:

Step: 1 10:15am Sep 9,2026

Miscellaneous Weight	800	Kgf
Offset from Element Centerline	2000	mm

Element#	7/12
Element From Node	70
Element To Node	80
Element Type	Cylinder
Description	Shell 2
Distance "FROM" to "TO"	2500 mm
Inside Diameter	2700 mm
Element Thickness	22.5 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	24 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	240 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	70
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2700 mm
Liquid Density	0.0009984 kg./cm³

Element From Node	70
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	2500 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm³

Element From Node	70
Detail Type	Lining
Detail ID	cladding
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	2500 mm
Thickness of Lining	3 mm
Density	0.008 kg./cm³

Element From Node	70
Detail Type	Nozzle
Detail ID	N3-Shell inlet
Dist. from "FROM" Node / Offset dist	1500 mm
Nozzle Diameter	36 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	1794.8 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2

Element From Node	70
Detail Type	Nozzle
Detail ID	N4-Shell outlet



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 20 of 262

Input Echo:

Step: 1 10:15am Sep 9,2026

Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	36	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	1573.1	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	

Element From Node	70	
Detail Type	Nozzle	
Detail ID	Drain	
Dist. from "FROM" Node / Offset dist	500	mm
Nozzle Diameter	2	in.
Nozzle Schedule	XXS	
Nozzle Class	2500	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	27.44	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element From Node	70	
Detail Type	Nozzle	
Detail ID	Vent	
Dist. from "FROM" Node / Offset dist	500	mm
Nozzle Diameter	2	in.
Nozzle Schedule	XXS	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	12.015	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	

Element#	8/12	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description	Shell 3	
Distance "FROM" to "TO"	1900	mm
Inside Diameter	2700	mm
Element Thickness	14.5	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1000	KPa.
Design Temperature Internal Pressure	240	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Saddle	
Detail ID	RH fixed saddle	
Dist. from "FROM" Node / Offset dist	640	mm
Width of Saddle	585	mm
Height of Saddle at Bottom	1981.2	mm



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 21 of 262

Input Echo:

Step: 1 10:15am Sep 9,2026

Saddle Contact Angle	140.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	875	mm
Thickness of Wear Plate	20	mm
Contact Angle, Wear Plate (degrees)	152.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm
Element From Node	80	
Detail Type	Liquid	
Detail ID	Liquid: 80	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2700	mm
Liquid Density	0.0009984	kg./cm ³
Element From Node	80	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1900	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³
Element From Node	80	
Detail Type	Lining	
Detail ID	Cladding	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	1900	mm
Thickness of Lining	3	mm
Density	0.008	kg./cm ³
Element From Node	80	
Detail Type	Weight	
Detail ID	Fireproofing we	
Dist. from "FROM" Node / Offset dist	640	mm
Miscellaneous Weight	800	Kgf
Offset from Element Centerline	2000	mm

Element#	9/12	
Element From Node	90	
Element To Node	100	
Element Type	Flange	
Description	Rhs Shell Flg-1	
Distance "FROM" to "TO"	310	mm
Flange Inside Diameter	2710	mm
Element Thickness	255	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	263	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1000	KPa.
Design Temperature Internal Pressure	240	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	136.4	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 22 of 262

Input Echo:

Step: 1 10:15am Sep 9,2026

P Number Thickness	32.004	mm
Yield Stress, Operating	206	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	90	
Detail Type	Liquid	
Detail ID	Liquid: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2700	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	90	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	272	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	90	
Detail Type	Lining	
Detail ID	weldoverlay	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	272	mm
Thickness of Lining	6	mm
Density	0.008	kg./cm ³

Element#	10/12	
Element From Node	100	
Element To Node	110	
Element Type	Flange	
Description	Rhs Shell Flg-2	
Distance "FROM" to "TO"	310	mm
Flange Inside Diameter	2710	mm
Element Thickness	255	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	263	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1000	KPa.
Design Temperature Internal Pressure	240	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	100	
Detail Type	Liquid	
Detail ID	Liquid: 80	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2700	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	100	
Detail Type	Lining	
Detail ID	weldoverlay	
Dist. from "FROM" Node / Offset dist	0	mm



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 23 of 262

Input Echo:

Step: 1 10:15am Sep 9,2026

Height/Length of Lining	272	mm
Thickness of Lining	6	mm
Density	0.008	kg./cm ³

Element#	11/12
Element From Node	110
Element To Node	120
Element Type	Cylinder
Description	Rhs ch shell
Distance "FROM" to "TO"	700 mm
Inside Diameter	2700 mm
Element Thickness	14.5 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	240 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	218 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	110
Detail Type	Liquid
Detail ID	Liquid: 90
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2700 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	110
Detail Type	Lining
Detail ID	cladding
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	700 mm
Thickness of Lining	3 mm
Density	0.008 kg./cm ³

Element#	12/12
Element From Node	120
Element To Node	130
Element Type	Elliptical
Description	Dish End
Distance "FROM" to "TO"	50.8 mm
Inside Diameter	2700 mm
Element Thickness	13.6 mm
Internal Corrosion Allowance	0.8 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 24 of 262

Input Echo:

Step: 1 10:15am Sep 9,2026

Design Internal Pressure	1000	KPa.
Design Temperature Internal Pressure	240	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	120
Detail Type	Liquid
Detail ID	Liquid: 100
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2700 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	120
Detail Type	Lining
Detail ID	cladding
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	725.8 mm
Thickness of Lining	3 mm
Density	0.008 kg./cm ³

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 25 of 262

XY Coordinate Calculations:

Step: 2 10:15am

Sep 9,2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
LHS Ch Cover		191.35	...	185	...
LHS Ch Flg-1		506.35	...	315	...
LHS Ch Shell		1706.35	...	1200	...
LHS Ch Flg-2		2022.7	...	310	...
LHS Shell Flg-1		2469.7	...	307	...
Shell-1		4369.7	...	1900	...
Shell 2		6869.7	...	2500	...
Shell 3		8769.7	...	1900	...
Rhs Shell Flg-1		9086.05	...	310	...
Rhs Shell Flg-2		9396.05	...	310	...
Rhs ch shell		10096	...	700	...
Dish End		10146.8	...	50.8	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 26 of 262

Flg Calc [Int P]: New Flange

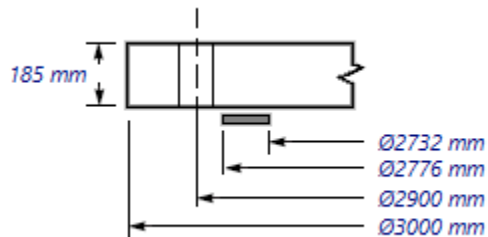
Flng: 31 10:15am Sep 9,2026

Flange Input Data Values

Description: New Flange :

LHS Ch Cover

Flange Type	TEMA Channel Cover		
Design Pressure	P	1000.00	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	3000.000	mm
Flange Thickness	t	185.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	9110.619	mm
Youngs Modulus for Blind Flange Matl.	Sy	197846.66	N./mm ²
Allowed Channel Cover Deflection	dfAllow	3.3750	mm
Diameter of Bolt Circle	C	2900.000	mm
Nominal Bolt Diameter	a	47.6250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		56	
Flange Face Outside Diameter	Fod	2787.000	mm
Flange Face Inside Diameter	Fid	2700.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2776.000	mm
Gasket Inside Diameter	Gi	2732.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm



ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Go - Gi)/2$	22.000 mm
Basic Gasket Width,	$bo = N/2$	11.000 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	8.358 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	2759.285 mm



PV Elite 28 Licensee: SL User
 FileName : 197-C_Channel MOC= CS Page 27 of 262
 Flg Calc [Int P]: New Flange Flng: 31 10:15am Sep 9,2026

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2759^2 \cdot 1000$$

$$= 609788.875 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.36 \cdot \pi \cdot 2759 \cdot 3 \cdot 1000$$

$$= 44327.938 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(609789 + 44328 + 0, 0)$$

$$= 654116.812 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 8.36 \cdot \pi \cdot 2759 + 0 \cdot 0 \cdot 0$$

$$= 509372.312 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((654117 + 0 + 4 \cdot 0 / 2759) / 172, 509372 / 172)$$

$$= 372.961 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	372.961	832.411	
Radial Distance between Bolts and Edge:	47.625	50.000	
Circ. Spacing between the Bolts:	101.600	162.604	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 832 \cdot 172 / (68.9 \cdot \pi (2776 + 2732))$$

$$= 12.000 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 832$$

$$= 1459922.62 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o$$

$$= 654116.81 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2900 - 2759) / 2$$

$$= 70.3576 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot W_{m1} \cdot h_G \cdot B_{sc} / (t^2 \cdot G) + C \cdot Z \cdot P_{eq} \cdot G^2 / t^2$$

$$= 1.9 \cdot 654117 \cdot 70.4 \cdot 1 / (182^2 \cdot 2759) + 0.3 \cdot 1 \cdot 1000 \cdot 2759^2 / 182^2$$

$$= 78.34 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 28 of 262

Flg Calc [Int P]: New Flange

Flng: 31 10:15am

Sep 9, 2026

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 1459922 \cdot 70.4 \cdot 1 / (182^2 \cdot 2759)$$

$$= 20.94 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 654117 / 832$$

$$= 77.06 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (509372 / 832)$$

$$= 60.01 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Tangential Flange	78.34	138.00	20.94	138.00
Bolting	77.06	172.00	60.01	172.00

TEMA Channel Cover Deflection Calculations:

Actual deflection at center per TEMA formula 3.2794 mm

Allowed deflection at center of Cover 3.3750 mm

Reqd. Blind Flange Thickness at Center 140.130 mm

Reqd. Blind Flange Thickness at Gasket 73.897 mm

Min. Required Flange Thickness per TEMA 180.265 mm

Estimated M.A.W.P. (Operating) 1045 KPa.

Estimated Finished Weight of Flange at given Thk. 10135.2 kg.

Estimated Unfinished Weight of Forging at given Thk 10135.2 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.9, Temperature Reduction per Fig. UCS 66.1 = 5 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 15 °C

Min Metal Temp. at Required thickness (UCS 66.1) 10 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Actual	Gasket Seating Allowed
Tangential Flange	75.82	138.00	20.27	138.00
Bolting	77.06	172.00	60.01	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 179.222 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 29 of 262

Flg Calc [Int P]: New Flange

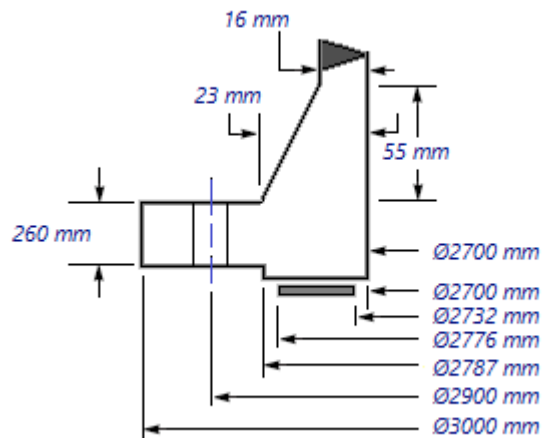
Flng: 32 10:15am Sep 9,2026

Flange Input Data Values

Description: New Flange :

LHS Ch Flg-1

Flange Type	Integral Weld Neck		
Design Pressure	P	1026.47	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2700.000	mm
Flange Outside Diameter	A	3000.000	mm
Flange Thickness	t	260.0000	mm
Thickness of Hub at Small End	go	16.0000	mm
Thickness of Hub at Large End	gl	23.0000	mm
Length of Hub	h	55.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2900.000	mm
Nominal Bolt Diameter	a	47.6250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		56	
Flange Face Outside Diameter	Fod	2787.000	mm
Flange Face Inside Diameter	Fid	2700.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2776.000	mm
Gasket Inside Diameter	Gi	2732.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm





PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 30 of 262

Flg Calc [Int P]: New Flange

Flng: 32 10:15am

Sep 9, 2026

ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1026 \cdot (2700/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 1026) + Ca$$

$$= 13.1097 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 13) / (1353 + 0.6 \cdot 13)$$

$$= 1318.242 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 16) / (1350 + 0.6 \cdot 16)$$

$$= 1623.883 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	257.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2706.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	20.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	13.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	77.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	22.000 mm
Basic Gasket Width, $b_o = N/2$	11.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	8.358 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2759.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2759^2 \cdot 1026$$

$$= 625927.188 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.36 \cdot \pi \cdot 2759 \cdot 3 \cdot 1026$$

$$= 45501.098 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2706^2 \cdot 1026$$

$$= 601985.938 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 625927 - 601986$$

$$= 23941.291 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o,1 + W_o,2 + H'p, 0)$$

$$= \max(625927 + 45501 + 0, 0)$$

$$= 671428.312 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 8.36 \cdot \pi \cdot 2759 + 0 \cdot 0$$

$$= 509372.312 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G)/S_b, W_g/S_a)$$

$$= \max((671428 + 0 + 4 \cdot 0/2759)/172, 509372/172)$$

$$= 382.832 \text{ cm}^2$$



PV Elite 28 Licensee: SL User
 FileName : 197-C_Channel MOC= CS Page 31 of 262
 Flg Calc [Int P]: New Flange Flng: 32 10:15am Sep 9,2026

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 47.6 + 6 \cdot 257/(3 + 0.5)$$

$$= 535.821 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2900 \cdot \sin(\pi / 56)$$

$$= 162.604 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{163/(2 \cdot 47.6 + 257)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	382.832	832.411	
Radial Distance between Hub and Bolts:	60.325	77.000	
Radial Distance between Bolts and Edge:	47.625	50.000	
Circ. Spacing between the Bolts:	101.600	162.604	535.821

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 832 \cdot 172 / (68.9 \cdot \pi(2776 + 2732))$$

$$= 12.000 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 832$$

$$= 1459922.62 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 671428 - 625927$$

$$= 45501.12 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2900 - 2759) / 2$$

$$= 70.3576 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((2900 - 2706) / 2 + 70.4)$$

$$= 83.6788 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2900 - 2706 - 20) / 2$$

$$= 87.0000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
---------	-------	----------	-----------	--------



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 32 of 262

Flg Calc [Int P]: New Flange

Flng: 32 10:15am

Sep 9, 2026

End Pressure, Md	601986.	87.0000	1.0000	52374.
Face Pressure, Mt	23941.	83.6788	1.0000	2003.
Gasket Load, Mg	45501.	70.3576	1.0000	3201.
Gasket Seating, Matm	1459923.	70.3576	1.0000	102719.

Total Moment for Operation, Mo 57579. Kg-m.
Total Moment for Gasket seating, Mg 102719. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 187.558 mm
Hub Ratio, h/h0 = HL / H0 0.293
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.538

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.882
Factor V per 4.16.5 0.366
Factor f per 4.16.5 1.255

Stress factor from Table 4.16.4 [K]:

= A / B
= 3000/2706
= 1.109

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169 * (K² * log10(K))) / (K² - 1)) / (K - 1)
= (0.66845 + (5.7169 * (1.11² * log10(1.11))) / (1.11² - 1)) / (1.11 - 1)
= 18.797

Stress factor from Table 4.16.4 [T]:

= (K² * (1 + 8.55246 * log10(K)) - 1) / ((1.0472 + 1.9448 * K²) * (K - 1))
= (1.11² * (1 + 8.55246 * log10(1.11)) - 1) / ((1.0472 + 1.9448 * 1.11²) * (1.11 - 1))
= 1.874

Stress factor from Table 4.16.4 [U]:

= (K² * (1 + 8.55246 * log10(K)) - 1) / (1.36136 * (K² - 1) * (K - 1))
= (1.11² * (1 + 8.55246 * log10(1.11)) - 1) / (1.36136 * (1.11² - 1) * (1.11 - 1))
= 20.657

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (1.11² + 1) / (1.11² - 1)
= 9.730

Stress factor from Table 4.16.4 for integral flanges [d]:

= U * g0² * h0 / V
= 20.7 * 13² * 188 / 0.37
= 1789969.000 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.88/188
= 0.005 1/mm

Stress factor from Table 4.16.4 [L]:

= (t * e + 1) / T + t³/d
= (257 * 0.0047 + 1) / 1.87 + 257³/1789969
= 10.661

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f * Mop / Bcor) / (L * g1²)
= (1.25 * 57579 / 2706) / (10.7 * 20²)
= 61.41 N./mm²

Longitudinal Hub Stress, Seating [SHa]:



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 33 of 262

Flg Calc [Int P]: New Flange

Flng: 32 10:15am

Sep 9, 2026

$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g1^2)$$

$$= (1.25 \cdot 102719 / 2706) / (10.7 \cdot 20^2)$$

$$= 109.55 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.61 \cdot 57579 / 2706) / (10.7 \cdot 257^2)$$

$$= 0.77 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.61 \cdot 102719 / 2706) / (10.7 \cdot 257^2)$$

$$= 1.38 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (18.8 \cdot 57579 / (257^2 \cdot 2706)) - 9.73 \cdot 0.77$$

$$= 51.86 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (18.8 \cdot 102719 / (257^2 \cdot 2706)) - 9.73 \cdot 1.38$$

$$= 92.51 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (61.4 + \max(0.77, 51.9)) / 2$$

$$= 56.63 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (110 + \max(1.38, 92.5)) / 2$$

$$= 101.03 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 671428 / 832$$

$$= 79.10 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (509372 / 832)$$

$$= 60.01 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	61.41	207.00	109.55	207.00
Radial Flange	0.77	138.00	1.38	138.00
Tangential Flange	51.86	138.00	92.51	138.00
Maximum Average	56.63	138.00	101.03	138.00
Bolting	79.10	172.00	60.01	172.00

Minimum Required Flange Thickness [Rigidity] 254.381 mm

Estimated M.A.W.P. (Operating) 1318 KPa.

Estimated Finished Weight of Flange at given Thk. 2777.4 kg.

Estimated Unfinished Weight of Forging at given Thk 3278.9 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 34 of 262

Flg Calc [Int P]: New Flange

Flng: 32 10:15am

Sep 9, 2026

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	441.40	499.99	0.883	441.40	499.99	0.883
Tangential Flange	St	374.99	374.99	1.000	374.99	374.99	1.000
Tangential Flange	Sto	121.33	250.00	0.485	121.33	250.00	0.485
Combined	Sh/f + St	728.81	749.99	0.972	728.81	749.99	0.972
Combined	Sr + St	380.59	749.99	0.507	380.59	749.99	0.507
Combined	Sh + Sto	562.72	749.99	0.750	562.72	749.99	0.750
Radial Flange	Sr	5.59	499.99	0.011	5.59	499.99	0.011
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.53/1.23)

= 1.2475

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)

= max(0.25; 0.4/1.16) if h/ho < 0.35, else max (0.25; 0.28/0.77)

= 0.3429

In this case, the hub ratio h/ho, 0.2932 was < 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 35 of 262

Flg Calc [Int P]: New Flange

Flng: 32 10:15am

Sep 9, 2026

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	697.2 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	14.8645 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.682 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (2776^2 - 2732^2)$$

$$= 1903.4 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 207 \cdot 1903 / (56 \cdot 14.9)$$

$$= 473.3 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(473; 508)$$

$$= 473.3 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(473; 290)$$

$$= 473.3 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(473; 697)$$

$$= 473.3 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$\therefore Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$\therefore 473 \geq 140 (1903 / (56 \cdot 14.9))$$

$$\therefore 473.3 \geq 320.1 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$\therefore Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$\therefore 473 \geq (97 \cdot 1903 + \pi/4 \cdot 1026 \cdot 2732^2) / (0.7 \cdot 14.9 \cdot 56)$$

$$\therefore 473.3 \geq 420.1 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$\therefore Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$\therefore 473 \leq 276 (1903 / (56 \cdot 14.9))$$

$$\therefore 473.3 \leq 631.1 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$\therefore Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$\therefore 473 \leq 697 (1/0.68)$$



PV Elite 28 Licensee: SL User
 FileName : 197-C_Channel MOC= CS Page 36 of 262
 Flg Calc [Int P]: New Flange Flng: 32 10:15am Sep 9,2026

:: 473.3 ≤ 1022.5 N./mm² Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 473 \cdot 0.2 \cdot 14.9 \cdot 47.6$$

$$= 683.4 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 102719 / 1 \cdot 9807 \cdot 0.37 / (10 \cdot 201334 \cdot 13^2 \cdot 188 \cdot 0.3)$
 $= 0.999$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 57579 / 1 \cdot 9807 \cdot 0.37 / (10 \cdot 197847 \cdot 13^2 \cdot 188 \cdot 0.3)$
 $= 0.570$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 102719 / 1 \cdot 9807 \cdot 0.37 / (10.7 \cdot 201334 \cdot 13^2 \cdot 188 \cdot 0.3)$
 $= 0.941$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 57579 / 1 \cdot 9807 \cdot 0.37 / (10.7 \cdot 197847 \cdot 13^2 \cdot 188 \cdot 0.3)$
 $= 0.536$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.64, Temperature Reduction per Fig. UCS 66.1 = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -34 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	62.47	207.00	109.68	207.00
Radial Flange	1.03	138.00	1.81	138.00
Tangential Flange	48.06	138.00	84.38	138.00
Maximum Average	55.27	138.00	97.03	138.00
Bolting	79.10	172.00	60.01	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 244.500 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 37 of 262

Flg Calc [Int P]: New Flange

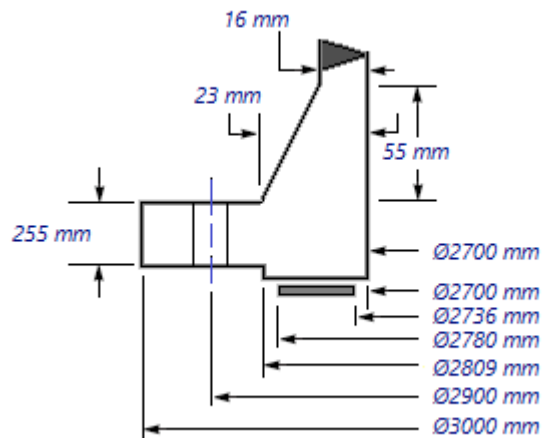
Flng: 33 10:15am Sep 9,2026

Flange Input Data Values

Description: New Flange :

LHS Ch Flg-2

Flange Type	Integral Weld Neck		
Design Pressure	P	1026.47	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2700.000	mm
Flange Outside Diameter	A	3000.000	mm
Flange Thickness	t	255.0000	mm
Thickness of Hub at Small End	go	16.0000	mm
Thickness of Hub at Large End	g1	23.0000	mm
Length of Hub	h	55.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2900.000	mm
Nominal Bolt Diameter	a	47.6250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		56	
Flange Face Outside Diameter	Fod	2809.000	mm
Flange Face Inside Diameter	Fid	2700.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2780.000	mm
Gasket Inside Diameter	Gi	2736.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm





PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 38 of 262

Flg Calc [Int P]: New Flange

Flng: 33 10:15am

Sep 9, 2026

ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1026 \cdot (2700/2 + 3)) / (138 \cdot 1 - 0.6 \cdot 1026) + Ca$$

$$= 13.1097 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 13) / (1353 + 0.6 \cdot 13)$$

$$= 1318.242 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 16) / (1350 + 0.6 \cdot 16)$$

$$= 1623.883 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	252.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2706.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	20.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	13.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	77.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	22.000 mm
Basic Gasket Width, $bo = N/2$	11.000 mm
Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$	8.358 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2763.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2763^2 \cdot 1026$$

$$= 627743.250 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.36 \cdot \pi \cdot 2763 \cdot 3 \cdot 1026$$

$$= 45567.059 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2706^2 \cdot 1026$$

$$= 601985.938 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 627743 - 601986$$

$$= 25757.336 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(627743 + 45567 + 0, 0)$$

$$= 673310.312 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 8.36 \cdot \pi \cdot 2763 + 0 \cdot 0$$

$$= 510110.719 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((673310 + 0 + 4 \cdot 0/2763)/172, 510111/172)$$

$$= 383.905 \text{ cm}^2$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 39 of 262

Flg Calc [Int P]: New Flange

Flng: 33 10:15am

Sep 9, 2026

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 47.6 + 6 \cdot 252/(3 + 0.5)$$

$$= 527.250 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2900 \cdot \sin(\pi / 56)$$

$$= 162.604 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{163/(2 \cdot 47.6 + 252)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	383.905	832.411	
Radial Distance between Hub and Bolts:	60.325	77.000	
Radial Distance between Bolts and Edge:	47.625	50.000	
Circ. Spacing between the Bolts:	101.600	162.604	527.250

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 832 \cdot 172 / (68.9 \cdot \pi (2780 + 2736))$$

$$= 11.983 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm

Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 832$$

$$= 1459922.62 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 673310 - 627743$$

$$= 45567.07 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2900 - 2763) / 2$$

$$= 68.3577 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((2900 - 2706) / 2 + 68.4)$$

$$= 82.6788 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2900 - 2706 - 20) / 2$$

$$= 87.0000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
---------	-------	----------	-----------	--------



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 40 of 262

Flg Calc [Int P]: New Flange

Flng: 33 10:15am

Sep 9, 2026

End Pressure, Md	601986.	87.0000	1.0000	52374.
Face Pressure, Mt	25757.	82.6788	1.0000	2130.
Gasket Load, Mg	45567.	68.3577	1.0000	3115.
Gasket Seating, Matm	1459923.	68.3577	1.0000	99799.

Total Moment for Operation, Mo 57618. Kg-m.
Total Moment for Gasket seating, Mg 99799. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 187.558 mm
Hub Ratio, h/h0 = HL / H0 0.293
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.538

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.882
Factor V per 4.16.5 0.366
Factor f per 4.16.5 1.255

Stress factor from Table 4.16.4 [K]:

= A / B
= 3000/2706
= 1.109

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169 * (K^2 * log10(K))) / (K^2 - 1)) / (K - 1)
= (0.66845 + (5.7169 * (1.11^2 * log10(1.11))) / (1.11^2 - 1)) / (1.11 - 1)
= 18.797

Stress factor from Table 4.16.4 [T]:

= (K^2 * (1 + 8.55246 * log10(K)) - 1) / ((1.0472 + 1.9448 * K^2) * (K - 1))
= (1.11^2 * (1 + 8.55246 * log10(1.11)) - 1) / ((1.0472 + 1.9448 * 1.11^2) * (1.11 - 1))
= 1.874

Stress factor from Table 4.16.4 [U]:

= (K^2 * (1 + 8.55246 * log10(K)) - 1) / (1.36136 * (K^2 - 1) * (K - 1))
= (1.11^2 * (1 + 8.55246 * log10(1.11)) - 1) / (1.36136 * (1.11^2 - 1) * (1.11 - 1))
= 20.657

Stress factor from Table 4.16.4 [Z]:

= (K^2 + 1) / (K^2 - 1)
= (1.11^2 + 1) / (1.11^2 - 1)
= 9.730

Stress factor from Table 4.16.4 for integral flanges [d]:

= U * g0^2 * h0 / V
= 20.7 * 13^2 * 188 / 0.37
= 1789969.000 mm^3

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.88 / 188
= 0.005 1/mm

Stress factor from Table 4.16.4 [L]:

= (t * e + 1) / (T + t^3/d)
= (252 * 0.0047 + 1) / (1.87 + 252^3 / 1789969)
= 10.106

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f * Mop / Bcor) / (L * g1^2)
= (1.25 * 57618 / 2706) / (10.1 * 20^2)
= 64.82 N./mm^2

Longitudinal Hub Stress, Seating [SHa]:



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 41 of 262

Flg Calc [Int P]: New Flange

Flng: 33 10:15am

Sep 9, 2026

$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g1^2)$$

$$= (1.25 \cdot 99799 / 2706) / (10.1 \cdot 20^2)$$

$$= 112.28 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.58 \cdot 57618 / 2706) / (10.1 \cdot 252^2)$$

$$= 0.84 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.58 \cdot 99799 / 2706) / (10.1 \cdot 252^2)$$

$$= 1.45 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (18.8 \cdot 57618 / (252^2 \cdot 2706)) - 9.73 \cdot 0.84$$

$$= 53.64 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (18.8 \cdot 99799 / (252^2 \cdot 2706)) - 9.73 \cdot 1.45$$

$$= 92.91 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (64.8 + \max(0.84, 53.6)) / 2$$

$$= 59.23 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (112 + \max(1.45, 92.9)) / 2$$

$$= 102.60 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 673310 / 832$$

$$= 79.32 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (510111 / 832)$$

$$= 60.10 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	64.82	207.00	112.28	207.00
Radial Flange	0.84	138.00	1.45	138.00
Tangential Flange	53.64	138.00	92.91	138.00
Maximum Average	59.23	138.00	102.60	138.00
Bolting	79.32	172.00	60.10	172.00

Minimum Required Flange Thickness [Rigidity] 251.638 mm

Estimated M.A.W.P. (Operating) 1318 KPa.

Estimated Finished Weight of Flange at given Thk. 2725.4 kg.

Estimated Unfinished Weight of Forging at given Thk 3226.8 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 42 of 262

Flg Calc [Int P]: New Flange

Flng: 33 10:15am

Sep 9, 2026

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	450.48	499.99	0.901	450.48	499.99	0.901
Tangential Flange	St	374.99	374.99	1.000	374.99	374.99	1.000
Tangential Flange	Sto	123.82	250.00	0.495	123.82	250.00	0.495
Combined	Sh/f + St	736.09	749.99	0.981	736.09	749.99	0.981
Combined	Sr + St	380.86	749.99	0.508	380.86	749.99	0.508
Combined	Sh + Sto	574.30	749.99	0.766	574.30	749.99	0.766
Radial Flange	Sr	5.87	499.99	0.012	5.87	499.99	0.012
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.53/1.23)

= 1.2475

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if h/h0 < 0.35, else max (0.25; fn3)

= max(0.25; 0.4/1.16) if h/h0 < 0.35, else max (0.25; 0.28/0.77)

= 0.3429

In this case, the hub ratio h/h0, 0.2932 was < 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 43 of 262

Flg Calc [Int P]: New Flange

Flng: 33 10:15am

Sep 9, 2026

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	694.2 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	14.8645 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.696 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (2780^2 - 2736^2)$$

$$= 1906.2 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 207 \cdot 1906 / (56 \cdot 14.9)$$

$$= 474.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(474; 508)$$

$$= 474.0 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(474; 290)$$

$$= 474.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(474; 694)$$

$$= 474.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 474 \geq 140 (1906 / (56 \cdot 14.9))$$

$$:: 474.0 \geq 320.6 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 474 \geq (97 \cdot 1906 + \pi/4 \cdot 1026 \cdot 2736^2) / (0.7 \cdot 14.9 \cdot 56)$$

$$:: 474.0 \geq 420.9 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 474 \leq 276 (1906 / (56 \cdot 14.9))$$

$$:: 474.0 \leq 632.0 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$:: 474 \leq 694 (1/0.7)$$



PV Elite 28 Licensee: SL User
 FileName : 197-C_Channel MOC= CS Page 44 of 262
 Flg Calc [Int P]: New Flange Flng: 33 10:15am Sep 9, 2026

:: 474.0 ≤ 997.6 N./mm² Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 474 \cdot 0.2 \cdot 14.9 \cdot 47.6$$

$$= 684.4 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 118 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 99799 / 1 \cdot 9807 \cdot 0.37 / (9.74 \cdot 201334 \cdot 13^2 \cdot 188 \cdot 0.3)$$

$$= 1.000 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 57618 / 1 \cdot 9807 \cdot 0.37 / (9.74 \cdot 197847 \cdot 13^2 \cdot 188 \cdot 0.3)$$

$$= 0.587 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 99799 / 1 \cdot 9807 \cdot 0.37 / (10.1 \cdot 201334 \cdot 13^2 \cdot 188 \cdot 0.3)$$

$$= 0.964 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 57618 / 1 \cdot 9807 \cdot 0.37 / (10.1 \cdot 197847 \cdot 13^2 \cdot 188 \cdot 0.3)$$

$$= 0.566 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.64, Temperature Reduction per Fig. UCS 66.1 = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-34 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	65.78	207.00	112.13	207.00
Radial Flange	1.11	138.00	1.90	138.00
Tangential Flange	49.59	138.00	84.53	138.00
Maximum Average	57.69	138.00	98.33	138.00
Bolting	79.32	172.00	60.10	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 241.808 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 45 of 262

Flg Calc [Int P]: New Flange

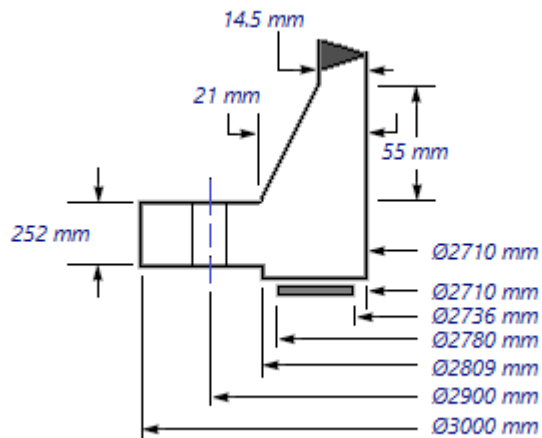
Flng: 34 10:15am Sep 9,2026

Flange Input Data Values

Description: New Flange :

LHS Shell Flg-1

Flange Type	Integral Weld Neck		
Design Pressure	P	1026.47	KPa.
Design Temperature		240	°C
Internal Corrosion Allowance	ci	0.8000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2710.000	mm
Flange Outside Diameter	A	3000.000	mm
Flange Thickness	t	252.0000	mm
Thickness of Hub at Small End	go	14.5000	mm
Thickness of Hub at Large End	g1	21.0000	mm
Length of Hub	h	55.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	136.40	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2900.000	mm
Nominal Bolt Diameter	a	47.6250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		56	
Flange Face Outside Diameter	Fod	2809.000	mm
Flange Face Inside Diameter	Fid	2710.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2780.000	mm
Gasket Inside Diameter	Gi	2736.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm





PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 46 of 262

Flg Calc [Int P]: New Flange

Flng: 34 10:15am

Sep 9, 2026

ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1026 \cdot (2710/2 + 0.8)) / (138 \cdot 1 - 0.6 \cdot 1026) + Ca$$

$$= 10.9306 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 13.7) / (1356 + 0.6 \cdot 13.7)$$

$$= 1385.944 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 14.5) / (1355 + 0.6 \cdot 14.5)$$

$$= 1467.220 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	251.200 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2711.600 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	20.200 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	13.700 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	74.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	22.000 mm
Basic Gasket Width, $b_o = N/2$	11.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	8.358 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2763.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2763^2 \cdot 1026$$

$$= 627743.250 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.36 \cdot \pi \cdot 2763 \cdot 3 \cdot 1026$$

$$= 45567.059 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2712^2 \cdot 1026$$

$$= 604480.188 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 627743 - 604480$$

$$= 23263.045 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o, 1 + W_o, 2 + H'p, 0)$$

$$= \max(627743 + 45567 + 0, 0)$$

$$= 673310.312 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 8.36 \cdot \pi \cdot 2763 + 0 \cdot 0$$

$$= 510110.719 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G)/S_b, W_g/S_a)$$

$$= \max((673310 + 0 + 4 \cdot 0/2763)/172, 510111/172)$$

$$= 383.905 \text{ cm}^2$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 47 of 262

Flg Calc [Int P]: New Flange

Flng: 34 10:15am

Sep 9, 2026

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 47.6 + 6 \cdot 251/(3 + 0.5)$$

$$= 525.879 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2900 \cdot \sin(\pi / 56)$$

$$= 162.604 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{163/(2 \cdot 47.6 + 251)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	383.905	832.411	
Radial Distance between Hub and Bolts:	60.325	74.000	
Radial Distance between Bolts and Edge:	47.625	50.000	
Circ. Spacing between the Bolts:	101.600	162.604	525.879

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 832 \cdot 172 / (68.9 \cdot \pi(2780 + 2736))$$

$$= 11.983 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm

Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 832$$

$$= 1459922.62 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 673310 - 627743$$

$$= 45567.07 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2900 - 2763) / 2$$

$$= 68.3577 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5(C - B) / 2 + hg$$

$$= 0.5(2900 - 2712) / 2 + 68.4$$

$$= 81.2788 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2900 - 2712 - 20.2) / 2$$

$$= 84.1000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
---------	-------	----------	-----------	--------



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 48 of 262

Flg Calc [Int P]: New Flange

Flng: 34 10:15am

Sep 9, 2026

End Pressure, Md	604480.	84.1000	1.0000	50838.
Face Pressure, Mt	23263.	81.2788	1.0000	1891.
Gasket Load, Mg	45567.	68.3577	1.0000	3115.
Gasket Seating, Matm	1459923.	68.3577	1.0000	99799.

Total Moment for Operation, Mo 55844. Kg-m.
Total Moment for Gasket seating, Mg 99799. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 192.741 mm
Hub Ratio, h/h0 = HL / H0 0.285
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.474

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.885
Factor V per 4.16.5 0.381
Factor f per 4.16.5 1.177

Stress factor from Table 4.16.4 [K]:

= A / B
= 3000/2712
= 1.106

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169 * (K² * log10(K))) / (K² - 1)) / (K - 1)
= (0.66845 + (5.7169 * (1.11² * log10(1.11))) / (1.11² - 1)) / (1.11 - 1)
= 19.176

Stress factor from Table 4.16.4 [T]:

= (K² * (1 + 8.55246 * log10(K)) - 1) / ((1.0472 + 1.9448 * K²) * (K - 1))
= (1.11² * (1 + 8.55246 * log10(1.11)) - 1) / ((1.0472 + 1.9448 * 1.11²) * (1.11 - 1))
= 1.875

Stress factor from Table 4.16.4 [U]:

= (K² * (1 + 8.55246 * log10(K)) - 1) / (1.36136 * (K² - 1) * (K - 1))
= (1.11² * (1 + 8.55246 * log10(1.11)) - 1) / (1.36136 * (1.11² - 1) * (1.11 - 1))
= 21.073

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (1.11² + 1) / (1.11² - 1)
= 9.927

Stress factor from Table 4.16.4 for integral flanges [d]:

= U * g0² * h0 / V
= 21.1 * 13.7² * 193 / 0.38
= 2002180.875 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.89 / 193
= 0.005 1/mm

Stress factor from Table 4.16.4 [L]:

= (t * e + 1) / T + t³ / d
= (251 * 0.0046 + 1) / 1.87 + 251³ / 2002180
= 9.066

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f * Mop / Bcor) / (L * g1²)
= (1.18 * 55844 / 2712) / (9.07 * 20.2²)
= 64.25 N./mm²

Longitudinal Hub Stress, Seating [SHa]:



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 49 of 262

Flg Calc [Int P]: New Flange

Flng: 34 10:15am

Sep 9, 2026

$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g^{12})$$

$$= (1.18 \cdot 99799 / 2712) / (9.07 \cdot 20.2^2)$$

$$= 114.83 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.54 \cdot 55844 / 2712) / (9.07 \cdot 251^2)$$

$$= 0.90 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.54 \cdot 99799 / 2712) / (9.07 \cdot 251^2)$$

$$= 1.60 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (19.2 \cdot 55844 / (251^2 \cdot 2712)) - 9.93 \cdot 0.9$$

$$= 52.48 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (19.2 \cdot 99799 / (251^2 \cdot 2712)) - 9.93 \cdot 1.6$$

$$= 93.79 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (64.3 + \max(0.9, 52.5)) / 2$$

$$= 58.37 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (115 + \max(1.6, 93.8)) / 2$$

$$= 104.31 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 673310 / 832$$

$$= 79.32 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (510111 / 832)$$

$$= 60.10 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	64.25	204.60	114.83	207.00
Radial Flange	0.90	136.40	1.60	138.00
Tangential Flange	52.48	136.40	93.79	138.00
Maximum Average	58.37	136.40	104.31	138.00
Bolting	79.32	172.00	60.10	172.00

Minimum Required Flange Thickness [Rigidity] 250.190 mm

Estimated M.A.W.P. (Operating) 1386 KPa.

Estimated Finished Weight of Flange at given Thk. 2605.0 kg.

Estimated Unfinished Weight of Forging at given Thk 3094.5 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 50 of 262

Flg Calc [Int P]: New Flange

Flng: 34 10:15am

Sep 9, 2026

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	419.78	470.86	0.892	419.78	499.99	0.840
Tangential Flange	St	346.16	353.14	0.980	346.16	374.99	0.923
Tangential Flange	Sto	113.85	235.43	0.484	113.85	250.00	0.455
Combined	Sh/f + St	706.29	706.29	1.000	706.29	749.99	0.942
Combined	Sr + St	352.07	706.29	0.498	352.07	749.99	0.469
Combined	Sh + Sto	533.63	706.29	0.756	533.63	749.99	0.712
Radial Flange	Sr	5.91	470.86	0.013	5.91	499.99	0.012
Maximum Stress Ratio				1.000			0.942

Coefficients for f and f_2

	f	f_2	f_3
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g_1}{g_0} \right) + e \left(\frac{h}{h_0} \right) + g \left(\frac{g_1}{g_0} \right)^2 + i \left(\frac{h}{h_0} \right)^2 + k \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}{\left[1 + b \left(\frac{g_1}{g_0} \right) + d \left(\frac{h}{h_0} \right) + f \left(\frac{g_1}{g_0} \right)^2 + h \left(\frac{h}{h_0} \right)^2 + j \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.43/1.22)

= 1.1656

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h_0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.37/1.16) if $h/h_0 < 0.35$, else max (0.25; 0.26/0.77)

= 0.3161

In this case, the hub ratio h/h_0 , 0.2854 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 51 of 262

Flg Calc [Int P]: New Flange

Flng: 34 10:15am

Sep 9, 2026

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	634.8 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	14.8645 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	206.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.943 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (2780^2 - 2736^2)$$

$$= 1906.2 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 207 \cdot 1906 / (56 \cdot 14.9)$$

$$= 474.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(474; 508)$$

$$= 474.0 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(474; 290)$$

$$= 474.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(474; 635)$$

$$= 474.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 474 \geq 140 (1906 / (56 \cdot 14.9))$$

$$:: 474.0 \geq 320.6 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 474 \geq (97 \cdot 1906 + \pi/4 \cdot 1026 \cdot 2736^2) / (0.7 \cdot 14.9 \cdot 56)$$

$$:: 474.0 \geq 420.9 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 474 \leq 276 (1906 / (56 \cdot 14.9))$$

$$:: 474.0 \leq 632.0 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$:: 474 \leq 635 (1/0.94)$$



PV Elite 28 Licensee: SL User
 FileName : 197-C_Channel MOC= CS Page 52 of 262
 Flg Calc [Int P]: New Flange Flng: 34 10:15am Sep 9,2026

:: 474.0 ≤ 673.4 N./mm² Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 474 \cdot 0.2 \cdot 14.9 \cdot 47.6$$

$$= 684.4 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 91 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 99799 / 1 \cdot 9807 \cdot 0.38 / (8.89 \cdot 201334 \cdot 13.7^2 \cdot 193 \cdot 0.3)$$

$$= 1.000 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 55844 / 1 \cdot 9807 \cdot 0.38 / (8.89 \cdot 188345 \cdot 13.7^2 \cdot 193 \cdot 0.3)$$

$$= 0.598 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 99799 / 1 \cdot 9807 \cdot 0.38 / (9.07 \cdot 201334 \cdot 13.7^2 \cdot 193 \cdot 0.3)$$

$$= 0.980 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 55844 / 1 \cdot 9807 \cdot 0.38 / (9.07 \cdot 188345 \cdot 13.7^2 \cdot 193 \cdot 0.3)$$

$$= 0.586 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.75, Temperature Reduction per Fig. UCS 66.1 = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -17 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -31 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	64.59	207.00	114.91	207.00
Radial Flange	0.97	138.00	1.72	138.00
Tangential Flange	51.38	138.00	91.41	138.00
Maximum Average	57.98	138.00	103.16	138.00
Bolting	79.32	172.00	60.10	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 247.548 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 53 of 262

Flg Calc [Int P]: New Flange

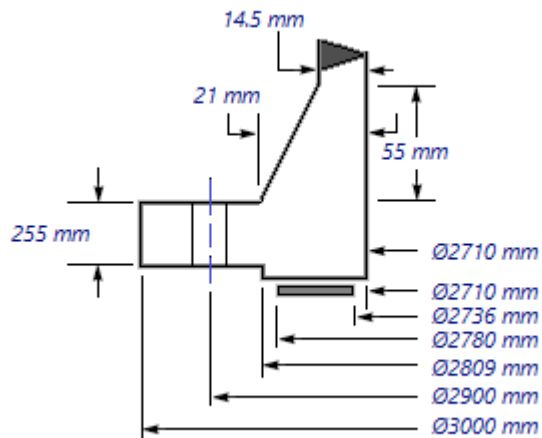
Flng: 35 10:15am Sep 9,2026

Flange Input Data Values

Description: New Flange :

Rhs Shell Flg-1

Flange Type	Integral Weld Neck		
Design Pressure	P	1026.47	KPa.
Design Temperature		240	°C
Internal Corrosion Allowance	ci	0.8000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2710.000	mm
Flange Outside Diameter	A	3000.000	mm
Flange Thickness	t	255.0000	mm
Thickness of Hub at Small End	go	14.5000	mm
Thickness of Hub at Large End	g1	21.0000	mm
Length of Hub	h	55.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	136.40	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2900.000	mm
Nominal Bolt Diameter	a	47.6250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		56	
Flange Face Outside Diameter	Fod	2809.000	mm
Flange Face Inside Diameter	Fid	2710.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2780.000	mm
Gasket Inside Diameter	Gi	2736.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm





PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 54 of 262

Flg Calc [Int P]: New Flange

Flng: 35 10:15am

Sep 9, 2026

ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1026 \cdot (2710/2 + 0.8)) / (138 \cdot 1 - 0.6 \cdot 1026) + Ca$$

$$= 10.9306 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 13.7) / (1356 + 0.6 \cdot 13.7)$$

$$= 1385.944 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 14.5) / (1355 + 0.6 \cdot 14.5)$$

$$= 1467.220 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	254.200 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2711.600 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	20.200 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	13.700 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	74.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	22.000 mm
Basic Gasket Width, $b_o = N/2$	11.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	8.358 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2763.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2763^2 \cdot 1026$$

$$= 627743.250 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.36 \cdot \pi \cdot 2763 \cdot 3 \cdot 1026$$

$$= 45567.059 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2712^2 \cdot 1026$$

$$= 604480.188 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 627743 - 604480$$

$$= 23263.045 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o, 1 + W_o, 2 + H'p, 0)$$

$$= \max(627743 + 45567 + 0, 0)$$

$$= 673310.312 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 8.36 \cdot \pi \cdot 2763 + 0 \cdot 0$$

$$= 510110.719 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((W_o + Fa + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((673310 + 0 + 4 \cdot 0 / 2763) / 172, 510111 / 172)$$

$$= 383.905 \text{ cm}^2$$



PV Elite 28 Licensee: SL User
 FileName : 197-C_Channel MOC= CS Page 55 of 262
 Flg Calc [Int P]: New Flange Flng: 35 10:15am Sep 9,2026

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 47.6 + 6 \cdot 254/(3 + 0.5)$$

$$= 531.021 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2900 \cdot \sin(\pi / 56)$$

$$= 162.604 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{163/(2 \cdot 47.6 + 254)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	383.905	832.411	
Radial Distance between Hub and Bolts:	60.325	74.000	
Radial Distance between Bolts and Edge:	47.625	50.000	
Circ. Spacing between the Bolts:	101.600	162.604	531.021

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 832 \cdot 172 / (68.9 \cdot \pi (2780 + 2736))$$

$$= 11.983 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 832$$

$$= 1459922.62 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 673310 - 627743$$

$$= 45567.07 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2900 - 2763) / 2$$

$$= 68.3577 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((2900 - 2712) / 2 + 68.4)$$

$$= 81.2788 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2900 - 2712 - 20.2) / 2$$

$$= 84.1000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
---------	-------	----------	-----------	--------



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 56 of 262

Flg Calc [Int P]: New Flange

Flng: 35 10:15am

Sep 9, 2026

End Pressure, Md	604480.	84.1000	1.0000	50838.
Face Pressure, Mt	23263.	81.2788	1.0000	1891.
Gasket Load, Mg	45567.	68.3577	1.0000	3115.
Gasket Seating, Matm	1459923.	68.3577	1.0000	99799.

Total Moment for Operation, Mo 55844. Kg-m.
Total Moment for Gasket seating, Mg 99799. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 192.741 mm
Hub Ratio, h/h0 = HL / H0 0.285
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.474

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.885
Factor V per 4.16.5 0.381
Factor f per 4.16.5 1.177

Stress factor from Table 4.16.4 [K]:

= A / B
= 3000/2712
= 1.106

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1))/(K - 1)
= (0.66845 + (5.7169(1.11² • log10(1.11))) / (1.11² - 1))/(1.11 - 1)
= 19.176

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
= (1.11²(1 + 8.55246 • log10(1.11)) - 1) / ((1.0472 + 1.9448 • 1.11²) • (1.11 - 1))
= 1.875

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
= (1.11²(1 + 8.55246 • log10(1.11)) - 1) / (1.36136(1.11² - 1) • (1.11 - 1))
= 21.073

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (1.11² + 1) / (1.11² - 1)
= 9.927

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 21.1 • 13.7² • 193/0.38
= 2002180.875 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.89/193
= 0.005 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (254 • 0.0046 + 1) / 1.87 + 254³/2002180
= 9.360

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.18•55844/2712) / (9.36•20.2²)
= 62.23 N./mm²

Longitudinal Hub Stress, Seating [SHa]:



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 57 of 262

Flg Calc [Int P]: New Flange

Flng: 35 10:15am

Sep 9, 2026

$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g^{12})$$

$$= (1.18 \cdot 99799 / 2712) / (9.36 \cdot 20.2^2)$$

$$= 111.22 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.56 \cdot 55844 / 2712) / (9.36 \cdot 254^2)$$

$$= 0.85 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.56 \cdot 99799 / 2712) / (9.36 \cdot 254^2)$$

$$= 1.53 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (19.2 \cdot 55844 / (254^2 \cdot 2712)) - 9.93 \cdot 0.85$$

$$= 51.46 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (19.2 \cdot 99799 / (254^2 \cdot 2712)) - 9.93 \cdot 1.53$$

$$= 91.96 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (62.2 + \max(0.85, 51.5)) / 2$$

$$= 56.85 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (111 + \max(1.53, 92)) / 2$$

$$= 101.59 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 673310 / 832$$

$$= 79.32 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (510111 / 832)$$

$$= 60.10 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	62.23	204.60	111.22	207.00
Radial Flange	0.85	136.40	1.53	138.00
Tangential Flange	51.46	136.40	91.96	138.00
Maximum Average	56.85	136.40	101.59	138.00
Bolting	79.32	172.00	60.10	172.00

Minimum Required Flange Thickness [Rigidity] 250.190 mm

Estimated M.A.W.P. (Operating) 1386 KPa.

Estimated Finished Weight of Flange at given Thk. 2635.2 kg.

Estimated Unfinished Weight of Forging at given Thk 3124.8 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 58 of 262

Flg Calc [Int P]: New Flange

Flng: 35 10:15am

Sep 9, 2026

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	417.24	470.86	0.886	417.24	499.99	0.834
Tangential Flange	St	348.34	353.14	0.986	348.34	374.99	0.929
Tangential Flange	Sto	113.16	235.43	0.481	113.16	250.00	0.453
Combined	Sh/f + St	706.29	706.29	1.000	706.29	749.99	0.942
Combined	Sr + St	354.12	706.29	0.501	354.12	749.99	0.472
Combined	Sh + Sto	530.40	706.29	0.751	530.40	749.99	0.707
Radial Flange	Sr	5.78	470.86	0.012	5.78	499.99	0.012
Maximum Stress Ratio				1.000			0.942

Coefficients for f and f_2

	f	f_2	f_3
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.43/1.22)

= 1.1656

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h_0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.37/1.16) if $h/h_0 < 0.35$, else max (0.25; 0.26/0.77)

= 0.3161

In this case, the hub ratio h/h_0 , 0.2854 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, S_{fmax} , until it exceeds the limits provided by WRC 538.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 59 of 262

Flg Calc [Int P]: New Flange

Flng: 35 10:15am

Sep 9,2026

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	651.5 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	14.8645 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	206.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.937 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (2780^2 - 2736^2)$$

$$= 1906.2 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 207 \cdot 1906 / (56 \cdot 14.9)$$

$$= 474.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(474; 508)$$

$$= 474.0 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(474; 290)$$

$$= 474.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(474; 651)$$

$$= 474.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$\therefore Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$\therefore 474 \geq 140 (1906 / (56 \cdot 14.9))$$

$$\therefore 474.0 \geq 320.6 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$\therefore Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$\therefore 474 \geq (97 \cdot 1906 + \pi/4 \cdot 1026 \cdot 2736^2) / (0.7 \cdot 14.9 \cdot 56)$$

$$\therefore 474.0 \geq 420.9 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$\therefore Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$\therefore 474 \leq 276 (1906 / (56 \cdot 14.9))$$

$$\therefore 474.0 \leq 632.0 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$\therefore Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$\therefore 474 \leq 651 (1/0.94)$$



PV Elite 28 Licensee: SL User
 FileName : 197-C_Channel MOC= CS Page 60 of 262
 Flg Calc [Int P]: New Flange Flng: 35 10:15am Sep 9,2026

:: 474.0 ≤ 695.3 N./mm² Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 474 \cdot 0.2 \cdot 14.9 \cdot 47.6$$

$$= 684.4 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 91 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 99799 / 1 \cdot 9807 \cdot 0.38 / (8.89 \cdot 201334 \cdot 13.7^2 \cdot 193 \cdot 0.3)$$

$$= 1.000 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 55844 / 1 \cdot 9807 \cdot 0.38 / (8.89 \cdot 188345 \cdot 13.7^2 \cdot 193 \cdot 0.3)$$

$$= 0.598 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 99799 / 1 \cdot 9807 \cdot 0.38 / (9.36 \cdot 201334 \cdot 13.7^2 \cdot 193 \cdot 0.3)$$

$$= 0.950 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 55844 / 1 \cdot 9807 \cdot 0.38 / (9.36 \cdot 188345 \cdot 13.7^2 \cdot 193 \cdot 0.3)$$

$$= 0.568 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.75, Temperature Reduction per Fig. UCS 66.1 = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -17 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -31 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	62.58	207.00	111.34	207.00
Radial Flange	0.92	138.00	1.64	138.00
Tangential Flange	50.40	138.00	89.68	138.00
Maximum Average	56.49	138.00	100.51	138.00
Bolting	79.32	172.00	60.10	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 247.548 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 61 of 262

Flg Calc [Int P]: New Flange

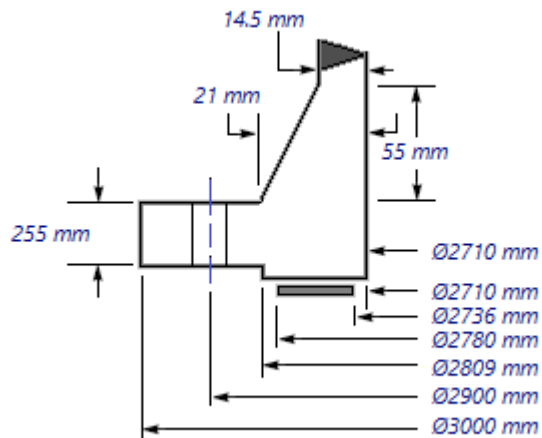
Flng: 36 10:15am Sep 9,2026

Flange Input Data Values

Description: New Flange :

Rhs Shell Flg-2

Flange Type	Integral Weld Neck		
Design Pressure	P	1026.47	KPa.
Design Temperature		240	°C
Internal Corrosion Allowance	ci	0.8000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2710.000	mm
Flange Outside Diameter	A	3000.000	mm
Flange Thickness	t	255.0000	mm
Thickness of Hub at Small End	go	14.5000	mm
Thickness of Hub at Large End	g1	21.0000	mm
Length of Hub	h	55.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	136.40	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2900.000	mm
Nominal Bolt Diameter	a	47.6250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		56	
Flange Face Outside Diameter	Fod	2809.000	mm
Flange Face Inside Diameter	Fid	2710.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2780.000	mm
Gasket Inside Diameter	Gi	2736.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm





PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 62 of 262

Flg Calc [Int P]: New Flange

Flng: 36 10:15am

Sep 9, 2026

ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1026 \cdot (2710/2 + 0.8)) / (138 \cdot 1 - 0.6 \cdot 1026) + Ca$$

$$= 10.9306 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 13.7) / (1356 + 0.6 \cdot 13.7)$$

$$= 1385.944 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 14.5) / (1355 + 0.6 \cdot 14.5)$$

$$= 1467.220 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	254.200 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2711.600 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	20.200 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	13.700 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	74.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	22.000 mm
Basic Gasket Width, $bo = N/2$	11.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{bo}$	8.358 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2763.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2763^2 \cdot 1026$$

$$= 627743.250 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 8.36 \cdot \pi \cdot 2763 \cdot 3 \cdot 1026$$

$$= 45567.059 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2712^2 \cdot 1026$$

$$= 604480.188 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 627743 - 604480$$

$$= 23263.045 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(627743 + 45567 + 0, 0)$$

$$= 673310.312 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 8.36 \cdot \pi \cdot 2763 + 0 \cdot 0$$

$$= 510110.719 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/S_b, Wg/S_a)$$

$$= \max((673310 + 0 + 4 \cdot 0/2763)/172, 510111/172)$$

$$= 383.905 \text{ cm}^2$$



PV Elite 28 Licensee: SL User
 FileName : 197-C_Channel MOC= CS Page 63 of 262
 Flg Calc [Int P]: New Flange Flng: 36 10:15am Sep 9,2026

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 47.6 + 6 \cdot 254/(3 + 0.5)$$

$$= 531.021 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2900 \cdot \sin(\pi / 56)$$

$$= 162.604 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{163/(2 \cdot 47.6 + 254)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	383.905	832.411	
Radial Distance between Hub and Bolts:	60.325	74.000	
Radial Distance between Bolts and Edge:	47.625	50.000	
Circ. Spacing between the Bolts:	101.600	162.604	531.021

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 832 \cdot 172 / (68.9 \cdot \pi (2780 + 2736))$$

$$= 11.983 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172 \cdot 832$$

$$= 1459922.62 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 673310 - 627743$$

$$= 45567.07 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2900 - 2763) / 2$$

$$= 68.3577 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((2900 - 2712) / 2 + 68.4)$$

$$= 81.2788 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2900 - 2712 - 20.2) / 2$$

$$= 84.1000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
---------	-------	----------	-----------	--------



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 64 of 262

Flg Calc [Int P]: New Flange

Flng: 36 10:15am

Sep 9, 2026

End Pressure, Md	604480.	84.1000	1.0000	50838.
Face Pressure, Mt	23263.	81.2788	1.0000	1891.
Gasket Load, Mg	45567.	68.3577	1.0000	3115.
Gasket Seating, Matm	1459923.	68.3577	1.0000	99799.

Total Moment for Operation, Mo 55844. Kg-m.
Total Moment for Gasket seating, Mg 99799. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 192.741 mm
Hub Ratio, h/h0 = HL / H0 0.285
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.474

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.885
Factor V per 4.16.5 0.381
Factor f per 4.16.5 1.177

Stress factor from Table 4.16.4 [K]:

= A / B
= 3000/2712
= 1.106

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169 * (K² * log10(K))) / (K² - 1)) / (K - 1)
= (0.66845 + (5.7169 * (1.11² * log10(1.11))) / (1.11² - 1)) / (1.11 - 1)
= 19.176

Stress factor from Table 4.16.4 [T]:

= (K² * (1 + 8.55246 * log10(K)) - 1) / ((1.0472 + 1.9448 * K²) * (K - 1))
= (1.11² * (1 + 8.55246 * log10(1.11)) - 1) / ((1.0472 + 1.9448 * 1.11²) * (1.11 - 1))
= 1.875

Stress factor from Table 4.16.4 [U]:

= (K² * (1 + 8.55246 * log10(K)) - 1) / (1.36136 * (K² - 1) * (K - 1))
= (1.11² * (1 + 8.55246 * log10(1.11)) - 1) / (1.36136 * (1.11² - 1) * (1.11 - 1))
= 21.073

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (1.11² + 1) / (1.11² - 1)
= 9.927

Stress factor from Table 4.16.4 for integral flanges [d]:

= U * g0² * h0 / V
= 21.1 * 13.7² * 193 / 0.38
= 2002180.875 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.89 / 193
= 0.005 1/mm

Stress factor from Table 4.16.4 [L]:

= (t * e + 1) / T + t³ / d
= (254 * 0.0046 + 1) / 1.87 + 254³ / 2002180
= 9.360

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f * Mop / Bcor) / (L * g1²)
= (1.18 * 55844 / 2712) / (9.36 * 20.2²)
= 62.23 N./mm²

Longitudinal Hub Stress, Seating [SHa]:



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 65 of 262

Flg Calc [Int P]: New Flange

Flng: 36 10:15am

Sep 9, 2026

$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g1^2)$$

$$= (1.18 \cdot 99799 / 2712) / (9.36 \cdot 20.2^2)$$

$$= 111.22 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.56 \cdot 55844 / 2712) / (9.36 \cdot 254^2)$$

$$= 0.85 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.56 \cdot 99799 / 2712) / (9.36 \cdot 254^2)$$

$$= 1.53 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (19.2 \cdot 55844 / (254^2 \cdot 2712)) - 9.93 \cdot 0.85$$

$$= 51.46 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (19.2 \cdot 99799 / (254^2 \cdot 2712)) - 9.93 \cdot 1.53$$

$$= 91.96 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (62.2 + \max(0.85, 51.5)) / 2$$

$$= 56.85 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (111 + \max(1.53, 92)) / 2$$

$$= 101.59 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 673310 / 832$$

$$= 79.32 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (510111 / 832)$$

$$= 60.10 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	62.23	204.60	111.22	207.00
Radial Flange	0.85	136.40	1.53	138.00
Tangential Flange	51.46	136.40	91.96	138.00
Maximum Average	56.85	136.40	101.59	138.00
Bolting	79.32	172.00	60.10	172.00

Minimum Required Flange Thickness [Rigidity] 250.190 mm

Estimated M.A.W.P. (Operating) 1386 KPa.

Estimated Finished Weight of Flange at given Thk. 2635.2 kg.

Estimated Unfinished Weight of Forging at given Thk 3124.8 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 66 of 262

Flg Calc [Int P]: New Flange

Flng: 36 10:15am

Sep 9, 2026

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	417.24	470.86	0.886	417.24	499.99	0.834
Tangential Flange	St	348.34	353.14	0.986	348.34	374.99	0.929
Tangential Flange	Sto	113.16	235.43	0.481	113.16	250.00	0.453
Combined	Sh/f + St	706.29	706.29	1.000	706.29	749.99	0.942
Combined	Sr + St	354.12	706.29	0.501	354.12	749.99	0.472
Combined	Sh + Sto	530.40	706.29	0.751	530.40	749.99	0.707
Radial Flange	Sr	5.78	470.86	0.012	5.78	499.99	0.012
Maximum Stress Ratio				1.000			0.942

Coefficients for f and f_2

	f	f_2	f_3
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g_1}{g_0} \right) + e \left(\frac{h}{h_0} \right) + g \left(\frac{g_1}{g_0} \right)^2 + i \left(\frac{h}{h_0} \right)^2 + k \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}{\left[1 + b \left(\frac{g_1}{g_0} \right) + d \left(\frac{h}{h_0} \right) + f \left(\frac{g_1}{g_0} \right)^2 + h \left(\frac{h}{h_0} \right)^2 + j \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.43/1.22)

= 1.1656

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h_0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.37/1.16) if $h/h_0 < 0.35$, else max (0.25; 0.26/0.77)

= 0.3161

In this case, the hub ratio h/h_0 , 0.2854 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 67 of 262

Flg Calc [Int P]: New Flange

Flng: 36 10:15am

Sep 9,2026

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	651.5 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	14.8645 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	206.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.937 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (2780^2 - 2736^2)$$

$$= 1906.2 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 207 \cdot 1906 / (56 \cdot 14.9)$$

$$= 474.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(474; 508)$$

$$= 474.0 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(474; 290)$$

$$= 474.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(474; 651)$$

$$= 474.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 474 \geq 140 (1906 / (56 \cdot 14.9))$$

$$:: 474.0 \geq 320.6 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 474 \geq (97 \cdot 1906 + \pi/4 \cdot 1026 \cdot 2736^2) / (0.7 \cdot 14.9 \cdot 56)$$

$$:: 474.0 \geq 420.9 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 474 \leq 276 (1906 / (56 \cdot 14.9))$$

$$:: 474.0 \leq 632.0 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$:: 474 \leq 651 (1/0.94)$$



PV Elite 28 Licensee: SL User
 FileName : 197-C_Channel MOC= CS Page 68 of 262
 Flg Calc [Int P]: New Flange Flng: 36 10:15am Sep 9,2026

:: 474.0 ≤ 695.3 N./mm² Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 474 \cdot 0.2 \cdot 14.9 \cdot 47.6$$

$$= 684.4 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 91 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 99799 / 1 \cdot 9807 \cdot 0.38 / (8.89 \cdot 201334 \cdot 13.7^2 \cdot 193 \cdot 0.3)$$

$$= 1.000 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 55844 / 1 \cdot 9807 \cdot 0.38 / (8.89 \cdot 188345 \cdot 13.7^2 \cdot 193 \cdot 0.3)$$

$$= 0.598 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 99799 / 1 \cdot 9807 \cdot 0.38 / (9.36 \cdot 201334 \cdot 13.7^2 \cdot 193 \cdot 0.3)$$

$$= 0.950 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 55844 / 1 \cdot 9807 \cdot 0.38 / (9.36 \cdot 188345 \cdot 13.7^2 \cdot 193 \cdot 0.3)$$

$$= 0.568 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.75, Temperature Reduction per Fig. UCS 66.1 = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-17 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-31 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	62.58	207.00	111.34	207.00
Radial Flange	0.92	138.00	1.64	138.00
Tangential Flange	50.40	138.00	89.68	138.00
Maximum Average	56.49	138.00	100.51	138.00
Bolting	79.32	172.00	60.10	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 247.548 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 69 of 262

Internal Pressure Calculations: Step: 9 10:15am Sep 9,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
LHS Ch Cover		1000	193	3	3000	138	1
LHS Ch Flg-1		1026.5	268	3	2700	138	2
LHS Ch Shell		1026.5	16	3	2700	117.3	3
LHS Ch Flg-2		1026.5	263	3	2700	138	4
LHS Shell Flg-1		1026.5	260	0.8	2710	136.4	5
Shell-1		1026.5	16	0.8	2700	117.3	6
Shell 2		1026.5	24	0.8	2700	117.3	7
Shell 3		1026.5	16	0.8	2700	117.3	8
Rhs Shell Flg-1		1026.5	263	0.8	2710	136.4	9
Rhs Shell Flg-2		1026.5	263	0.8	2710	136.4	10
Rhs ch shell		1026.5	16	0.8	2700	117.3	11
Dish End		1026.5	16	0.8	2700	117.3	12

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
LHS Ch Cover		1000	1045	1155	185	180.265	1
LHS Ch Flg-1		1000	1292	1624	260	254.381	2
LHS Ch Shell		1000	1094	1380	16	14.9035	3
LHS Ch Flg-2		1000	1292	1624	255	251.638	4
LHS Shell Flg-1		1000	1359	1467	252	250.19	5
Shell-1		1000	1156	1252	14.5	12.6839	6
Shell 2		1000	1840	1935	22.5	12.6839	7
Shell 3		1000	1156	1252	14.5	12.6839	8
Rhs Shell Flg-1		1000	1359	1467	255	250.19	9
Rhs Shell Flg-2		1000	1359	1467	255	250.19	10
Rhs ch shell		1000	1156	1252	14.5	12.6839	11
Dish End		1000	1085	1180	13.6	12.6226	12

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 1156 KPa.
 Shell Side MAPnc = 1251 KPa.
 Channel Side MAWP = 1045 KPa.
 Channel Side MAPnc = 1155 KPa.

Elements Suitable for Design Internal Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 70 of 262

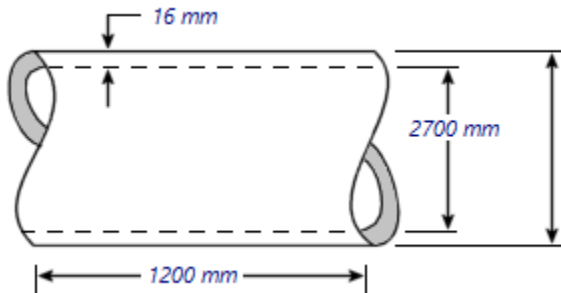
Element: LHS Ch Shell

Step: 10 10:15am

Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: LHS Ch Shell

Cylindrical Shell Design Information:			
Design Pressure P	1026.495 KPa.	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	82 °C
Length	1200.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2700.000 mm	Actual Stress	126.4 N./mm ²
Spec. Min. Thk t,e, or sa	16.0000 mm	Spec. Nominal Thk.	16.000 mm
Surface Area	102994.0 cm ²	SG of Contents	1.000
Element Volume	6871895.0 cm ³	Weight of Contents	6867.7 Kg
Empty Weight	1269.7 Kg	Operating Weight	8137.4 Kg
		Comp. All. Ope. Stress	108.6 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1026 \cdot 1353) / (138 \cdot 0.85 - 0.6 \cdot 1026) \\
 &= 11.9035 + 3.0000 = 14.9035 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 26.495 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 0.85 \cdot 13) / (1353 + 0.6 \cdot 13) \\
 &= 1121 - 26.5 = 1094 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 0.85 \cdot 16) / (1350 + 0.6 \cdot 16) \\
 &= 1380 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1026 \cdot (1353 + 0.6 \cdot 13)) / (0.85 \cdot 13) \\
 &= 126.419 \text{ N./mm}^2
 \end{aligned}$$



PV Elite 28 Licensee: SL User
 FileName : 197-C_Channel MOC= CS Page 71 of 262
 Element: LHS Ch Shell Step: 10 10:15am Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.589 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 16$, $t_r = 12.1$, $c = 3$ mm, $E^* = 0.85$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.79$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -26 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1045.152 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.793	Temperature Difference	12 °C
UCS-66(c) red.	No	Computed Min. Temp.	-26 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

EXTERNAL PRESSURE RESULTS for: LHS Ch Shell

Cylindrical Shell From 30 to 40 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	2732.00	1200.00	210.15	0.4392	0.0010505	86.80

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 86.8) / (3 \cdot 210) = 551 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.220	2732.00	1200.00	439.24	0.4392	0.0003408	34.07

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 34.1) / (3 \cdot 439) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	2732.00	7034.16	210.15	2.5747	0.0001631	16.31

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 16.3) / (3 \cdot 210) = 103 \text{ KPa.}$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

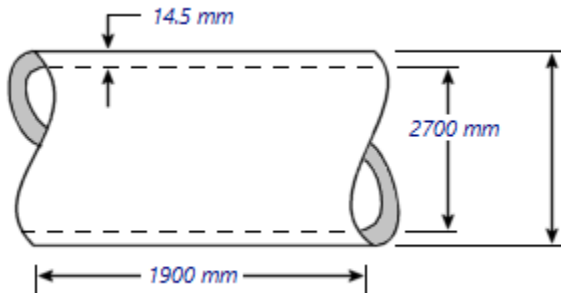
Page 72 of 262

Element: Shell-1

Step: 11 10:15am Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:			
Design Pressure P	1026.473 KPa.	Int. Design Temperature	240 °C
Material	SA-516 70	Ext. Design Temperature	40 °C
Length	1900.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	0.8000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2700.000 mm	Actual Stress	119.8 N./mm ²
Spec. Min. Thk t,e, or sa	14.5000 mm	Spec. Nominal Thk.	16.000 mm
Surface Area	163073.8 cm ²	SG of Contents	1.000
Element Volume	6139097.5 cm ³	Weight of Contents	6135.4 Kg
Empty Weight	2010.4 Kg	Operating Weight	8145.8 Kg
		Comp. All. Ope. Stress	90.8 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1026 \cdot 1351) / (138 \cdot 0.85 - 0.6 \cdot 1026) \\
 &= 11.8839 + 0.8000 = 12.6839 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 26.473 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 0.85 \cdot 13.7) / (1351 + 0.6 \cdot 13.7) \\
 &= 1182 - 26.5 = 1156 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 0.85 \cdot 14.5) / (1350 + 0.6 \cdot 14.5) \\
 &= 1252 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1026 \cdot (1351 + 0.6 \cdot 13.7)) / (0.85 \cdot 13.7) \\
 &= 119.801 \text{ N./mm}^2
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 73 of 262

Element: Shell-1

Step: 11 10:15am

Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.589 %**Minimum Design Metal Temperature Results:**Govrn. thk, $t_g = 14.5$, $t_r = 13.4$, $c = 0.8$ mm, $E^* = 0.85$ Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.83$, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -17 °C

Min Metal Temp. at Required thickness (UCS 66.1) -27 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1155.914 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	14.500 mm	Basic MDMT	-17 °C
Stress/Thk. Ratio	0.831	Temperature Difference	9 °C
UCS-66(c) red.	No	Computed Min. Temp.	-27 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 74 of 262

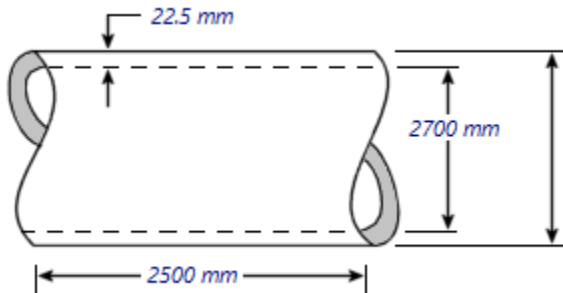
Element: Shell 2

Step: 12 10:15am

Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell 2

Cylindrical Shell Design Information:			
Design Pressure P	1026.473 KPa.	Int. Design Temperature	240 °C
Material	SA-516 70	Ext. Design Temperature	40 °C
Length	2500.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	0.8000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2700.000 mm	Actual Stress	75.9 N./mm ²
Spec. Min. Thk t,e, or sa	22.5000 mm	Spec. Nominal Thk.	24.000 mm
Surface Area	215827.4 cm ²	SG of Contents	1.000
Element Volume	8077760.5 cm ³	Weight of Contents	8072.8 Kg
Empty Weight	3979.6 Kg	Operating Weight	12052.4 Kg
		Comp. All. Ope. Stress	100.0 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1026 \cdot 1351) / (138 \cdot 0.85 - 0.6 \cdot 1026) \\
 &= 11.8839 + 0.8000 = 12.6839 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 26.473 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 0.85 \cdot 21.7) / (1351 + 0.6 \cdot 21.7) \\
 &= 1866 - 26.5 = 1840 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 0.85 \cdot 22.5) / (1350 + 0.6 \cdot 22.5) \\
 &= 1935 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1026 \cdot (1351 + 0.6 \cdot 21.7)) / (0.85 \cdot 21.7) \\
 &= 75.902 \text{ N./mm}^2
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 75 of 262

Element: Shell 2

Step: 12 10:15am

Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.881 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 22.5$, $t_r = 13.4$, $c = 0.8$ mm, $E^* = 0.85$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.52$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -4 °C

Min Metal Temp. at Required thickness (UCS 66.1) -33 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1155.914 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	22.500 mm	Basic MDMT	-4 °C
Stress/Thk. Ratio	0.525	Temperature Difference	29 °C
UCS-66(c) red.	No	Computed Min. Temp.	-33 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 76 of 262

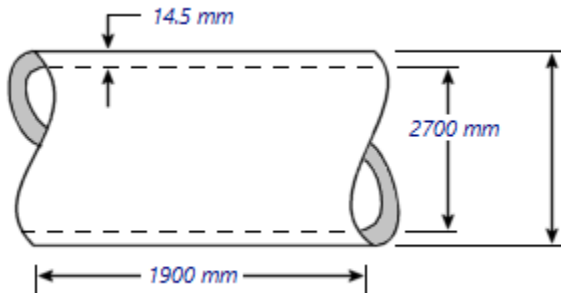
Element: Shell 3

Step: 13 10:15am

Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell 3

Cylindrical Shell Design Information:			
Design Pressure P	1026.473 KPa.	Int. Design Temperature	240 °C
Material	SA-516 70	Ext. Design Temperature	40 °C
Length	1900.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	0.8000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2700.000 mm	Actual Stress	119.8 N./mm ²
Spec. Min. Thk t,e, or sa	14.5000 mm	Spec. Nominal Thk.	16.000 mm
Surface Area	163073.8 cm ²	SG of Contents	1.000
Element Volume	6630709.5 cm ³	Weight of Contents	6626.7 Kg
Empty Weight	2010.4 Kg	Operating Weight	8637.1 Kg
		Comp. All. Ope. Stress	90.8 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1026 \cdot 1351) / (138 \cdot 0.85 - 0.6 \cdot 1026) \\
 &= 11.8839 + 0.8000 = 12.6839 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 26.473 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 0.85 \cdot 13.7) / (1351 + 0.6 \cdot 13.7) \\
 &= 1182 - 26.5 = 1156 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 0.85 \cdot 14.5) / (1350 + 0.6 \cdot 14.5) \\
 &= 1252 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1026 \cdot (1351 + 0.6 \cdot 13.7)) / (0.85 \cdot 13.7) \\
 &= 119.801 \text{ N./mm}^2
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 77 of 262

Element: Shell 3

Step: 13 10:15am

Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.589 %**Minimum Design Metal Temperature Results:**Govrn. thk, $t_g = 14.5$, $t_r = 13.4$, $c = 0.8$ mm, $E^* = 0.85$ Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.83$, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -17 °C

Min Metal Temp. at Required thickness (UCS 66.1) -27 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

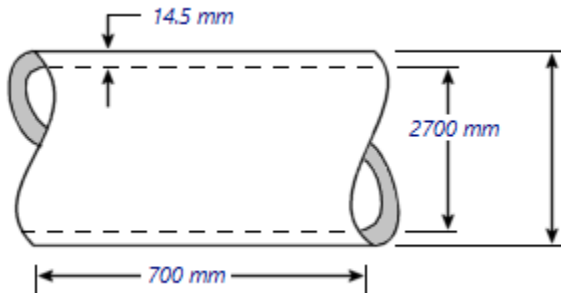
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1155.914 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	14.500 mm	Basic MDMT	-17 °C
Stress/Thk. Ratio	0.831	Temperature Difference	9 °C
UCS-66(c) red.	No	Computed Min. Temp.	-27 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Rhs ch shell

Cylindrical Shell Design Information:			
Design Pressure P	1026.473 KPa.	Int. Design Temperature	240 °C
Material	SA-516 70	Ext. Design Temperature	40 °C
Length	700.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	0.8000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2700.000 mm	Actual Stress	119.8 N./mm ²
Spec. Min. Thk t _e , or sa	14.5000 mm	Spec. Nominal Thk.	16.000 mm
Surface Area	60079.8 cm ²	SG of Contents	1.000
Element Volume	4008605.0 cm ³	Weight of Contents	4006.2 Kg
Empty Weight	740.7 Kg	Operating Weight	4746.8 Kg
		Comp. All. Ope. Stress	90.8 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1026 \cdot 1351) / (138 \cdot 0.85 - 0.6 \cdot 1026) \\
 &= 11.8839 + 0.8000 = 12.6839 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 26.473 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 0.85 \cdot 13.7) / (1351 + 0.6 \cdot 13.7) \\
 &= 1182 - 26.5 = 1156 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 0.85 \cdot 14.5) / (1350 + 0.6 \cdot 14.5) \\
 &= 1252 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1026 \cdot (1351 + 0.6 \cdot 13.7)) / (0.85 \cdot 13.7) \\
 &= 119.801 \text{ N./mm}^2
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 79 of 262

Element: Rhs ch shell

Step: 14 10:15am

Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.589 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 14.5$, $t_r = 12.1$, $c = 0.8$ mm, $E^* = 0.85$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.75$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -17 °C

Min Metal Temp. at Required thickness (UCS 66.1) -31 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1045.152 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	14.500 mm	Basic MDMT	-17 °C
Stress/Thk. Ratio	0.751	Temperature Difference	14 °C
UCS-66(c) red.	No	Computed Min. Temp.	-31 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 80 of 262

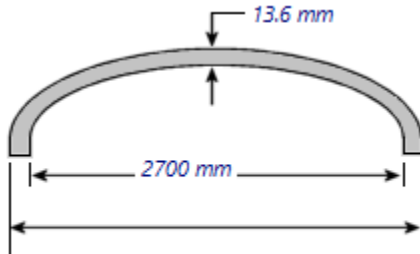
Element: Dish End

Step: 15 10:15am

Sep 9,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Dish End

Elliptical Head Design Information:					
Design Pressure	P	1026.473 KPa.	Int. Design Temperature	240 °C	
Material		SA-516 70	Ext. Design Temperature	40 °C	
Length		50.800 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	0.8000 mm	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	2700.000 mm	Actual Stress	127.5 N./mm ²	
Spec. Min. Thk t,e, or sa		13.6000 mm	Spec. Nominal Thk.	16.000 mm	
Surface Area		85266.9 cm ²	SG of Contents	1.000	
Element Volume		2867870.5 cm ³	Weight of Contents	2866.1 Kg	
Empty Weight		1148.6 Kg	Operating Weight	4014.7 Kg	
			Comp. All. Ope. Stress	90.8 N./mm ²	
			Aspect Ratio	2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 33.750 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 16.875 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (1026 \cdot 2702 \cdot 1) / (2 \cdot 138 \cdot 0.85 - 0.2 \cdot 1026)$$

$$= 11.8226 + 0.8000 = 12.6226 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 26.473 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 12.8) / (1 \cdot 2702 + 0.2 \cdot 12.8)$$

$$= 1111 - 26.5 = 1085 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 0.85 \cdot 13.6) / (1 \cdot 2700 + 0.2 \cdot 13.6)$$

$$= 1180 \text{ KPa.}$$



PV Elite 28 Licensee: SL User
 FileName : 197-C_Channel MOC= CS Page 81 of 262
 Element: Dish End Step: 15 10:15am Sep 9,2026

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (1026 \cdot (1 \cdot 2702 + 0.2 \cdot 12.8)) / (2 \cdot 0.85 \cdot 12.8)$$

$$= 127.469 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (1026 \cdot 1351) / (138 \cdot 0.85 - 0.6 \cdot 1026) + 0.8$$

$$= 12.684 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 26.473 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 0.85 \cdot 15.2) / (1351 + 0.6 \cdot 15.2)$$

$$= 1311 - 26.5 = 1285 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (2702 / (2 \cdot 676)))^2 / 6$$

$$= 0.999211$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 2.570 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 13.6, tr = 12, c = 0.8 mm, E* = 0.85
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.8$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-30 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 16, tr = 12.1, c = 0.8 mm, E* = 0.85
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.68$, Temp. Reduction = 18 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-32 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1045.152 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.677	Temperature Difference	18 °C
UCS-66(c) red.	No	Computed Min. Temp.	-32 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 33.750 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 16.875 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 82 of 262

Shop Pressure Test Results:

Step: 16 10:15am Sep 9, 2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1502.688 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1300.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1596.923 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1271.505 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1430.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1430.000 KPa.
App 27-4	= M.A.W.P.	1155.914 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1358.698 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1300.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1472.169 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1149.668 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1430.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1430.000 KPa.
App 27-4	= M.A.W.P.	1045.152 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	184.6	234.0	0.789	1498.63
Shell-1	189.5	234.0	0.810	1623.39
Shell 2	120.0	234.0	0.513	1623.39
Shell 3	189.5	234.0	0.810	1623.39
Rhs ch shell	174.9	234.0	0.747	1498.63
Dish End	186.1	234.0	0.795	1498.63

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	118.00	118.00	1.000
N1-Tube Inlet	Pad	138.00	138.00	1.000
N2-Tube outlet	Nozzle	118.00	118.00	1.000
N2-Tube outlet	Pad	138.00	138.00	1.000
N3-Shell inlet	Nozzle	138.00	136.40	1.012
N4-Shell outlet	Nozzle	138.00	136.40	1.012
Drain	Nozzle	138.00	136.40	1.012
Drain	Pad	138.00	138.00	1.000
Vent	Nozzle	138.00	136.40	1.012
Vent	Pad	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Ch Cover	138.00	138.00	1.000
LHS Ch Flg-1	138.00	138.00	1.000
LHS Ch Shell	138.00	138.00	1.000



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 83 of 262

Shop Pressure Test Results:

Step: 16 10:15am Sep 9,2026

LHS Ch Flg-2	138.00	138.00	1.000
LHS Shell Flg-1	138.00	136.40	1.012
Shell-1	138.00	138.00	1.000
Shell 2	138.00	138.00	1.000
Shell 3	138.00	138.00	1.000
Rhs Shell Flg-1	138.00	136.40	1.012
Rhs Shell Flg-2	138.00	136.40	1.012
Rhs ch shell	138.00	138.00	1.000
Dish End	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Floating Head Matl.	187.00	174.00	1.075
Backing Ring Matl.	137.90	137.90	1.000
Flange Material	187.00	174.00	1.075
Tube Material	187.00	170.80	1.095
Tubesheet Material	187.00	170.80	1.095

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	15.94	216.00	0.074
N2-Tube outlet	15.94	216.00	0.074
N3-Shell inlet	39.70	179.40	0.221
N4-Shell outlet	44.92	179.40	0.250
Drain	5.42	225.00	0.024
Vent	5.42	225.00	0.024

Elements Suitable for Test Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 197-C_Channel MOC= CS Page 84 of 262
 Field Pressure Test Results: Step: 17 10:15am Sep 9,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1502.688 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1300.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1596.923 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1271.505 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1430.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1430.000 KPa.
App 27-4	= M.A.W.P.	1155.914 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1358.698 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1300.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1472.169 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1149.668 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1430.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1430.000 KPa.
App 27-4	= M.A.W.P.	1045.152 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 1000 • 1
 = 1300.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 1000 • 1
 = 1300.000 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS Ch Shell	163.4	234.0	0.698	1326.47
Shell-1	154.8	234.0	0.662	1326.47
Shell 2	98.1	234.0	0.419	1326.47
Shell 3	154.8	234.0	0.662	1326.47
Rhs ch shell	154.8	234.0	0.662	1326.47
Dish End	164.7	234.0	0.704	1326.47

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	118.00	118.00	1.000
N1-Tube Inlet	Pad	138.00	138.00	1.000
N2-Tube outlet	Nozzle	118.00	118.00	1.000
N2-Tube outlet	Pad	138.00	138.00	1.000
N3-Shell inlet	Nozzle	138.00	136.40	1.012
N4-Shell outlet	Nozzle	138.00	136.40	1.012
Drain	Nozzle	138.00	136.40	1.012
Drain	Pad	138.00	138.00	1.000
Vent	Nozzle	138.00	136.40	1.012
Vent	Pad	138.00	138.00	1.000



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 85 of 262

Field Pressure Test Results:

Step: 17 10:15am

Sep 9, 2026

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Ch Cover	138.00	138.00	1.000
LHS Ch Flg-1	138.00	138.00	1.000
LHS Ch Shell	138.00	138.00	1.000
LHS Ch Flg-2	138.00	138.00	1.000
LHS Shell Flg-1	138.00	136.40	1.012
Shell-1	138.00	138.00	1.000
Shell 2	138.00	138.00	1.000
Shell 3	138.00	138.00	1.000
Rhs Shell Flg-1	138.00	136.40	1.012
Rhs Shell Flg-2	138.00	136.40	1.012
Rhs ch shell	138.00	138.00	1.000
Dish End	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Floating Head Matl.	187.00	174.00	1.075
Backing Ring Matl.	137.90	137.90	1.000
Flange Material	187.00	174.00	1.075
Tube Material	187.00	170.80	1.095
Tubesheet Material	187.00	170.80	1.095

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	14.11	216.00	0.065
N2-Tube outlet	14.11	216.00	0.065
N3-Shell inlet	32.44	179.40	0.181
N4-Shell outlet	36.71	179.40	0.205
Drain	4.42	225.00	0.020
Vent	4.42	225.00	0.020

Elements Suitable for Test Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 86 of 262

ASME TS Calc: Pressure Test

Case: 1 10:15a Sep 9,2026

Input Echo, Tubesheet Number 1, Description: Fixed Tubeheet

Shell Data:

Main Shell Description: Shell-1

Shell Test Pressure		1300.00 KPa.
Shell Thickness	ts	16.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2700.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		1300.00 KPa.
Channel Thickness	tc	16.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2700.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	4924
Tube Wall Thickness	et	1.6510 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	6550.00 mm
Straight Tube Length (bet. inner tubsht faces) L		6270.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-789
Tube Material UNS Number		S32205
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		187.00 N./mm ²
Tube Allowable Stress At Ambient		187.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	479.99 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	1.6510 mm
Groove Weld Leg	ag	1.6510 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	1207.750 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	42498.0 cm ²
Perimeter of Tube Layout	Cp	7703.8 mm



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 87 of 262

ASME TS Calc: Pressure Test

Case: 1 10:15a Sep 9,2026

Tubesheet Data:

Tubesheet TYPE: Stationary, Gasketed both Sides, Conf. d

Exchanger TYPE: With an Immersed Floating Head, Conf. a

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-182 F60
Tubesheet Material UNS Number		S32205
Tubesheet Allowable Stress at Temperature	S	187.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	187.00 N./mm ²
Thickness of Tubesheet	h	140.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	2806.000 mm
Dimension G for the Channel Side	Gc	2763.285 mm
Area of the Untubed Lanes	AL	428.6 cm ²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	1068.500 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Channel Side Min. Design Pressure	Ptd,min	-103.0000 KPa.
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Both
Flange Outside Diameter	A	3000.000 mm
Flange Inside Diameter	B	2710.000 mm
Flange Face Outside Diameter	Fod	2809.000 mm
Flange Face Inside Diameter	Fid	2710.000 mm
Gasket Outside Diameter	Go	2780.000 mm
Gasket Inside Diameter	Gi	2736.000 mm
Small end Hub thk.	g0	14.5000 mm
Large end Hub thk.	g1	21.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	6.3500 mm
Full face Gasket Flange Option		Program Selects

Bolting Information:

Diameter of Bolt Circle	C	2900.000 mm
Nominal Bolt Diameter	dB	47.6250 mm
Thread Series		TEMA Thread Series
Number of Bolts	n	56
Bolt Material		SA-193 B7
Bolt Material UNS Number		G41400
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm
Alternate Flange Operating Bolt Load, Wm1		852717.94 KgF
Alternate Flange Seating Bolt Load, Wm2		510110.72 KgF
Alternate Flange Design Bolt Load, W		1459922.62 KgF

Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Integral with	None
Tubesheet Extended as Flange	No



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

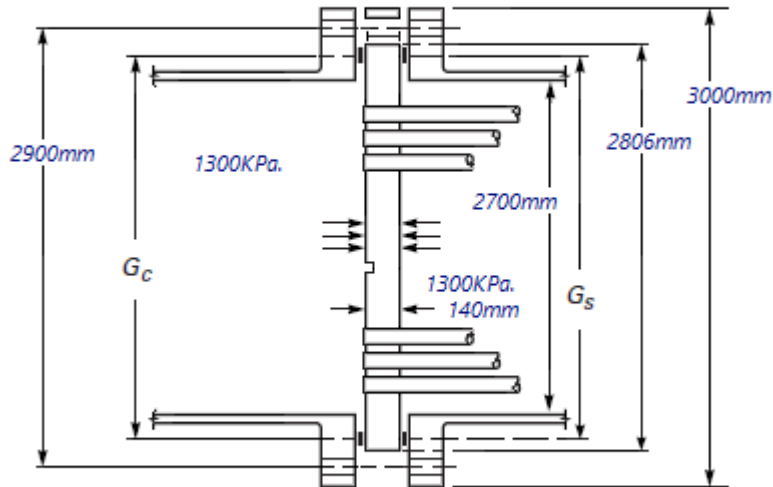
Page 88 of 262

ASME TS Calc: Pressure Test

Case: 1 10:15a Sep 9,2026

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

Tubesheet Gasketed w/Shell and Channel



Configuration d:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1

Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1

Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Low Alloy Steels, Group 2 Group 1

Tubes - TM-1 Material Group H

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000125990 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.19994E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.19994E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.19994E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19994E+09 KPa.

TubeSheet - TE-1 Low Alloy Steels, Group 2 Group 1

TubeSheet - TM-1 Material Group H

Elastic Mod. at Design Temperature	21.1 °C	0.19994E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19994E+09 KPa.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 89 of 262

ASME TS Calc: Pressure Test

Case: 1 10:15a Sep 9, 2026

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1300 \cdot (25.4/2 - 0)) / (187 \cdot 1 + 0.4 \cdot 1300)$$

$$= 0.0881 + 0.0000 = 0.0881 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-5 at 21.11 °C
Elastic Modulus for Material 195117184.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6510	25.40	6270.00	15.38	50.0000	0.0046475	183.26
EMAWP = (4*B)/(3*(D/T)) = (4 *183)/(3 *15.4) = 15881 KPa.						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.5437	25.40	6270.00	46.72	50.0000	0.0005039	49.17
EMAWP = (4*B)/(3*(D/T)) = (4 *49.2)/(3 *46.7) = 1403 KPa.						

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.5437 mm
Allowable Internal Pressure at Corroded thickness	25641.51 KPa.
Required Internal Design Pressure	1300.00 KPa.
Allowable External Pressure at Corroded thickness	15881.18 KPa.
Required External Design Pressure	1403.00 KPa.
Required Thickness due to Shell Side pressure	0.5437 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Gasketed Tubesheets:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	22.000 mm
Basic Gasket Width,	$b0 = N / 2$	11.000 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) * 2.5$	8.355 mm
Gasket Reaction Diameter,	$G = Go - 2.0 * b$	2763.291 mm

Results for ASME Stationary Tubesheet Calculations for Configuration d:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.9.4 Step 1:

Tube expansion depth ratio [ρ]:

$$= ltx / h \text{ (modified for corrosion if present)}$$

$$= 0/140$$

$$= 0.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole diameter [d*]:

$$= \max(dt - 2 \cdot tt(Et/E) \cdot (St/T/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(25.4 - 2 \cdot 1.65 (199943392/199943392) \cdot$$

$$(187/187) \cdot (0); 25.4 - 2 \cdot 1.65)$$

$$= 25.4000 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 90 of 262

ASME TS Calc: Pressure Test

Case: 1 10:15a Sep 9, 2026

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned}
 &= 2 \cdot r_o + dt \\
 &= 2 \cdot 1208 + 25.4 \\
 &= 2440.900 \text{ mm}
 \end{aligned}$$

Determine the basic ligament efficiency for shear [μ]:

$$\begin{aligned}
 &= (p - dt) / p \\
 &= (31.8 - 25.4) / 31.8 \\
 &= 0.200
 \end{aligned}$$

Equivalent outer tube limit radius [ao]:

$$\begin{aligned}
 &= Do / 2 \\
 &= 2441 / 2 \\
 &= 1220.4500 \text{ mm}
 \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned}
 &= p / \sqrt{1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2) } \\
 &= 31.8 / \sqrt{1 - 4 \cdot \min(429 \cdot 100 ; 4 \cdot 2441 \cdot 31.8) / (\pi \cdot 2441^2) } \\
 &= 31.8964 \text{ mm}
 \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned}
 &= (p^* - d^*) / p^* \\
 &= (31.9 - 25.4) / 31.9 \\
 &= 0.2037
 \end{aligned}$$

Ratio [ps]:

$$\begin{aligned}
 &= a_s / a_o \\
 &= 1382 / 1220 \\
 &= 1.1321
 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned}
 &= a_c / a_o \\
 &= 1382 / 1220 \\
 &= 1.1321
 \end{aligned}$$

Compute parameter [xt]:

$$\begin{aligned}
 &= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot a_o))^2 \\
 &= 1 - 4924 ((25.4 - 2 \cdot 1.65) / (2 \cdot 1220))^2 \\
 &= 0.5964
 \end{aligned}$$

Determine parameter [xs]:

$$\begin{aligned}
 &= 1 - Nt(dt / (2 \cdot a_o))^2 \\
 &= 1 - 4924 (25.4 / (2 \cdot 1220))^2 \\
 &= 0.4668
 \end{aligned}$$

Determine the value [h'g]:

$$\begin{aligned}
 &= \max((h_g - CATC); 0) \text{ (For pressure only cases)} \\
 &= \max((8 - 0); 0) \\
 &= 8.000 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.9.4 Step 2:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

β_s , k_s , λ_s , δ_s are all = 0

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

β_C , k_C , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.9.4 Step 3:

$h/p = 4.409449$; $\mu^* = 0.203672$; $E^* = 31928848 \text{ KPa}$.

$E^*/E = 0.159689$; $v^* = 0.453250$

Note: As h/p (4.409) is > 2, data values for $h/p = 2$ were used.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 91 of 262

ASME TS Calc: Pressure Test

Case: 1 10:15a Sep 9, 2026

Tube Bundle Stiffness Factor [Xa]:

$$= \left(\left(24(1 - \nu^2) Nt \cdot Et \cdot tt (dt - tt) ao^2 \right) / (E \cdot L \cdot H^3) \right)^{1/4}$$

$$= \left(\left(24(1 - 0.45^2) 4924 \cdot 199943392 \cdot 1.65 \right. \right. \\ \left. \left. (25.4 - 1.65) 1220^2 \right) / (31928848 \cdot 6270 \cdot 140^3) \right)^{1/4}$$

$$= 6.6841$$

Values from Table 4.18.3

$$Zd = 0.004960 ; Zv = 0.022184 ; Zm = 0.211940$$

$$Za = 0.198049E+03 ; Zw = 0.022184$$

ASME Section VIII-2 Part 4.18.9.4 Step 4:

Diameter ratio [K]:

$$= A / Do$$

$$= 2806/2441$$

$$= 1.1496$$

Compute coefficient [F]:

$$= (1 - \nu) / E \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.45) / 31928848 \cdot (0 + 0 + 199943392 \cdot \ln(1.15))$$

$$= 0.4773$$

Compute parameter [ϕ]:

$$= (1 + \nu) F$$

$$= (1 + 0.45) 0.48$$

$$= 0.6936$$

Compute parameter [Q1]:

$$= (ps - 1 - \phi \cdot Zv) / (1 + \phi \cdot Zm)$$

$$= (1.13 - 1 - 0.69 \cdot 0.022) / (1 + 0.69 \cdot 0.21)$$

$$= 0.101735$$

ASME Section VIII-2 Part 4.18.9.4 Step 5:

Determine factor [γ_b]:

$$= (Gc - Gs) / Do \text{ (config d)}$$

$$= (2763 - 2763) / 2441 = 0$$

Calculate parameter [ω_s]:

$$= ps \cdot ks \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.13 \cdot 0 \cdot 0 \cdot 0.000 (1 + 140 \cdot 0)$$

$$= 0.0000 \text{ mm}^2$$

Calculate parameter [ω_s]:

$$= ao^2 (ps^2 - 1) \cdot (ps - 1) / 4 - \omega_s$$

$$= 1220^2 (1.13^2 - 1) \cdot (1.13 - 1) / 4 - 0$$

$$= 13849.4170 \text{ mm}^2$$

Parameter [ω_c]:

$$= pc \cdot kc [\delta C (CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) - \\ \zeta c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2)]$$

$$= 1.13 \cdot 0 [0.000 (0 + 0 \cdot 0 \cdot 140) - \\ 0.000 (0/140 + 0 \cdot 0/2)]$$

$$= 0.0000 \text{ mm}^2$$

Parameter [ω_c]:

$$= ao^2 [(pc^2 + 1) \cdot (pc - 1) / 4 - (ps - 1) / 2] - \omega_c$$

$$= 1220^2 [(1.13^2 + 1) \cdot (1.13 - 1) / 4 - (1.13 - 1) / 2] - 0$$

$$= 13849.4150 \text{ mm}^2$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 92 of 262

ASME TS Calc: Pressure Test

Case: 1 10:15a Sep 9, 2026

Pressure [P**C*]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.9.4 Step 6:

$P_e = P_s - P_t$

= 1300 - 1300 = 0 KPa.

ASME Section VIII-2 Part 4.18.9.4 Step 7:

Determine factor [Q2]:

$$= \left[\left(\omega_s \cdot P_s - \omega_c \cdot P_t \right) - \left(\omega_s \cdot P_s - \omega_c \cdot P_c \right) \right] + \frac{W \cdot \gamma_b}{(2 \cdot \pi) \cdot (1 + \phi \cdot Z_m)}$$

$$= \left[\left(13849 \cdot 1300 - 13849 \cdot 1300 \right) - \left(0 \cdot 0 - 0 \cdot 0 \right) \right] \cdot 0.0001 + \frac{1459922 \cdot 0}{(2 \cdot \pi) \cdot (1 + 0.69 \cdot 0.21)}$$

= 0.0003 Kgf

Tubesheet Bending Stress - original thickness [σ]:

$$= 6 \cdot Q2 / (\mu \cdot (H - h'g)^2)$$

$$= 6 \cdot 0.00029 / (0.2 \cdot (140 - 8)^2)$$

= 0.0000 N./mm²

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta_T \cdot S_y = 1.5 \cdot 0.95 \cdot 480 = 684 \text{ N./mm}^2, 4.18.233$$

Could not converge on a min. required tubesheet thk. to satisfy the Bending Stress criteria.



Use the best Estimated Required Tubesheet Thickness value as design guidance & Compare the Actual Stresses versus Allowable Stresses.

Tubesheet thickness = 19.0133 mm

Tubesheet Bending Stress = 0.0008 N./mm²

ASME Section VIII-2 Part 4.18.9.4 Step 8:

Shear Stress check [τ_{limit}]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot (0.8 \cdot \beta_T \cdot S_y)$$

$$= (2 \cdot 0.2 \cdot 140 / 1220) \cdot 365$$

= 16.7384 N./mm²

Shear Stress is not required to be computed; [P_e] <= τ_{limit}

Tubesheet Shear Stress - final thickness [τ_f]:

$$= (A_p \cdot P_e) / (\mu \cdot h \cdot C_p)$$

$$= (42498 \cdot 0) / (0.2 \cdot 19 \cdot 7704)$$

= 0.0000 N./mm²

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$= H + C_{ats} + C_{atc}$$

$$= 19 + 0 + 0$$

= 19.0133 mm

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(19; 19)$$

= 19.0133 mm

ASME Section VIII-2 Part 4.18.9.4 Step 9:



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 93 of 262

ASME TS Calc: Pressure Test

Case: 1 10:15a Sep 9, 2026

The Ftmin and Ftmax Coefficients from Table 4.18.4:

$$F_{tmin} = -6.42, F_{tmax} = 22.1$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$\begin{aligned} &= ((P_s \cdot x_s - P_t \cdot x_t) - 2 \cdot Q_2 \cdot F_{tmin} / a_o^2) / (X_t - X_s) \\ &= ((1300 \cdot 0.47 - 1300 \cdot 0.6) - 2 \cdot 0.00064 \cdot -6.42 / 1220^2) / (0.6 - 0.47) \\ &= -1.3001 \text{ N./mm}^2 \end{aligned}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$\begin{aligned} &= ((P_s \cdot x_s - P_t \cdot x_t) - 2 \cdot Q_2 \cdot F_{tmax} / a_o^2) / (X_t - X_s) \\ &= ((1300 \cdot 0.47 - 1300 \cdot 0.6) - 2 \cdot 0.00064 \cdot 22.1 / 1220^2) / (0.6 - 0.47) \\ &= -1.3001 \text{ N./mm}^2 \end{aligned}$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 1.3 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= \beta_T \cdot S_{y,t} = 455.9918 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= 1.2500 \text{ (as } P_e \text{ is 0.0)}$$

Determine the factor [rt]:

$$\begin{aligned} &= (\sqrt{d_t^2 + (d_t - 2 \cdot t_t)^2}) / 4 \\ &= (\sqrt{25.4^2 + (25.4 - 2 \cdot 1.65)^2}) / 4 \\ &= 8.4168 \text{ mm} \end{aligned}$$

Determine the factor [Ct]:

$$\begin{aligned} &= \sqrt{2 \cdot \pi^2 \cdot E_t / S_{y,t}} \\ &= \sqrt{2 \cdot \pi^2 \cdot 199943392 / 480} \\ &= 98.6283 \end{aligned}$$

Determine the factor [Ft]:

$$\begin{aligned} &= k \cdot L / r \\ &= 1 \cdot 1068 / 8.42 \\ &= 126.9484 \end{aligned}$$

Buckling Allowable Stress [Stb]:

$$\begin{aligned} &= (1 / F_s) \cdot (\pi^2 \cdot E_t / F_t^2) (C_t \leq F_t) \\ &= (1 / 1.25) \cdot (\pi^2 \cdot 199943392 / 127^2) \\ &= 455.992 \text{ N./mm}^2 \text{ (Never greater than } S_{ot}) \end{aligned}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$\begin{aligned} &= \sigma_{t,max} \cdot \text{Tube Area} \\ &= 1.3 \cdot 1.23 \\ &= 16.33 \text{ Kg} \end{aligned}$$

Tube Metal Area [At]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (25.4 - 1.65) 1.65 \\ &= 1.232 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$\begin{aligned} &= \pi \cdot t (d_o - t) S \\ &= \pi \cdot 1.65 (25.4 - 1.65) 456 \\ &= 5727.573 \text{ Kg} \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 94 of 262

ASME TS Calc: Pressure Test

Case: 1 10:15a Sep 9, 2026

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft)$$

$$= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 456$$

$$= 5433.6445 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / Sw \text{ where } Sw = \min(S \text{ and } St)$$

$$= 456/456$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft)$$

$$= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 456$$

$$= 3515.8875 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t)fw \cdot fd} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65)1 \cdot 1} - 0.75 \cdot 25.4 \right)$$

$$= 2.142 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0711 mm

Minimum Required Groove Weld Leg [agr]: 1.0711 mm

Tube-Tubesheet Jt allowable, 5728 is $\geq$ tube strength 5728 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Computations completed for ASME tubesheet configuration: d

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	0.0 $\leq$	684.0 N./mm ²	Pass
Tubesheet Shear Stress	0.0 $\leq$	364.8 N./mm ²	Pass
Maximum Stress in the Tubes	1.3 $\leq$	456.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-1.3 $\leq$	-456.0 N./mm ²	Pass
Maximum Force on any one Tube	16.3 $\leq$	5727.6 Kgf	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	19.013	140.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.071	1.651	Ok
Tube-Tubesheet Groove Weld Leg :	1.071	1.651	Ok

Stationary Tubesheet results per ASME Div. 2 Part 4.18.9 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/ Fail	
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd			
D1uc	88.867	5.944	271	684	26	365	Ps+Pt-Th	D1	Ok
D2uc	92.147	6.415	292	684	28	365	Ps+Pt-Th	D2	Ok
D3uc	19.013	...	...	684	...	365	Ps+Pt-Th	D3	Ok



PV Elite 28 Licensee: SL User
 FileName : 197-C_Channel MOC= CS Page 95 of 262
 ASME TS Calc: Pressure Test Case: 1 10:15a Sep 9,2026

Max: 92.1468 6.415 mm 0.427 0.076 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses						Tube Loads		Shell Stress		Channel Stress		Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	70	455	-17	-455	877	5727	...	...	...	...	...	...	Ok
D2uc	77	455	-77	-455	963	5727	...	...	...	...	...	...	Ok
D3uc	1	455	-1	-455	16	5727	...	...	...	...	...	...	Ok
Max RATIO	0.168		0.168		0.168		...		...		...		

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	92.147	140.000	Ok
Tubesheet thickness flanged extension :	6.415	124.000	Ok
Tube thickness :	0.544	1.651	Ok
Tube-tubesheet fillet weld leg :	1.071	1.651	Ok
Tube-tubesheet groove weld leg :	1.071	1.651	Ok

Note: This is a full strength Tube to Tubesheet Joint.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 96 of 262

ASME Fl-TS Calc: Pressure Test

Case: 1 10:15a Sep 9, 2026

Input Echo, Tubesheet Number 1, Description: Floatating tube

Shell Data:

Main Shell Description: Shell-1

Shell Test Pressure		1300.00 KPa.
Shell Thickness	ts	16.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2700.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		1300.00 KPa.
Channel Thickness	tc	16.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2700.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	4924
Tube Wall Thickness	et	1.6510 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	6550.00 mm
Straight Tube Length (bet. inner tubsht faces) L		6270.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-789
Tube Material UNS Number		S32205
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		187.00 N./mm ²
Tube Allowable Stress At Ambient		187.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	479.99 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	1.6510 mm
Groove Weld Leg	ag	1.6510 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	1207.750 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	42498.0 cm ²
Perimeter of Tube Layout	Cp	7703.8 mm



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 97 of 262

ASME Fl-TS Calc: Pressure Test

Case: 1 10:15a Sep 9, 2026

Tubesheet Data:

Tubesheet TYPE: Floating, Tubesheet Gasketed, Conf. B
Exchanger TYPE: With an Immersed Floating Head, Conf. a

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-182 F60
Tubesheet Material UNS Number		S32205
Tubesheet Allowable Stress at Temperature	S	187.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	187.00 N./mm ²
Thickness of Tubesheet	h	140.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	2650.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		No
Area of the Untubed Lanes	AL	428.6 cm ²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	1068.500 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Channel Side Min. Design Pressure	Ptd,min	-103.0000 KPa.
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Channel
Flange Outside Diameter	A	2650.000 mm
Flange Inside Diameter	B	2454.000 mm
Flange Face Outside Diameter	Fod	2486.000 mm
Flange Face Inside Diameter	Fid	2454.000 mm
Gasket Outside Diameter	Go	2483.000 mm
Gasket Inside Diameter	Gi	2457.000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	4.5000 mm
Full face Gasket Flange Option		Program Selects
Length of Partition Gasket	lp	2700.000 mm
Width of Partition Gasket	wp	22.0000 mm
Partition Gasket Factor,	mPart	3.0000
Partition Gasket Design Seating Stress	yPart	68.95 N./mm ²
Partition Gasket Facing Sketch		Code Sketch 1a
Partition Gasket Column for Gasket Seating		Code Column I

Bolting Information:

Diameter of Bolt Circle	C	2550.000 mm
Nominal Bolt Diameter	dB	47.6250 mm
Thread Series		UNC Thread Series
Number of Bolts	n	40
Bolt Material		SA-193 B7
Bolt Material UNS Number		G41400
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Extended as Flange	Yes
------------------------------	-----



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 98 of 262

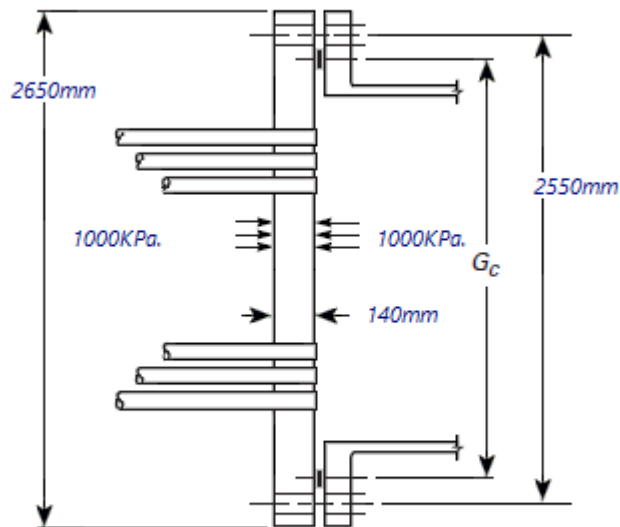
ASME Fl-TS Calc: Pressure Test

Case: 1 10:15a Sep 9, 2026

Thickness of Extended Portion of Tubesheet Tf 114.0000 mm
Is Bolt Load Transferred to the Tubesheet Yes

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

Tubesheet Gasketed, Extended as a Flange



Configuration B:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Low Alloy Steels, Group 2 Group 1
Tubes - TM-1 Material Group H

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000125990 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.19994E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.19994E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.19994E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19994E+09 KPa.

TubeSheet - TE-1 Low Alloy Steels, Group 2 Group 1
TubeSheet - TM-1 Material Group H



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 99 of 262

ASME Fl-TS Calc: Pressure Test

Case: 1 10:15a Sep 9, 2026

Elastic Mod. at Design Temperature	21.1 °C	0.19994E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19994E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$$
 per Appendix 1-1 (a)(1)

$$= (1300 \cdot (25.4/2 - 0)) / (187 \cdot 1 + 0.4 \cdot 1300)$$

$$= 0.0881 + 0.0000 = 0.0881 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	HA-5	at	21.11 °C
Elastic Modulus for Material			195117184.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6510	25.40	6270.00	15.38	50.0000	0.0046475	183.26

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 183) / (3 \cdot 15.4) = 15881 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.5437	25.40	6270.00	46.72	50.0000	0.0005039	49.17

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 49.2) / (3 \cdot 46.7) = 1403 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.5437 mm
Allowable Internal Pressure at Corroded thickness	25641.51 KPa.
Required Internal Design Pressure	1300.00 KPa.
Allowable External Pressure at Corroded thickness	15881.18 KPa.
Required External Design Pressure	1403.00 KPa.
Required Thickness due to Shell Side pressure	0.5437 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm
Basic Gasket Width,	$b0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	2470.156 mm

Results for ASME Floating Tubesheet Calculations for Configuration B:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.9.4 Step 1:

Tube expansion depth ratio [ρ]:

$= ltx / h$ (modified for corrosion if present)
 $= 0/140$
 $= 0.0000$ (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter [d^*]:

$= \max(dt - 2 \cdot tt(Et/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 100 of 262

ASME Fl-TS Calc: Pressure Test

Case: 1 10:15a Sep 9, 2026

$$= \max(25.4 - 2 \cdot 1.65 (199943392/199943392) \cdot (187/187) \cdot (0); 25.4 - 2 \cdot 1.65)$$

$$= 25.4000 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + d_t$$

$$= 2 \cdot 1208 + 25.4$$

$$= 2440.900 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - d_t) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

Equivalent outer tube limit radius [ao]:

$$= D_o / 2$$

$$= 2441/2$$

$$= 1220.4500 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{(1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2))}$$

$$= 31.8 / \sqrt{(1 - 4 \cdot \min(429 \cdot 100; 4 \cdot 2441 \cdot 31.8) / (\pi \cdot 2441^2))}$$

$$= 31.8964 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (31.9 - 25.4) / 31.9$$

$$= 0.2037$$

Ratio [ps]:

$$= a_s / a_o$$

$$= 1235/1220$$

$$= 1.0120$$

Ratio [pc]:

$$= a_c / a_o$$

$$= 1235/1220$$

$$= 1.0120$$

Compute parameter [xt]:

$$= 1 - Nt((d_t - 2 \cdot t_t) / (2 \cdot a_o))^2$$

$$= 1 - 4924((25.4 - 2 \cdot 1.65) / (2 \cdot 1220))^2$$

$$= 0.5964$$

Determine parameter [xs]:

$$= 1 - Nt(d_t / (2 \cdot a_o))^2$$

$$= 1 - 4924(25.4 / (2 \cdot 1220))^2$$

$$= 0.4668$$

Determine the value [h'g]:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((8 - 0); 0)$$

$$= 8.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.9.4 Step 2:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

β_s , k_s , λ_s , δ_s are all = 0

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

β_C , k_C , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.9.4 Step 3:

$h/p = 4.409449$; $\mu^* = 0.203672$; $E^* = 31928848 \text{ KPa}$.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 101 of 262

ASME Fl-TS Calc: Pressure Test

Case: 1 10:15a Sep 9, 2026

$$E^*/E = 0.159689 ; v^* = 0.453250$$

Note: As h/p (4.409) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [Xa]:

$$\begin{aligned} &= \left(\left(24(1 - v^*)Nt \cdot Et \cdot tt(dt - tt)ao^2 \right) / (E^* \cdot L \cdot H^3) \right)^{1/4} \\ &= \left(\left(24(1 - 0.45^2)4924 \cdot 199943392 \cdot 1.65 \right. \right. \\ &\quad \left. \left. (25.4 - 1.65)1220^2 \right) / (31928848 \cdot 6270 \cdot 140^3) \right)^{1/4} \\ &= 6.6841 \end{aligned}$$

Values from Table 4.18.3

$$\begin{aligned} Z_d &= 0.004960 ; Z_v = 0.022184 ; Z_m = 0.211940 \\ Z_a &= 0.198049E+03 ; Z_w = 0.022184 \end{aligned}$$

ASME Section VIII-2 Part 4.18.9.4 Step 4:

Diameter ratio [K]:

$$\begin{aligned} &= A / Do \\ &= 2650/2441 \\ &= 1.0857 \end{aligned}$$

Compute coefficient [F]:

$$\begin{aligned} &= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.45) / 31928848 \cdot (0 + 0 + \\ &\quad 199943392 \cdot \ln(1.09)) \\ &= 0.2814 \end{aligned}$$

Compute parameter [ϕ]:

$$\begin{aligned} &= (1 + v^*)F \\ &= (1 + 0.45)0.28 \\ &= 0.4090 \end{aligned}$$

Compute parameter [Q1]:

$$\begin{aligned} &= (\rho_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m) \\ &= (1.01 - 1 - 0.41 \cdot 0.022) / (1 + 0.41 \cdot 0.21) \\ &= 0.002681 \end{aligned}$$

ASME Section VIII-2 Part 4.18.9.4 Step 5:

Determine factor [γ_b]:

$$\begin{aligned} &= (G_c - C) / Do \text{ (config b)} \\ &= (2470 - 2550) / 2441 = -0.033 \end{aligned}$$

Calculate parameter [ω_s]:

$$\begin{aligned} &= \rho_s \cdot k_s \cdot \beta_s \cdot \delta S(1 + h \cdot \beta_s) \\ &= 1.01 \cdot 0 \cdot 0 \cdot 0.000(1 + 140 \cdot 0) \\ &= 0.0000 \text{ mm}^2 \end{aligned}$$

Calculate parameter [ω^*s]:

$$\begin{aligned} &= ao^2(\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega_s \\ &= 1220^2(1.01^2 - 1) \cdot (1.01 - 1) / 4 - 0 \\ &= 107.6267 \text{ mm}^2 \end{aligned}$$

Parameter [ω_c]:

$$\begin{aligned} &= \rho_c \cdot k_c [\delta C(C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \\ &\quad \zeta c(C_m \theta + h \cdot C_q \theta \cdot h \cdot \beta C / 2)] \\ &= 1.01 \cdot 0 [0.000(0 + 0 \cdot 0 \cdot 140) - \\ &\quad 0.000(0/140 + 0 \cdot 0/2)] \\ &= 0.0000 \text{ mm}^2 \end{aligned}$$

Parameter [ω^*c]:

$$\begin{aligned} &= ao^2[(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - \omega_c \\ &= 1220^2[(1.01^2 + 1) \cdot (1.01 - 1) / 4 - (1.01 - 1) / 2] - 0 \\ &= 107.6262 \text{ mm}^2 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 102 of 262

ASME Fl-TS Calc: Pressure Test

Case: 1 10:15a Sep 9, 2026

Pressure [P*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.9.4 Step 6:

$P_e = P_s - P_t$

= 1300 - 1300 = 0 KPa.

ASME Section VIII-2 Part 4.18.9.4 Step 7:

Determine factor [Q2]:

$$= [((\omega_s \cdot P_s - \omega_c \cdot P_t) - (\omega_s \cdot P_s - \omega_c \cdot P_c)) + W \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((108 \cdot 1300 - 108 \cdot 1300) - (0 \cdot 0 - 0 \cdot 0)) \cdot 0.0001 + 923767 \cdot -0.033 / (2 \cdot \pi)] / (1 + 0.41 \cdot 0.21)$$

= -4425.6455 Kgf

Tubesheet Bending Stress - original thickness [σ]:

$$= 6 \cdot Q2 / (\mu \cdot (H - h'g)^2)$$

$$= 6 \cdot -4426 / (0.2 \cdot (140 - 8)^2)$$

= -73.3798 N./mm²

Allowable Tubesheet Bending Stress [σ allowed]:

= $1.5 \cdot \beta T \cdot S_y = 1.5 \cdot 0.95 \cdot 480 = 684 \text{ N./mm}^2$, 4.18.233

Tubesheet Bending Stress - final thickness [σ_f]:

$$= 6 \cdot Q2 / (\mu \cdot (H - h'g)^2)$$

$$= 6 \cdot -4623 / (0.2 \cdot (52.2 - 8)^2)$$

= -683.9709 N./mm²

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

= $h + C_{ats} + C_{atc}$

= 52.2 + 0 + 0

= 52.1891 mm

H reqd. = HReqB = 52.1891 mm

Computations completed for ASME tubesheet configuration: B

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	73.4 <=	684.0 N./mm ²	Pass

Floating Tubesheet results per ASME Div. 2 Part 4.18.9 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/Fail
D1uc	66.828 30.557	176 684	...	Ps+Pt-Th D1	Ok
D2uc	50.604 ...	132 684	...	Ps+Pt-Th D2	Ok
D3uc	52.189 ...	73 684	...	Ps+Pt-Th D3	Ok
Max:	66.8282 30.557 mm	0.257	...	(Str. Ratio)	

Load Case Definitions:



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 103 of 262

ASME Fl-TS Calc: Pressure Test

Case: 1 10:15a Sep 9, 2026

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and
Operating Temperatures

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	66.828	140.000	Ok
Tubesheet thickness flanged extension :	30.557	114.000	Ok
Tube thickness :	0.544	1.651	Ok
Tube-tubesheet fillet weld leg :	0.000	1.651	Ok
Tube-tubesheet groove weld leg :	0.000	1.651	Ok

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 104 of 262

Floating Head: Operating

Case: 1 10:15a Sep 9,2026

Input Echo, Floating Head Item 1, Description: Operating

Floating Head Type		D	
Tube Side (Internal) Design Pressure	Pts	1000.00	KPa.
Shell Side (External) Design Pressure	Pss	1103.00	KPa.
Design Temperature for Spherical Head	Temp	200.00	°C
Head Material		SA-240	
Head Material UNS Number		S32205	
Head Allowable Stress at Temperature	Sho	187.00	N./mm ²
Head Allowable Stress at Ambient	Sha	187.00	N./mm ²
Crown Radius for Spherical Head	L	2208.600	mm
Head Thickness	Th	30.000	mm
Tube Side (Internal) Corrosion Allowance	Cats	0.0000	mm
Shell Side (External) Corrosion Allowance	Cass	0.0000	mm
Flange Material		SA-182 F60	
Flange Material UNS Number		S32205	
Flange Allowable Stress at Temperature	Sfo	187.00	N./mm ²
Flange Allowable Stress at Ambient	Sfa	187.00	N./mm ²
Flange Outside Diameter	Fod	2650.000	mm
Flange Inside Diameter	Fid	2454.000	mm
Flange Thickness	Tf	150.0000	mm
Bolt Material		SA-193 B7	
Bolt Material UNS Number		G41400	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2550.000	mm
Nominal Bolt Diameter	dB	47.6250	mm
Type of Threads		UNC Thread Series	
Number of Bolts		40	
Full face Gasket Flange Option		Program Selects	
Flange Face Outside Diameter	Ffod	2486.000	mm
Flange Face Inside Diameter	Ffid	2454.000	mm
Gasket Outside Diameter	Go	2483.000	mm
Gasket Inside Diameter	Gi	2457.000	mm
Gasket Factor,	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Flange Facing Sketch		Code Sketch 1a	
Column for Gasket Seating		Code Column I	
Gasket Thickness		4.5000	mm
Width of raised face		0.0000	mm
Length of Partition Gasket		2700.000	mm
Width of Partition Gasket		22.0000	mm
Partition Gasket Factor,	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²
Partition Gasket Facing Sketch		Code Sketch 1a	
Partition Gasket Column for Gasket Seating		Code Column I	
Dist. from Head Centerline to Flange Centroid	hr	27.0400	mm

The Flange is not Slotted.

This unit does not have a Backing Ring.

Use Full Bolt Load in Flange Design Yes



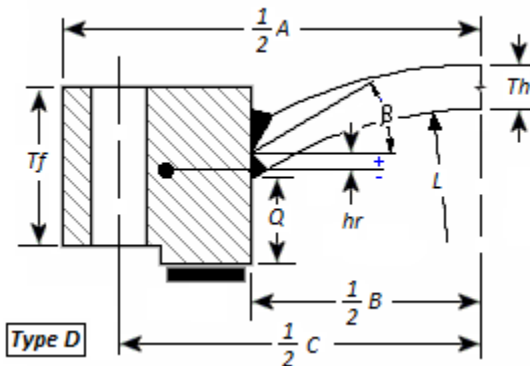
PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 105 of 262

Floating Head: Operating

Case: 1 10:15a Sep 9,2026



Internal Pressure results for Spherical Heads:

ASME Code, Section VIII, Division 1, 2025, App. 1-6

Thickness Due to Internal Pressure[thr]:

$$= 5PL / 6Sho$$

$$= (5 \cdot 1000 \cdot 2209) / (6 \cdot 187)$$

$$= 9.8430 \text{ mm}$$

Maximum Allowable Working Pressure at Given Thickness[Pa]:

$$= 6Sho \cdot (Th - Cass - Cats) / 5L$$

$$= (6 \cdot 187 \cdot 30) / (5 \cdot 2209)$$

$$= 3047.85 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold[Pnc]:

$$= 6Sha / 5L$$

$$= (6 \cdot 187 \cdot 30) / (5 \cdot 2209)$$

$$= 3047.85 \text{ KPa.}$$

Actual Stress at given Pressure and Thickness[Sact]:

$$= 5PL / 6(Th - Cass - Cats)$$

$$= (5 \cdot 1000 \cdot 2209) / (6 \cdot 30)$$

$$= 61. \text{ N./mm}^2$$

Spherical head Percent Elongation $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$: 1.012 %

External Pressure results, Spherical Head

ASME Code, Section VIII Division 1, 2025

External Pressure Chart HA-5 at 200.00 °C
Elastic Modulus for Material 183008944.00 KPa.

Results for Maximum Allowable External Pressure:

Corroded Thickness of Shell	TCA	30.0000	mm
Outside Crown Radius of Shell	OD/2	2238.5999	mm
Crown Radius / Thickness Ratio	$(OD/2)/T$	74.6200	
Geometry Factor, A	f(D/T, LD)	0.0016752	
Materials Factor, B	f(A, Chart)	113.1410	N./mm ²
Maximum Allowable Working Pressure		1516.14	KPa.
EMAWP = $B / ((OD/T)/2) = 113 / (149/2.0) = 1516$			

Results for Req'd Thickness for Ext. Pressure (Tca):

Corroded Thickness of Head	TCA	23.2605	mm
Outside Diameter of Shell	ODCA	4477.200	mm
Diameter / Thickness Ratio	(D/T)	192.4807	
Geometry Factor, A	f(DT, LD)	0.0012988	
Materials Factor, B	f(A, Chart)	106.1707	N./mm ²
Maximum Allowable Working Pressure		1103.12	KPa.
EMAWP = $B / ((D/T)/2) = 106 / (192/2) = 1103$			



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 106 of 262

Floating Head: Operating

Case: 1 10:15a Sep 9,2026

Intermediate Calculations for Flanged portion:

ASME Code, Section VIII, Div. 1, 2025 App. 2

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	13.000 mm
Basic Gasket Width,	$b_0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b_0) * 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = G_o - 2.0 * b$	2470.156 mm

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsm_{ax}]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 * 47.6 + 6 * 150 / (3 + 0.5)$$

$$= 352.393 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C * \sin(\pi / n)$$

$$= 2550 * \sin(\pi / 40)$$

$$= 200.071 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{200 / (2 * 47.6 + 150)}, 1)$$

$$= 1.0000$$

Bolting Information for UNC Thread Series (Non Mandatory):

Total Area of Bolts 526.709 cm²

	Minimum	Actual	Maximum
Radial Distance between Bolts and Edge:	47.625	50.000	
Circ. Spacing between the Bolts:	101.600	200.071	352.393

Results for the Internal Pressure case:

Basic Flange and Bolt loads:

Hydrostatic End Load due to Pressure	H	488691.8	Kgf
Contact Load on Gasket Surfaces	Hp	44295.3	Kgf
Hydrostatic End Load at Flange ID	Hd	482320.3	Kgf
Pressure Force on Flange Face	Ht	6371.4	Kgf
Radial Component of Head Membrane Force	Hr	727362.9	Kgf
Operating Bolt Load:	Wm1	532987.1	Kgf
Gasket Seating Bolt Load	Wm2	508996.8	Kgf
Required Bolt Area	Am	303.896	cm ²

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calculation]:

$$= Ab * Sb / (y * PI * (Go + Gi))$$

$$= 527 * 172 / (68.9 * 3.14 * (2483 + 2457))$$

$$= 8.466 \text{ mm}$$

Note: Recommended Min. Width for Sheet and Composite Gaskets per table 2-4 :

= 38.100 mm [Note: Exceeds actual gasket width, 13.000]

Flange Design Bolt Load (Seating)	W	923766.8	Kgf
Gasket Seating Force (Operating)	Hg	44295.3	Kgf
Distance to Gasket Load Reaction	hg	39.9222	mm
Distance to Face Pressure Reaction	ht	43.9611	mm
Distance to End Pressure Reaction	hd	48.0000	mm

Summary of Moments for Internal Pressure:

Loading	Force	Distance	Bolt Corr	Moment
End Pressure,	Md	482320.	48.0000	1.0000
				23152. Kg-m.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 107 of 262

Floating Head: Operating

Case: 1 10:15a Sep 9, 2026

Face Pressure,	Mt	6371.	43.9611	1.0000	280. Kg-m.
Gasket Load,	Mg	44295.	39.9222	1.0000	1768. Kg-m.
Floating Hd. Load,	Mh	727363.	27.0400	1.0000	-19668. Kg-m.
Gasket Seating,	Ma	923767.	39.9222	1.0000	36880. Kg-m.

Total Moment for Operation (Internal Pressure) 5532.041 Kg-m.

Total Moment for Gasket Seating (Int. Pressure) 36879.508 Kg-m.

Results for the External Pressure case:

Basic Flange and Bolt loads:

Hydrostatic End Load due to Pressure	H	539027.1	Kgf
Hydrostatic End Load at Flange ID	Hd	531999.3	Kgf
Pressure Force on Flange Face	Ht	7027.7	Kgf
Radial Component of Head Membrane Force	Hr	802281.2	Kgf
Gasket Seating Bolt Load	Wm2	508996.8	Kgf
Required Bolt Area	Am	290.217	cm ²

Flange Design Bolt Load (Seating)	W	923766.8	Kgf
Distance to Gasket Load Reaction	hg	39.9222	mm
Distance to Face Pressure Reaction	ht	43.9611	mm
Distance to End Pressure Reaction	hd	48.0000	mm

Summary of Moments for External Pressure:

Loading		Force	Distance	Bolt Corr	Moment
End Pressure,	Md	531999.	8.0778	1.0000	4297. Kg-m.
Face Pressure,	Mt	7028.	4.0389	1.0000	28. Kg-m.
Floating Hd. Load,	Mh	802281.	27.0400	1.0000	-21694. Kg-m.
Gasket Seating,	Ma	923767.	39.9222	1.0000	36880. Kg-m.

Total Moment for Operation (External Pressure) 17368.260 Kg-m.

Total Moment for Gasket Seating (Ext. Pressure) 36879.508 Kg-m.

Factor for integral type flanges [F]:

$$= P \cdot B \cdot \sqrt{(4 \cdot L \cdot L - B \cdot B) / (8 \cdot S(A - B))}$$

$$= 1000 \cdot 2454 \cdot \sqrt{(4 \cdot 2209 \cdot 2209 - 2454 \cdot 2454) / (8 \cdot 187(2650 - 2454))}$$

$$= 30.7410 \text{ mm}$$

Factor [J]:

$$= Mo(A + B) / (S \cdot B)(A - B)$$

$$= 5532(2650 + 2454) / (187 \cdot 2454)(2650 - 2454)$$

$$= 3078.5886 \text{ mm}^2$$

Required thickness for Main Flange, Internal Operating Condition [T]:

$$= F + \sqrt{(F \cdot F + J)} \text{ per 1-6(g)}$$

$$= 30.7 + \sqrt{(30.7 \cdot 30.7 + 3079)}$$

$$= 94.1729 \text{ mm}$$

Factor [J]:

$$= Mo(A + B) / (S \cdot B)(A - B)$$

$$= 36880(2650 + 2454) / (187 \cdot 2454)(2650 - 2454)$$

$$= 20523.4980 \text{ mm}^2$$

Required thickness for Main Flange, Internal Bolt-up Condition [T]:

$$= F + \sqrt{(F \cdot F + J)} \text{ per 1-6(g)}$$

$$= 0 + \sqrt{(0 \cdot 0 + 20523)}$$

$$= 143.2602 \text{ mm}$$

Factor for integral type flanges [F]:

$$= P \cdot B \cdot \sqrt{(4 \cdot L \cdot L - B \cdot B) / (8 \cdot S(A - B))}$$

$$= 1103 \cdot 2454 \cdot \sqrt{(4 \cdot 2209 \cdot 2209 - 2454 \cdot 2454) / (8 \cdot 187(2650 - 2454))}$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 108 of 262

Floating Head: Operating

Case: 1 10:15a Sep 9, 2026

= 33.9074 mm

Factor [J]:

= $Mo(A + B)/(S \cdot B)(A - B)$

= $17368(2650+2454)/(187 \cdot 2454)(2650-2454)$

= 9665.4600 mm²

Required thickness for Main Flange, External Operating Condition [T]:

= $F + \sqrt{(F \cdot F + J)}$ per 1-6(g)

= $33.9 + \sqrt{(33.9 \cdot 33.9 + 9665)}$

= 137.9034 mm

Factor [J]:

= $Mo(A + B)/(S \cdot B)(A - B)$

= $36880(2650+2454)/(187 \cdot 2454)(2650-2454)$

= 20523.4980 mm²

Required thickness for Main Flange, External Bolt-up Condition [T]:

= $F + \sqrt{(F \cdot F + J)}$ per 1-6(g)

= $0 + \sqrt{(0 \cdot 0 + 20523)}$

= 143.2602 mm

Summary of Required Thicknesses(mm):

	Head	Flange
Tubeside (Internal) Pressure	9.8430	94.1729
Shellside (External) Pressure	23.2605	137.9034
Tubeside Gasket Seating Load		143.2602
Shellside Gasket Seating Load		143.2602
Maximum + Corrosion Allowance	23.2605	143.2602
Actual Thickness as Given	30.0000	150.0000

Floating Head MAWP and MAPnc (KPa.):

Stress Condition	Tubeside		Shellside	
	MAWP	MAPnc	MAWP	MAPnc
Flange Stress	1827.37	1828.69	1250.28	1251.02
Floating head Stress	3047.75	3047.75	1516.04	1852.72
Bolt Stress	1733.09	1733.09	...	...
Minimum MAWP & MAPnc	1733.09	1733.09	1250.28	1251.02

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 197-C_Channel MOC= CS Page 109 of 262
 External Pressure Calculations: Step: 18 10:15am Sep 9,2026

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	...	182	No Calc	No Calc
20	30	No Calc	...	257	No Calc	No Calc
30	40	1200	2732	13	0.0010505	86.8005
40	50	No Calc	...	252	No Calc	No Calc
50	60	No Calc	...	251.2	No Calc	No Calc
60	70	6300	2729	13.7	0.00019682	19.6773
70	80	6300	2745	21.7	0.0003903	39.0208
80	90	6300	2729	13.7	0.00019682	19.6773
90	100	No Calc	...	254.2	No Calc	No Calc
100	110	No Calc	...	254.2	No Calc	No Calc
110	120	975.8	2729	13.7	0.0014211	95.3711
120	130	No Calc	2727.2	12.8	0.00065187	65.1723

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	185	138.227	103.419	No Calc
20	30	260	254.381	103.419	No Calc
30	40	16	9.21987	103.419	551
40	50	255	251.638	103.419	No Calc
50	60	252	250.19	...	No Calc
60	70	14.5	No Calc	...	132
70	80	22.5	No Calc	...	411
80	90	14.5	No Calc	...	132
90	100	255	250.19	...	No Calc
100	110	255	250.19	...	No Calc
110	120	14.5	No Calc	...	638
120	130	13.6	No Calc	...	340

Minimum

132

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	40	1200	7034.16	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc
60	70	6300	No Calc	No Calc	No Calc
70	80	6300	No Calc	No Calc	No Calc
80	90	6300	No Calc	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc
100	110	No Calc	No Calc	No Calc	No Calc
110	120	975.8	No Calc	No Calc	No Calc
120	130	No Calc	No Calc	No Calc	No Calc



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 110 of 262

External Pressure Calculations: Step: 18 10:15am Sep 9, 2026

[Elements Suitable for External Pressure.](#)

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 111 of 262

Element and Detail Weights:

Step: 19 10:15am Sep 9,2026

Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	10573.5	...	10573.5	...	1057.35
20	30	2860.67	1856513	2796.9	1857915	286.067
30	40	1269.73	6871895	1032.79	6902471	126.973
40	50	2808.63	1827753	2745.84	1829154	280.863
50	60	2685.6	1819030	2668.97	1819405	268.56
60	70	2010.41	6139097	1910.45	6151997	201.041
70	80	3979.6	8077760	3848.07	8094733	397.96
80	90	2010.41	6630709	1910.45	6643609	201.041
90	100	2715.84	1836358	2699.05	1836733	271.584
100	110	2715.84	1836358	2699.05	1836733	271.584
110	120	740.677	4008605	703.849	4013357	74.0677
120	130	1148.59	2867870	1091.16	2872800	114.859
Total		35519	43771952.00	34680	43858908.00	3551



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Liqd	...	92.5	1500	...	Liquid: 10
20	Liqd	1855.38	157.5	...	...	Liquid: 20
30	Liqd	6867.7	600	...	...	Liquid: 30
30	Nozl	224.498	600	1531.79	...	N1-Tube Inlet
30	Nozl	228.425	600	-1531.79	...	N2-Tube outlet
30	Wght	400	600	...	...	Pass partition
40	Liqd	1826.64	155	...	...	Liquid: 40
50	Liqd	1817.23	153.5	5	...	Liquid: 50
50	Insl	59.6378	136	...	...	Insulation
50	Lini	111.04	136	...	...	cladd
60	Sadl	2103.22	1260	1648.35	...	Left Saddle flo
60	Liqd	6135.35	950	...	...	Liquid: 60
60	Insl	595.096	950	...	...	Insulation
60	Lini	386.81	950	...	...	Cladding
60	Wght	800	1260	-2000	...	Fireproofing we
70	Liqd	8072.83	1250	...	...	Liquid: 60
70	Insl	787.45	1250	...	...	Insulation
70	Lini	508.96	1250	...	...	cladding
70	Nozl	1974.31	1500	1807.2	...	N3-Shell inlet
70	Nozl	1730.36	1500	-1807.2	...	N4-Shell outlet
70	Nozl	30.1839	500	-1369.09	...	Drain
70	Nozl	13.2166	500	1369.09	...	Vent
80	Sadl	2103.22	640	1648.35	...	RH fixed saddle
80	Liqd	6626.67	950	...	...	Liquid: 80
80	Insl	595.096	950	...	...	Insulation
80	Lini	386.81	950	...	...	Cladding
80	Wght	800	640	-2000	...	Fireproofing we
90	Liqd	1834.54	155	5	...	Liquid: 70
90	Insl	59.6473	136	...	...	Insulation
90	Lini	111.04	136	...	...	weldoverlay
100	Liqd	1834.54	155	5	...	Liquid: 80
100	Lini	111.04	136	...	...	weldoverlay



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 112 of 262

Element and Detail Weights:

Step: 19 10:15am

Sep 9, 2026

110 Liqd	4006.16	350	...	...
110 Lini	142.509	350	...	...
120 Liqd	2866.12	275.8	...	...
120 Lini	219.523	362.9	...	...
40 FTsh	4434.45	368.35	...	...
40 Tube	34112.6	3573.35	...	...
40 FLTs	3630.87	6848.35	...	...
40 FLHd	2333.43	6923.35	...	...

Liquid: 90
cladding
Liquid: 100
cladding
Fixed Tubeheet
Tubes

Total Weight of Each Detail Type:

Saddles	4206.4
Liquid	43743.2
Insulation	2096.9
Lining	695.2
Cladding & Overlay	1282.6
Nozzles	4201.0
Weights	2000.0
Exchanger Components	44511.4
Liquid in Tubes	12364.2

Sum of the Detail Weights 115100.9 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	39071.5	39071.5	39071.5	39071.5	39071.5	39071.5
Saddles	4206.4	4206.4	4206.4	4206.4	4206.4	4206.4
Cladding	1282.6	1282.6	1282.6	1282.6	1282.6	1282.6
Nozzles	4201.0	4201.0	4201.0	4201.0	4201.0	4201.0
Wld Weights	2000.0	2000.0	2000.0	2000.0	2000.0	2000.0
Exchanger	44511.4	44511.4	44511.4	44511.4	44511.4	44511.4
Insulation	...	...	...	...	2096.9	2096.9 *
Lining	695.2	695.2	695.2	695.2	695.2	695.2
Ope. Liquid	...	...	...	...	...	43743.2
Tube Ope Lqd	...	...	...	...	...	12364.2
Test Liquid	...	43498.4	...	...	...	...
Tube Tst Lqd	...	12364.3	...	...	...	...
Totals	95968.0	151830.7	95968.0	95968.0	98065.0	154172.3

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	95968.0 kg.
Shop Test Wt.	- Fabricated weight + water (full)	151830.7 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	95968.0 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	95968.0 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	98065.0 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	154172.3 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	153249.0 kg.
Field Test Wt.	- Empty weight + test medium (full)	153091.2 kg.

Exchanger Tube Data

Volume of Exchanger tubes : 12371804.0 cm³



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 113 of 262

Element and Detail Weights:

Step: 19 10:15am Sep 9,2026

Weight of Ope Liq in tubes : 12364.2 kg.
Weight of Water in tubes : 12364.3 kg.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	88875.7	...
20	30	43433.5	27397.8
30	40	102994	101788
40	50	42962.2	26973.7
50	60	42264.9	26818.2
60	70	163074	161164
70	80	215827	212058
80	90	163074	161164
90	100	42547.7	27073.6
100	110	42547.7	27073.6
110	120	60079.8	59376.1
120	130	85266.9	83331.5

Total 1092947.4 914217.3 cm²
Total 1176.4 984.1 ft²

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 114 of 262

Nozzle Flange MAWP:

Step: 20 10:15am Sep 9,2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating Ope.	Flange Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	UG-44	Max Pressure 50%	DNV
N1-Tube Inlet	1820.4	1960.0	82	150	GR 1.1	2050.0	1820.4	1820.4	680.6
N2-Tube outlet	1820.4	1960.0	82	150	GR 1.1	2050.0	1820.4	1820.4	680.6
N3-Shell inlet	4236.0	5100.0	240	300	GR 1.1	...	...	...	...
N4-Shell outle	4236.0	5100.0	240	300	GR 1.1	953.6	3706.1	4236.0	4236.0
Drain	35278.3	42510.0	240	2500	GR 1.1	...	...	...	...
Vent	1244.0	1960.0	240	150	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 1244.040 KPa.

Lowest Flange Pressure Rating was (Amb)[ShellSide]: 1960.006 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 1820.434 KPa.

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 1960.006 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
N1-Tube Inlet	442.976	1.20	3229	3059	1045.2	1820.4	Pass
N2-Tube outlet	442.976	1.20	3229	3059	1045.2	1820.4	Pass
N4-Shell outlet	981.202	0.10	16348	6884	1155.9	3706.1	Pass

Evaluating Flange on Nozzle [N1-Tube Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(3229) + 4 \cdot \text{abs}(3059) \cdot 442.976 \leq \pi \cdot 442.976^3 [(1820.434 - 1045.152) + 1.200 \cdot 1820.434]$$

$$= 0.57089E+05 \leq 0.82425E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-Tube Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1820.434, 1.200 \cdot 1820.434 + 1820.434 - 57089.2 / (\pi \cdot 442.976^3))$$

$$= \min(1820.434, 1954.935)$$

$$= 1820.434 \text{ KPa.}$$

Evaluating Flange on Nozzle [N2-Tube outlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(3229) + 4 \cdot \text{abs}(3059) \cdot 442.976 \leq \pi \cdot 442.976^3 [(1820.434 - 1045.152) + 1.200 \cdot 1820.434]$$

$$= 0.57089E+05 \leq 0.82425E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2-Tube outlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1820.434, 1.200 \cdot 1820.434 + 1820.434 - 57089.2 / (\pi \cdot 442.976^3))$$

$$= \min(1820.434, 1954.935)$$

$$= 1820.434 \text{ KPa.}$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 115 of 262

Nozzle Flange MAWP:

Step: 20 10:15am Sep 9, 2026

Evaluating Flange on Nozzle [N4-Shell outlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(16348) + 4 \cdot \text{abs}(6884) 981.202 \leq \pi \cdot 981.202^3 [(4236.041 - 1155.914) + .100 \cdot 4236.041]$$

$$= 0.28859\text{E}+06 \leq 0.10604\text{E}+07 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N4-Shell outlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4236.041, 0.100 \cdot 4236.041 + 4236.041 - 288588.0 / (\pi \cdot 981.202^3))$$

$$= \min(4236.041, 3706.085)$$

$$= 3706.085 \text{ KPa.}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 116 of 262

Wind Load Calculation:

Step: 21 10:15am Sep 9,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	9000 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q)/(1 + 1.7 \cdot gV \cdot I_{zbar})))$$

$$= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.22 \cdot 0.94)/(1 + 1.7 \cdot 3.4 \cdot 0.22)))$$

$$= \min(0.85, 0.89)$$

$$= 0.850$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.548
Structure Height to Diameter ratio	3.893

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Z_g = 274320 \text{ mm}$

Effective Height [z]:

= Centroid Height + Vessel Base Elevation
 = 1981 + 9000 = 10981 mm
 = 36 ft. Imperial Units

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\text{Because } z (36 \text{ ft.}) > 15 \text{ ft.}$$

$$= 2.01 (z / Z_g)^{2/\alpha}$$

$$= 2.01 (36/900)^{2/9.5}$$

$$= 1.02$$

Type of Hill: No Hill



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 117 of 262

Wind Load Calculation:

Step: 21 10:15am

Sep 9, 2026

Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0 • 0 • 0)²

= 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 1.02 • 1 • 1 • 1 • 60.4²)

= 16 psf [78.1] Kgs/m²

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16 • 0.85 • 0.55 • 7.17

= 53.4 lbs. [24.2] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
LHS Ch Cover	10981.2	0.000	0.000	0.000	1.021	1.000	78.120
LHS Ch Flg-1	10981.2	0.000	0.000	0.000	1.021	1.000	78.120
LHS Ch Shell	10981.2	0.000	0.000	0.000	1.021	1.000	78.120
LHS Ch Flg-2	10981.2	0.000	0.000	0.000	1.021	1.000	78.120
LHS Shell Flg-1	10981.2	0.000	0.000	0.000	1.021	1.000	78.120
Shell-1	10981.2	0.000	0.000	0.000	1.021	1.000	78.120
Shell 2	10981.2	0.000	0.000	0.000	1.021	1.000	78.120
Shell 3	10981.2	0.000	0.000	0.000	1.021	1.000	78.120
Rhs Shell Flg-1	10981.2	0.000	0.000	0.000	1.021	1.000	78.120
Rhs Shell Flg-2	10981.2	0.000	0.000	0.000	1.021	1.000	78.120
Rhs ch shell	10981.2	0.000	0.000	0.000	1.021	1.000	78.120
Dish End	10981.2	0.000	0.000	0.000	1.021	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Pass partitio	0.00	10981.20	78.12	0.00
Fireproofing	0.00	9000.00	78.12	0.00
Fireproofing	0.00	9000.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	10981.2	3600	6660	78.12	26.6687
20	30	10981.2	3240	10206	78.12	40.8679
30	40	10981.2	3278.4	39340.8	78.12	157.533
40	50	10981.2	3240	10044	78.12	40.2192
50	60	10981.2	3464.64	10636.4	78.12	42.5916
60	70	10981.2	3514.8	66781.2	78.12	267.412
70	80	10981.2	3534	88350	78.12	353.78
80	90	10981.2	3514.8	66781.2	78.12	267.412
90	100	10981.2	3462.58	10734	78.12	42.9822
100	110	10981.2	3252	10081.2	78.12	40.3682
110	120	10981.2	3274.8	22923.6	78.12	91.7931
120	130	10981.2	3272.64	16411.9	78.12	65.7182



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 118 of 262

Wind Load Calculation:

Step: 21 10:15am Sep 9, 2026

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = Ct • hnx where Ct = 0.020, x = 0.75 and hn = Structural Height (ft.)
 = 0.020(11.40.75)
 = 0.124 seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33)
 = 0.030

Check the Value of T which is the smaller of Cu•Ta and T:
 = Minimum Value of (1.7 • 0.12, 0.03) per 12.8.2
 = 0.030
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = 0.3 • Sds • W • I
 = 0.3 • 0.16 • 149966 • 1.25
 = 8817.995 Kgf

Vertical load per 12.4-4, [Ev]:
 = 0.2 • Sds • W
 = 0.2 • 0.16 • 149966
 = 4702.9 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	1500	10711.8	693.784



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 120 of 262

Earthquake Load Calculation:

Step: 22 10:15am

Sep 9, 2026

20	30	1350	10711.8	624.406
30	40	1350	10711.8	624.406
40	50	1350	10711.8	624.406
50	60	1355	10711.8	626.718
60	70	1350	10711.8	624.406
60	70	1350	10711.8	624.406
70	80	1350	10711.8	624.406
80	0	1350	10711.8	624.406
80	90	1350	10711.8	624.406
90	100	1355	10711.8	626.718
100	110	1355	10711.8	626.718
110	120	1350	10711.8	624.406
120	130	1350	10711.8	624.406

The Earthquake Weight does not include the weight of the saddles.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 121 of 262

Center of Gravity Calculation: Step: 23 10:15am Sep 9,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.
Lining is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	5619.700 mm
Center of Gravity of the Liquid	5398.357 mm
Center of Gravity of the Insulation	5618.900 mm
Center of Gravity of the Lining	6654.822 mm
Center of Gravity of the Nozzles	5345.818 mm
Center of Gravity of the Added Weights (Operating)	4717.030 mm
Center of Gravity of the Added Weights (Empty)	4717.030 mm
Center of Gravity of the Tubesheet(s)	4991.885 mm
Center of Gravity of the Tubes	5279.700 mm
Center of Gravity of the Floating Head	8636.050 mm
Center of Gravity of Bare Shell New and Cold	3617.421 mm
Center of Gravity of Bare Shell Corroded	3602.065 mm
Vessel CG in the Operating Condition	4958.104 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	4694.766 mm
Vessel CG in the Test Condition	4946.879 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 197-C_Channel MOC= CS Page 122 of 262
 Saddle Calcs: Bundle Pull New: Step: 24 10:15am Sep 9,2026

ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1357.25	mm
Shell Thickness used in this Case	t	14.500	mm
Stiffened Vessel Length per 4.15.6	L	7227.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1260.00	mm
Saddle Width	b1 or b	585.00	mm
Saddle Bearing Angle	delta or theta	140.00	degrees
Wear Plate Width	b2 or b1	875.00	mm
Wear Plate Bearing Angle	delta2 or theta1	152.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		136.40	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1981.20	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		48400.00	Kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	

Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		22398.23	Kgf
Pressure used in Analysis	P	0.000	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-1.76	-234.00
Long. Stress at Top of Midspan		1.76	138.00
Long. Stress at Bottom of Midspan		1.76	138.00
Long. Stress at Top of Saddles		1.71	138.00
Long. Stress at Bottom of Saddles		-0.97	-234.00
Long. Stress at Bottom of Saddles		0.97	138.00
Tangential Shear in Shell		6.54	110.40
Circ. Stress at Horn of Saddle		16.04	207.00
Circ. Compressive Stress in Shell		0.55	138.00

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
 = Bundle Pull Out Force • Impact Factor • B / Ls
 = 48400 • 1.5 • 1981/3780
 = 38051.6 Kgf

Load Combination Results for Tube Bundle Pull Out [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = -15653 + max(38052, 0, 0, 0)



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 123 of 262

Saddle Calcs: Bundle Pull New: Step: 24 10:15am Sep 9, 2026

= 22398.2 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	24501.45	Kgf
Transverse Shear Load	0.00	Kgf
Longitudinal Shear Load	72600.00	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= Baseplate Length

= 2349.500 mm

The Computed K values from Table 4.15.1:

K1 = 0.1412	K2 = 0.9003	K3 = 0.5922	K4 = 0.3269
K5 = 0.6971	K6 = 0.0379	K7 = 0.0338	K8 = 0.3167
K9 = 0.2372	K10 = 0.0432	K1* = 0.2486	K6p = 0.0305
K7p = 0.0272			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \frac{(1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))}{(1 + (4hm)/3L)} \right]$$

$$= -22398 \cdot 1260 \left[1 - \frac{(1 - 1260/7227 + (1357^2 - 0^2)/(2 \cdot 1260 \cdot 7227))}{(1 + (4 \cdot 0)/(3 \cdot 7227))} \right]$$

$$= -2065.8 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 \left(\frac{(1 + 2(Rm^2 - hm^2)/(L^2))}{(1 + (4hm^2)/(3L))} - 4a/L \right)$$

$$= 22398 \cdot 7227/4 \left(\frac{(1 + 2(1357^2 - 0^2)/(7227^2))}{(1 + (4 \cdot 0)/(3 \cdot 7227))} - 4 \cdot 1260/7227 \right)$$

$$= 15101.1 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$

$$= 0 \cdot 1357/(2 \cdot 14.5) - 15101/(\pi \cdot 1357^2 \cdot 14.5)$$

$$= -1.76 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 0 \cdot 1357/(2 \cdot 14.5) + 15101/(\pi \cdot 1357^2 \cdot 14.5)$$

$$= 1.76 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t)$$

$$= 0 \cdot 1357/(2 \cdot 14.5) - 2066/(0.14 \cdot \pi \cdot 1357^2 \cdot 14.5)$$

$$= 1.71 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm/(2t) + M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 0 \cdot 1357/(2 \cdot 14.5) + 2066/(0.25 \cdot \pi \cdot 1357^2 \cdot 14.5)$$

$$= -0.97 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot hm/3))$$

$$= 22398 \cdot (7227 - 2 \cdot 1260)/(7227 + (4 \cdot 0/3))$$

$$= 14588.1 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 0.9 \cdot 14588/(1357 \cdot 14.5)$$

$$= 6.54 \text{ N./mm}^2$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 124 of 262

Saddle Calcs: Bundle Pull New: Step: 24 10:15am Sep 9,2026

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1357 \cdot 14.5)}$$

$$= 109.423 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(585 + 1.56 \cdot \sqrt{(1357 \cdot 14.5)}, 2 \cdot 1260)$$

$$= 803.85 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

= 1.0000 Materials are the same, bundle pull case

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7 \cdot 22398 \cdot 0.1 / (804(14.5 + 1 \cdot 20))$$

$$= -0.55 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -22398 / (4(14.5 + 1 \cdot 20)804) -$$

$$12 \cdot 0.034 \cdot 22398 \cdot 1357 / (7227(14.5 + 1 \cdot 20)^2)$$

$$= -16.04 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2349.5000	mm
Baseplate Thickness	Bpthk	50.0000	mm
Baseplate Width	Bpwid	610.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	650.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1350 \cdot 14.5)} + 875)14.5^3) / 12$$

$$= 27.774 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 27.8 + 159(300 - 7.25)^2$$

$$= 135426.000 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	1093.3	14.5	7.2	158.5	114929.0	0.135E+06
Wearplate	875.0	20.0	24.5	175.0	428749.9	0.132E+06
Web	20.0	546.7	307.8	109.3	3366030.5	0.273E+05
BasePlate	610.0	50.0	606.2	305.0	18489098.0	0.288E+06
Totals	...	...	...	747.9	22398806.0	0.583E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 22398806/748$$

$$= 299.504 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 125 of 262

Saddle Calcs: Bundle Pull New: Step: 24 10:15am Sep 9,2026

$$= 180 - 140/2$$

$$= 110.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(110) - 0.5 \cdot \sin(110)^2) / (\pi - 1.92 + \sin(110) \cos(110))$$

$$= 0.2404$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.24 \cdot 22398$$

$$= 5385.2251 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.8961 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1981 - 1350 \cdot \sin(140/2) / 1.22$$

$$= 942.849 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 5077.5547 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.5595 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(22398 + 2103) \cdot 610 / (4 \cdot 2350 \cdot 173)}$$

$$= 16.430 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 140/2) (1350 + 14.5 + 20)$$

$$= 2602.009 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 2602 / (5 - 1)$$

$$= 650.502 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot B_{\text{pwid}} / 2$$

$$= 651 \cdot 610 / 2$$

$$= 1984.032 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 24501 / (2350 \cdot 610)$$

$$= 1.710 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1984 \cdot 1.71$$

$$= 3391.838 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 113 + 65.1$$

$$= 178.050 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 126 of 262

Saddle Calcs: Bundle Pull New: Step: 24 10:15am Sep 9,2026

= 3392/178
= 1.868 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	585.0	...	117.0	...	0.334E+05

Rib dimension [D]:
= Saddle Width - Web Thickness
= 585 - 20
= 565.000 mm

Distance to Centroid from Datum [ytot]:
= AY / A
= 0/178
= 0.000 mm

Distance to Centroid [C1]:
= Saddle Width / 2
= 585/2
= 292.500 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{33367/178}$
= 136.895 mm

Length of Outer Rib, step 1, [Lv]:
= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk
= 1981 - cos(140/2)(1350 + 14.5 + 20) - 50
= 1457.673 mm

Length of Outer Rib, step 2, [Lw]:
= 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)
= 2 • sin(140/2)(1350 + 14.5 + 20)
= 2602.009 mm

Length of Outer Rib, step 3, [Bd]:
= Baseplate length - clearance
= 2350 - 50.8
= 2298.700 mm

Length of Outer Rib, step 4, [Dd]:
= (Lw - Bd)/2
= (2602 - 2299)/2
= 151.654 mm

Length of Outer Rib [L]:
= $\sqrt{Lv^2 + Dd^2}$
= $\sqrt{1458^2 + 152^2}$
= 1465.541 mm

Intermediate Term [Cc]:
= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 199943008/260}$
= 123.211

Slenderness ratio [KL/r]:
= KL/r
= 1 • 1466/137
= 10.706



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 127 of 262

Saddle Calcs: Bundle Pull New: Step: 24 10:15am Sep 9,2026

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 72600 / (2 \cdot 2350) \cdot 651 \cdot 1466 / 2$$

$$= 7364.725 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($10.7 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (10.7)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (10.7) / (8 \cdot 123) - (10.7^3) / (8 \cdot 123^3))$$

$$= 152.4 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 1.87 / 152 + (7365 \cdot 292 / 333669344) / 173$$

$$= 0.378$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	565.0	0.0	113.0	0.0	0.334E+05
Web	650.5	20.0	0.0	130.1	0.0	43.4
Totals	...	...	...	243.1	...	0.334E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 243$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 585 / 2$$

$$= 292.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1981 - \cos(1385 / 2) (50 + 0 + 0) - 0$$

$$= 546.700 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (33409 / 243) }$$

$$= 117.230 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 547 / 117$$

$$= 4.663$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 72600 / (2 \cdot 2350)$$

$$= 15.450 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 15.5 \cdot 651 \cdot 547$$

$$= 5494.619 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 5495 / 1142186$$

$$= 47.176 \text{ N./mm}^2$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 128 of 262

Saddle Calcs: Bundle Pull New: Step: 24 10:15am Sep 9,2026

Compressive Allowable, $KL/r < C_c$ ($4.66 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (4.66)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (4.66) / (8 \cdot 123) - (4.66^3) / (8 \cdot 123^3))$$

$$= 154.6 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 2.74 / 155 + (5495 \cdot 292 / 33409) / 173$$

$$= 0.290$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	16.1400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	14.0743 cm ²
Saddle Load Q0 (Weight)	Q0	24501.4 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	72600.0 Kg
Maximum Transverse Force	Ft	0.0 Kg
Maximum Longitudinal Force	F1	72600.0 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 72600 / (14.1 \cdot 8)$$

$$= 63.2 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 0 / (14.1 \cdot 8)$$

$$= 0.0 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= (QL - Q0) / (Stba \cdot Rnbolts)$$

$$= (72600 - 24501) / (48.3 \cdot 8)$$

$$= 12.2076 \text{ cm}^2$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 72600 / (29 \cdot 8)$$

$$= 30.7104 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1981 \cdot 0 + 0$$

$$= 0.00 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / Q0$$

$$= 0 / 24501$$

$$= 0.00 \text{ mm} < B_{plen} / 6 \rightarrow \text{No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 129 of 262

Saddle Calcs: Bundle Pull New:

Step: 24 10:15am

Sep 9, 2026

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[12.2, 30.7, 0]

= 30.7104 cm²[Failed]

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 197-C_Channel MOC= CS Page 130 of 262
 Saddle Calcs: Bundle Pull Cor: Step: 25 10:16am Sep 9,2026

ASME Horizontal Vessel Analysis: Bundle pullout, Left Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1357.65	mm
Shell Thickness used in this Case	t	14.500	mm
Stiffened Vessel Length per 4.15.6	L	7227.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1260.00	mm
Saddle Width	b1 or b	585.00	mm
Saddle Bearing Angle	delta or theta	140.00	degrees
Wear Plate Width	b2 or b1	875.00	mm
Wear Plate Bearing Angle	delta2 or theta1	152.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		136.40	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1981.20	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		48400.00	Kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	

Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		21936.54	Kgf
Pressure used in Analysis	P	0.000	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-1.83	-234.00
Long. Stress at Top of Midspan		1.83	138.00
Long. Stress at Bottom of Midspan		1.83	138.00
Long. Stress at Top of Saddles		1.77	138.00
Long. Stress at Bottom of Saddles		-1.01	-234.00
Long. Stress at Bottom of Saddles		1.01	138.00
Tangential Shear in Shell		6.78	110.40
Circ. Stress at Horn of Saddle		16.43	207.00
Circ. Compressive Stress in Shell		0.56	138.00

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
 = Bundle Pull Out Force • Impact Factor • B / Ls
 = 48400 • 1.5 • 1981/3780
 = 38051.6 Kgf

Load Combination Results for Tube Bundle Pull Out [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = -16115 + max(38052, 0, 0, 0)



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 131 of 262

Saddle Calcs: Bundle Pull Cor: Step: 25 10:16am Sep 9, 2026

= 21936.5 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	24039.76	Kgf
Transverse Shear Load	0.00	Kgf
Longitudinal Shear Load	72600.00	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= Baseplate Length

= 2349.500 mm

The Computed K values from Table 4.15.1:

K1 = 0.1412	K2 = 0.9003	K3 = 0.5922	K4 = 0.3269
K5 = 0.6971	K6 = 0.0379	K7 = 0.0338	K8 = 0.3167
K9 = 0.2372	K10 = 0.0432	K1* = 0.2486	K6p = 0.0305
K7p = 0.0272			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(R_m^2 - h m^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4hm)}{3L} \right) \right]$$

$$= -21937 \cdot 1260 \left[1 - \left(1 - \frac{1260}{7227} + \frac{(1358^2 - 0^2)}{(2 \cdot 1260 \cdot 7227)} \right) / \left(1 + \frac{(4 \cdot 0)}{(3 \cdot 7227)} \right) \right]$$

$$= -2021.6 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{(1 + 2(R_m^2 - h m^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L}{(3 \cdot 7227)} - 4 \cdot 1260 / 7227 \right)$$

$$= 21937 \cdot 7227 / 4 \left(\frac{(1 + 2(1358^2 - 0^2)/(7227^2)) / (1 + (4 \cdot 0)/(3 \cdot 7227)) - 4 \cdot 1260 / 7227}{(3 \cdot 7227)} \right)$$

$$= 14791.5 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 t)$$

$$= 0 \cdot 1358 / (2 \cdot 13.7) - 14792 / (\pi \cdot 1358^2 \cdot 13.7)$$

$$= -1.83 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 1358 / (2 \cdot 13.7) + 14792 / (\pi \cdot 1358^2 \cdot 13.7)$$

$$= 1.83 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 0 \cdot 1358 / (2 \cdot 13.7) - 2022 / (0.14 \cdot \pi \cdot 1358^2 \cdot 13.7)$$

$$= 1.77 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 1358 / (2 \cdot 13.7) + 2022 / (0.25 \cdot \pi \cdot 1358^2 \cdot 13.7)$$

$$= -1.01 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm/3))$$

$$= 21937 \cdot (7227 - 2 \cdot 1260) / (7227 + (4 \cdot 0/3))$$

$$= 14287.4 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 0.9 \cdot 14287 / (1358 \cdot 13.7)$$

$$= 6.78 \text{ N./mm}^2$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 132 of 262

Saddle Calcs: Bundle Pull Cor: Step: 25 10:16am Sep 9,2026

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1358 \cdot 13.7)}$$

$$= 106.377 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(585 + 1.56 \cdot \sqrt{(1358 \cdot 13.7)}, 2 \cdot 1260)$$

$$= 797.75 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

= 1.0000 Materials are the same, bundle pull case

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ6,r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7 \cdot 21937 \cdot 0.1 / (798(13.7 + 1 \cdot 20))$$

$$= -0.56 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.29) [σ7,r*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -21937 / (4(13.7 + 1 \cdot 20)798) -$$

$$12 \cdot 0.034 \cdot 21937 \cdot 1358 / (7227(13.7 + 1 \cdot 20)^2)$$

$$= -16.43 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2349.5000	mm
Baseplate Thickness	Bpthk	50.0000	mm
Baseplate Width	Bpwid	610.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	650.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1351 \cdot 13.7)} + 875)13.7^3) / 12$$

$$= 23.297 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 23.3 + 149(303 - 6.85)^2$$

$$= 130283.680 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	1087.2	13.7	6.9	148.9	102029.9	0.130E+06
Wearplate	875.0	20.0	23.7	175.0	414749.9	0.136E+06
Web	20.0	546.7	307.1	109.3	3357284.2	0.273E+05
BasePlate	610.0	50.0	605.4	305.0	18464700.0	0.280E+06
Totals	...	...	...	738.3	22338764.0	0.574E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 22338764 / 738$$

$$= 302.575 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 133 of 262

Saddle Calcs: Bundle Pull Cor: Step: 25 10:16am Sep 9,2026

$$= 180 - 140/2$$

$$= 110.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(110) - 0.5 \cdot \sin(110)^2) / (\pi - 1.92 + \sin(110) \cos(110))$$

$$= 0.2404$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.24 \cdot 21937$$

$$= 5274.2217 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.8776 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1981 - 1351 \cdot \sin(140/2) / 1.22$$

$$= 942.234 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 4969.6475 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.5689 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(21937 + 2103) \cdot 610 / (4 \cdot 2350 \cdot 173)}$$

$$= 16.274 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 140/2) (1350 + 14.5 + 20)$$

$$= 2602.009 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2602 / (5 - 1)$$

$$= 650.502 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 651 \cdot 610 / 2$$

$$= 1984.032 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 24040 / (2350 \cdot 610)$$

$$= 1.677 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1984 \cdot 1.68$$

$$= 3327.925 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 113 + 65.1$$

$$= 178.050 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 134 of 262

Saddle Calcs: Bundle Pull Cor: Step: 25 10:16am Sep 9,2026

= 3328/178
= 1.833 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	585.0	...	117.0	...	0.334E+05

Rib dimension [D]:
= Saddle Width - Web Thickness
= 585 - 20
= 565.000 mm

Distance to Centroid from Datum [ytot]:
= AY / A
= 0/178
= 0.000 mm

Distance to Centroid [C1]:
= Saddle Width / 2
= 585/2
= 292.500 mm

Radius of Gyration [r]:
= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
= $\sqrt{33367/178}$
= 136.895 mm

Length of Outer Rib, step 1, [Lv]:
= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk
= 1981 - cos(140/2)(1350 + 14.5 + 20) - 50
= 1457.673 mm

Length of Outer Rib, step 2, [Lw]:
= 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)
= 2 • sin(140/2)(1350 + 14.5 + 20)
= 2602.009 mm

Length of Outer Rib, step 3, [Bd]:
= Baseplate length - clearance
= 2350 - 50.8
= 2298.700 mm

Length of Outer Rib, step 4, [Dd]:
= (Lw - Bd)/2
= (2602 - 2299)/2
= 151.654 mm

Length of Outer Rib [L]:
= $\sqrt{Lv^2 + Dd^2}$
= $\sqrt{1458^2 + 152^2}$
= 1465.541 mm

Intermediate Term [Cc]:
= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
= $\sqrt{2 \cdot \pi^2 \cdot 199943008/260}$
= 123.211

Slenderness ratio [KL/r]:
= KL/r
= 1 • 1466/137
= 10.706



PV Elite 28 Licensee: SL User
 FileName : 197-C_Channel MOC= CS Page 135 of 262
 Saddle Calcs: Bundle Pull Cor: Step: 25 10:16am Sep 9,2026

Bending Moment [Rm]:
 $= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$
 $= 72600 / (2 \cdot 2350) \cdot 651 \cdot 1466 / 2$
 $= 7364.725 \text{ Kg-m.}$

Compressive Allowable, $KL/r < C_c$ ($10.7 < 123$) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (10.7)^2 / (2 \cdot 123^2)) 260 /$
 $(5/3 + 3 \cdot (10.7) / (8 \cdot 123) - (10.7^3) / (8 \cdot 123^3))$
 $= 152.4 \text{ N./mm}^2$

AISC Unity Check of Outside Ribs (must be ≤ 1)
 $= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$
 $= 1.83 / 152 + (7365 \cdot 292 / 333669344) / 173$
 $= 0.377$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	565.0	0.0	113.0	0.0	0.334E+05
Web	650.5	20.0	0.0	130.1	0.0	43.4
Totals	...	...	...	243.1	...	0.334E+05

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 243$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$
 $= 585 / 2$
 $= 292.500 \text{ mm}$

Length of Inner Rib [L]:
 $= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$
 $= 1981 - \cos(1385 / 2) (50 + 0 + 0) - 0$
 $= 546.700 \text{ mm}$

Radius of Gyration [r]:
 $= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$
 $= \sqrt{ (33409 / 243) }$
 $= 117.230 \text{ mm}$

Slenderness ratio [KL/r]:
 $= KL / r$
 $= 1 \cdot 547 / 117$
 $= 4.663$

Unit Force [Force,u]:
 $= FI / (2 \cdot \text{Baseplate Length})$
 $= 72600 / (2 \cdot 2350)$
 $= 15.450 \text{ Kg/mm}$

Moment at base of inner Rib [Mbase,c]:
 $= \text{Unit Force} \cdot e \cdot L$
 $= 15.5 \cdot 651 \cdot 547$
 $= 5494.619 \text{ Kg-m.}$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 $= \text{Bending Moment} / \text{Section Modulus}$
 $= 5495 / 1142186$
 $= 47.176 \text{ N./mm}^2$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 136 of 262

Saddle Calcs: Bundle Pull Cor: Step: 25 10:16am Sep 9,2026

Compressive Allowable, $KL/r < C_c$ ($4.66 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (4.66)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (4.66) / (8 \cdot 123) - (4.66^3) / (8 \cdot 123^3))$$

$$= 154.6 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 2.69 / 155 + (5495 \cdot 292 / 33409) / 173$$

$$= 0.290$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	16.1400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	14.0743 cm ²
Saddle Load QO (Weight)	QO	24039.8 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	72600.0 Kg
Maximum Transverse Force	Ft	0.0 Kg
Maximum Longitudinal Force	F1	72600.0 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= F1 / (Bolt Area \cdot Number of Bolts)$$

$$= 72600 / (14.1 \cdot 8)$$

$$= 63.2 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (Bolt Area \cdot Number of Bolts)$$

$$= 0 / (14.1 \cdot 8)$$

$$= 0.0 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= (QL - QO) / (Stba \cdot Rnbolts)$$

$$= (72600 - 24040) / (48.3 \cdot 8)$$

$$= 12.3248 \text{ cm}^2$$

Bolt Area due to Shear Load [Bltarears]:

$$= F1 / (BoltShearAllowable \cdot Nbolts)$$

$$= 72600 / (29 \cdot 8)$$

$$= 30.7104 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1981 \cdot 0 + 0$$

$$= 0.00 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 0 / 24040$$

$$= 0.00 \text{ mm} < B_{plen} / 6 \rightarrow \text{No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 137 of 262

Saddle Calcs: Bundle Pull Cor: Step: 25 10:16am Sep 9, 2026

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[12.3, 30.7, 0]

= 30.7104 cm²[Failed]

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 138 of 262

Saddle Calcs: Road Transport:

Step: 26 10:16am

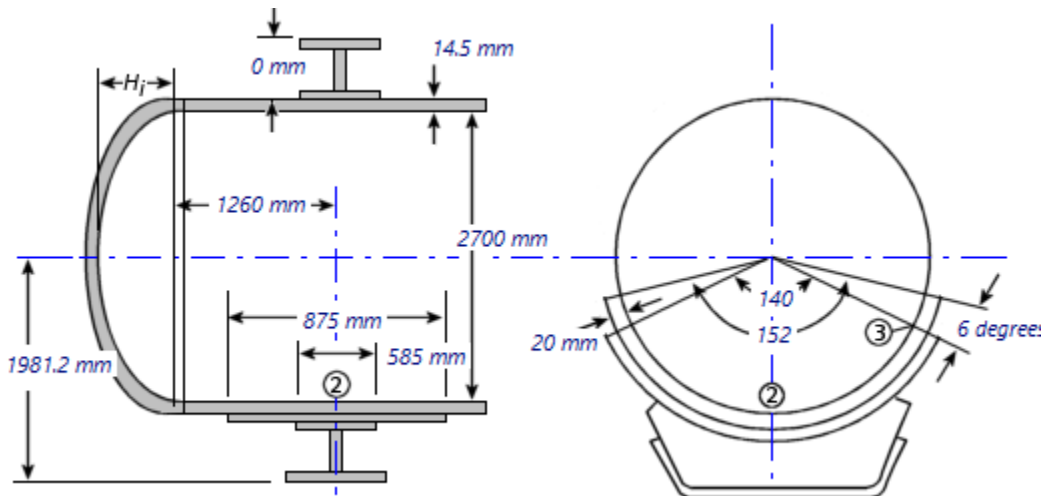
Sep 9, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Input and Calculated Values:

Vessel Mean Radius	Rm	1357.25	mm
Shell Thickness used in this Case	t	14.500	mm
Stiffened Vessel Length per 4.15.6	L	7227.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1260.00	mm
Saddle Width	b1 or b	585.00	mm
Saddle Bearing Angle	delta or theta	140.00	degrees
Wear Plate Width	b2 or b1	875.00	mm
Wear Plate Bearing Angle	delta2 or theta1	152.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		136.40	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1981.20	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		77757.15	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr



PV Elite 28 Licensee: SL User

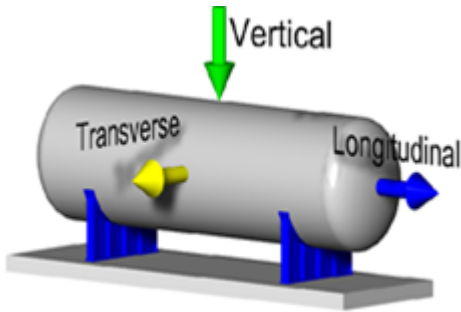
FileName : 197-C_Channel MOC= CS

Page 139 of 262

Saddle Calcs: Road Transport:

Step: 26 10:16am

Sep 9, 2026



Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-6.13	-234.00
Long. Stress at Top of Midspan	6.13	138.00
Long. Stress at Bottom of Midspan	6.13	138.00
Long. Stress at Top of Saddles	5.94	138.00
Long. Stress at Bottom of Saddles	-3.37	-234.00
Long. Stress at Bottom of Saddles	3.37	138.00
Tangential Shear in Shell	22.72	110.40
Circ. Stress at Horn of Saddle	55.69	207.00
Circ. Compressive Stress in Shell	1.92	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 386 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \\
 &= 3281.9 \text{ lbf} \\
 &= 1488.685 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [F_{lw}]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 8.95^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \cdot 1.2 \\
 &= 641.8 \text{ lbf} \\
 &= 291.1 \text{ Kgf}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [F_t]:

$$\begin{aligned}
 &= g_z \cdot \text{Shipping Weight} \cdot g_w + F_{tw} \\
 &= 0.1 \cdot 91066 \cdot 1 + 1489 \\
 &= 10595.3 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [F_l]:

$$\begin{aligned}
 &= g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw} \\
 &= 0.1 \cdot 91066 \cdot 1 + 291 \\
 &= 9397.7 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [F_{wt}]:

$$\begin{aligned}
 &= F_t / \text{Num of Saddles} \cdot B / E \\
 &= 10595/2 \cdot 1981/2350 \\
 &= 4467.2 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:

$$\begin{aligned}
 &= F_l \cdot B / L_s \\
 &= 9398 \cdot 1981/3780 \\
 &= 4925.6 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot g_y \\
 &= 69363 \cdot 0.05
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 140 of 262

Saddle Calcs: Road Transport: Step: 26 10:16am Sep 9, 2026

= 3468.168 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)

= 69363 + 3468 + max(4926, 4467)

= 77757.1 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 79860.37 Kgf

Transverse Shear Load 5297.66 Kgf

Longitudinal Shear Load 9397.75 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= Baseplate Length

= 2349.500 mm

The Computed K values from Table 4.15.1:

K1 = 0.1412 K2 = 0.9003 K3 = 0.5922 K4 = 0.3269

K5 = 0.6971 K6 = 0.0379 K7 = 0.0338 K8 = 0.3167

K9 = 0.2372 K10 = 0.0432 K1* = 0.2486 K6p = 0.0305

K7p = 0.0272

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q*a [1 - (1 - a/L + (Rm²-hm²)/(2a*L))/(1+(4hm)/3L)]

= -77757*1260[1-(1-1260/7227+(1357²-0²)/

(2*1260*7227))/(1+(4*0)/(3*7227))]

= -7171.6 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q*L/4*((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)

= 77757*7227/4*((1+2(1357²-0²)/(7227²))/(1+(4*0)/

(3*7227))-4*1260/7227)

= 52424.8 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P * Rm/(2t) - M2/(π*Rm²t)

= 0 * 1357/(2*14.5) - 52425/(π*1357²*14.5)

= -6.13 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P * Rm/(2t) + M2/(π * Rm² * t)

= 0 * 1357/(2 * 14.5) + 52425/(π * 1357² * 14.5)

= 6.13 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P * Rm/(2t) - M1/(K1*π*Rm²t)

= 0*1357/(2*14.5)-7172/(0.14*π*1357²*14.5)

= 5.94 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P * Rm/(2t) + M1/(K1*π * Rm² * t)

= 0*1357/(2*14.5)+7172/(0.25*π*1357²*14.5)

= -3.37 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4*hm/3))

= 77757 (7227 - 2 * 1260)/(7227 + (4 * 0/3))

= 50643.8 Kgf



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 141 of 262

Saddle Calcs: Road Transport:

Step: 26 10:16am

Sep 9, 2026

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.9 \cdot 50644 / (1357 \cdot 14.5)$$

$$= 22.72 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1357 \cdot 14.5)}$$

$$= 109.423 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(585 + 1.56 \cdot \sqrt{(1357 \cdot 14.5)}, 2 \cdot 1260)$$

$$= 803.85 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7 \cdot 77757 \cdot 0.1 / (804(14.5 + 1 \cdot 20))$$

$$= -1.92 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -77757 / (4(14.5 + 1 \cdot 20)804) -$$

$$12 \cdot 0.034 \cdot 77757 \cdot 1357 / (7227(14.5 + 1 \cdot 20)^2)$$

$$= -55.69 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2349.5000	mm
Baseplate Thickness	Bpthk	50.0000	mm
Baseplate Width	Bpwid	610.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	650.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1350 \cdot 14.5)} + 875)14.5^3) / 12$$

$$= 27.774 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 27.8 + 159(300 - 7.25)^2$$

$$= 135426.000 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	1093.3	14.5	7.2	158.5	114929.0	0.135E+06
Wearplate	875.0	20.0	24.5	175.0	428749.9	0.132E+06
Web	20.0	546.7	307.8	109.3	3366030.5	0.273E+05
BasePlate	610.0	50.0	606.2	305.0	18489098.0	0.288E+06
Totals	...	...	...	747.9	22398806.0	0.583E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 22398806/748$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 142 of 262

Saddle Calcs: Road Transport: Step: 26 10:16am Sep 9, 2026

= 299.504 mm

Angle [β]:

= 180 - Saddle Angle/2

= 180 - 140/2

= 110.0

Saddle Splitting Coefficient [K1]:

= (1 + cos(β) - 0.5•sin(β)²)/(π - β + sin(β)cos(β))

= (1 + cos(110) - 0.5•sin(110)²)/(π - 1.92 + sin(110)cos(110))

= 0.2404

Saddle Splitting Force [Fh]:

= K1 • Q

= 0.24 • 77757

= 18695.2168 Kgf

Tension Stress, St = (Fh/As) = 3.1110 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

= B - R • sin(theta/2)/(theta/2 in radians)

= 1981 - 1350 • sin(140/2)/1.22

= 942.849 mm

Bending Moment, M = Fh * d = 17627.1152 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 8.8855 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 173.3302 N./mm²

Minimum Thickness of Baseplate per Moss:

= √(3(Q + SaddleWt)BasePlateWidth / (4 • BasePlateLength • AllStress))

= √(3(77757 + 2103)610/(4 • 2350 • 173))

= 29.662 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

= 2 • cos(90 - Saddle Angle/2)(Inside Radius + Shell Thk + Wear Plate Thk)

= 2 • cos(90 - 140/2)(1350 + 14.5 + 20)

= 2602.009 mm

Distance between Ribs [e]:

= Web Length / (Nribs - 1)

= 2602/(5 - 1)

= 650.502 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2

= 651 • 610/2

= 1984.032 cm²

Bearing Pressure [Bp]:

= (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)

= 79860/(2350 • 610)

= 5.572 Kgf/cm²

Axial Load [P]:

= Ap • Bp

= 1984 • 5.57

= 11055.404 Kgf

Area of the Rib and Web [Ar]:

= Rib Area + Web Area

= 113 + 65.1



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 143 of 262

Saddle Calcs: Road Transport:

Step: 26 10:16am

Sep 9, 2026

$$= 178.050 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 11055/178$$

$$= 6.089 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	585.0	...	117.0	...	0.334E+05

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 585 - 20$$

$$= 565.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/178$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 585/2$$

$$= 292.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(33367/178)}$$

$$= 136.895 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1981 - \cos(140/2) (1350 + 14.5 + 20) - 50$$

$$= 1457.673 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(140/2) (1350 + 14.5 + 20)$$

$$= 2602.009 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2350 - 50.8$$

$$= 2298.700 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (2602 - 2299) / 2$$

$$= 151.654 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{(Lv^2 + Dd^2)}$$

$$= \sqrt{(1458^2 + 152^2)}$$

$$= 1465.541 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)}$$

$$= 123.211$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 144 of 262

Saddle Calcs: Road Transport:

Step: 26 10:16am

Sep 9, 2026

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1466/137$$

$$= 10.706$$

Bending Moment [Rm]:

$$= Fl / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 9398 / (2 \cdot 2350) \cdot 651 \cdot 1466 / 2$$

$$= 953.331 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($10.7 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (10.7)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (10.7) / (8 \cdot 123) - (10.7)^3 / (8 \cdot 123^3))$$

$$= 152.4 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 6.09/152 + (953 \cdot 292/333669344) / 173$$

$$= 0.087$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	565.0	0.0	113.0	0.0	0.334E+05
Web	650.5	20.0	0.0	130.1	0.0	43.4
Totals	...	...	...	243.1	...	0.334E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/243$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 585/2$$

$$= 292.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1981 - \cos(1385/2) (50 + 0 + 0) - 0$$

$$= 546.700 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (33409/243) }$$

$$= 117.230 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 547/117$$

$$= 4.663$$

Unit Force [Force,u]:

$$= Fl / (2 \cdot \text{Baseplate Length})$$

$$= 9398 / (2 \cdot 2350)$$

$$= 2.000 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 2 \cdot 651 \cdot 547$$

$$= 711.254 \text{ Kg-m.}$$



PV Elite 28 Licensee: SL User
 FileName : 197-C_Channel MOC= CS Page 145 of 262
 Saddle Calcs: Road Transport: Step: 26 10:16am Sep 9,2026

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 = Bending Moment / Section Modulus
 = 711/1142186
 = 6.107 N./mm²

Compressive Allowable, $KL/r < C_c$ (4.66 < 123) per AISC E2-1 [Sca]:
 = (1-(KL/r)²/(2• C_c^2)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c^3)
 = (1-(4.66)²/(2 • 123²))260/
 (5/3+3•(4.66)/(8• 123)-(4.66³)/(8•123³)
 = 154.6 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)
 = $S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$
 = 8.92/155 + (711 • 292/33409)/173
 = 0.093

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1357.25	mm
Shell Thickness used in this Case	t	14.500	mm
Stiffened Vessel Length per 4.15.6	L	7227.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1260.00	mm
Saddle Width	b1 or b	585.00	mm
Saddle Bearing Angle	delta or theta	140.00	degrees
Wear Plate Width	b2 or b1	875.00	mm
Wear Plate Bearing Angle	delta2 or theta1	152.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		136.40	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1981.20	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		28443.75	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top	of Midspan	-2.24	-234.00
Long. Stress at Top	of Midspan	2.24	138.00
Long. Stress at Bottom	of Midspan	2.24	138.00
Long. Stress at Top	of Saddles	2.17	138.00
Long. Stress at Bottom	of Saddles	-1.23	-234.00
Long. Stress at Bottom	of Saddles	1.23	138.00
Tangential Shear in Shell		8.31	110.40
Circ. Stress at Horn of Saddle		20.37	207.00
Circ. Compressive Stress in Shell		0.70	138.00



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 146 of 262

Saddle Calcs: Road Transport:

Step: 26 10:16am

Sep 9, 2026

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 386 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \\
 &= 3281.9 \text{ lbf} \\
 &= 1488.685 \text{ Kg}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 8.95^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1 \cdot 1.2 \\
 &= 641.8 \text{ lbf} \\
 &= 291.1 \text{ Kg}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [Ft]:

$$\begin{aligned}
 &= gz \cdot \text{Shipping Weight} \cdot gw + Ftw \\
 &= 0.1 \cdot 91066 \cdot 1 + 1489 \\
 &= 10595.3 \text{ Kg}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [Fl]:

$$\begin{aligned}
 &= gx \cdot \text{Shipping Weight} \cdot gw + Flw \\
 &= 0.1 \cdot 91066 \cdot 1 + 291 \\
 &= 9397.7 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [Fwt]:

$$\begin{aligned}
 &= Ft/\text{Num of Saddles} \cdot B / E \\
 &= 10595/2 \cdot 1981/2350 \\
 &= 4467.2 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [Fwl]:

$$\begin{aligned}
 &= Fl \cdot B / Ls \\
 &= 9398 \cdot 1981/3780 \\
 &= 4925.6 \text{ Kg}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot gy \\
 &= 22398 \cdot 0.05 \\
 &= 1119.911 \text{ Kg}
 \end{aligned}$$

Load Combination Results due to Transport Loading [Q]:

$$\begin{aligned}
 &= \text{Saddle Load} + Frv + \max(Fwl, Fwt) \\
 &= 22398 + 1120 + \max(4926, 4467) \\
 &= 28443.8 \text{ Kg}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	30546.97	Kgf
Transverse Shear Load	5297.66	Kgf
Longitudinal Shear Load	9397.75	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$\begin{aligned}
 &= \text{Baseplate Length} \\
 &= 2349.500 \text{ mm}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1412	K2 = 0.9003	K3 = 0.5922	K4 = 0.3269
K5 = 0.6971	K6 = 0.0379	K7 = 0.0338	K8 = 0.3167
K9 = 0.2372	K10 = 0.0432	K1* = 0.2486	K6p = 0.0305
K7p = 0.0272			

The suffix 'p' denotes the values for a wear plate if it exists.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 147 of 262

Saddle Calcs: Road Transport: Step: 26 10:16am Sep 9, 2026

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(Rm^2 - hm^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4hm)}{3L} \right) \right]$$

$$= -28444 \cdot 1260 \left[1 - \left(1 - \frac{1260}{7227} + \frac{(1357^2 - 0^2)}{(2 \cdot 1260 \cdot 7227)} \right) / \left(1 + \frac{(4 \cdot 0)}{(3 \cdot 7227)} \right) \right]$$

$$= -2623.4 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{(1 + 2(Rm^2 - hm^2)/(L^2))}{(1 + (4hm^2)/(3L))} - 4a/L \right)$$

$$= 28444 \cdot 7227 / 4 \left(\frac{(1 + 2(1357^2 - 0^2)/(7227^2))}{(1 + (4 \cdot 0)/(3 \cdot 7227))} - 4 \cdot 1260 / 7227 \right)$$

$$= 19177.1 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 0 \cdot 1357 / (2 \cdot 14.5) - 19177 / (\pi \cdot 1357^2 \cdot 14.5)$$

$$= -2.24 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 0 \cdot 1357 / (2 \cdot 14.5) + 19177 / (\pi \cdot 1357^2 \cdot 14.5)$$

$$= 2.24 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= 0 \cdot 1357 / (2 \cdot 14.5) - 2623 / (0.14 \cdot \pi \cdot 1357^2 \cdot 14.5)$$

$$= 2.17 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 0 \cdot 1357 / (2 \cdot 14.5) + 2623 / (0.25 \cdot \pi \cdot 1357^2 \cdot 14.5)$$

$$= -1.23 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 28444 (7227 - 2 \cdot 1260) / (7227 + (4 \cdot 0 / 3))$$

$$= 18525.6 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 0.9 \cdot 18526 / (1357 \cdot 14.5)$$

$$= 8.31 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1357 \cdot 14.5)}$$

$$= 109.423 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min \left(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a \right)$$

$$= \min (585 + 1.56 \cdot \sqrt{(1357 \cdot 14.5)}, 2 \cdot 1260)$$

$$= 803.85 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7 \cdot 28444 \cdot 0.1 / (804(14.5 + 1 \cdot 20))$$

$$= -0.70 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [σ_7, r_{*}]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$$

$$= -28444 / (4(14.5 + 1 \cdot 20)804) -$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 148 of 262

Saddle Calcs: Road Transport:

Step: 26 10:16am

Sep 9, 2026

$$12 \cdot 0.034 \cdot 28444 \cdot 1357 / (7227(14.5 + 1 \cdot 20)^2) \\ = -20.37 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2349.5000	mm
Baseplate Thickness	Bpthk	50.0000	mm
Baseplate Width	Bpwid	610.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	650.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12 \\ = ((1.56 \cdot \sqrt{(1350 \cdot 14.5)} + 875)14.5^3)/12 \\ = 27.774 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2 \\ = 27.8 + 159(300 - 7.25)^2 \\ = 135426.000 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	1093.3	14.5	7.2	158.5	114929.0	0.135E+06
Wearplate	875.0	20.0	24.5	175.0	428749.9	0.132E+06
Web	20.0	546.7	307.8	109.3	3366030.5	0.273E+05
BasePlate	610.0	50.0	606.2	305.0	18489098.0	0.288E+06
Totals	...	...	...	747.9	22398806.0	0.583E+06

Distance to Centroid [C1]:

$$= AY / A \\ = 22398806/748 \\ = 299.504 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2 \\ = 180 - 140/2 \\ = 110.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta)) \\ = (1 + \cos(110) - 0.5 \cdot \sin(110)^2) / (\pi - 1.92 + \sin(110)\cos(110)) \\ = 0.2404$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q \\ = 0.24 \cdot 28444 \\ = 6838.7554 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 1.1380 \text{ N./mm}^2 \\ \text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ = 1981 - 1350 \cdot \sin(140/2) / 1.22 \\ = 942.849 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 149 of 262

Saddle Calcs: Road Transport:

Step: 26 10:16am Sep 9, 2026

Bending Moment, $M = F_h \cdot d = 6448.0410 \text{ Kg-m.}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 3.2504 \text{ N./mm}^2$

Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$

$= \sqrt{(3(28444 + 2103) \cdot 610) / (4 \cdot 2350 \cdot 173)}$

$= 18.345 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$

$= 2 \cdot \cos(90 - 140/2) (1350 + 14.5 + 20)$

$= 2602.009 \text{ mm}$

Distance between Ribs [e]:

$= \text{Web Length} / (\text{Nr ribs} - 1)$

$= 2602 / (5 - 1)$

$= 650.502 \text{ mm}$

Baseplate Pressure Area [Ap]:

$= e \cdot \text{Bpwid} / 2$

$= 651 \cdot 610 / 2$

$= 1984.032 \text{ cm}^2$

Bearing Pressure [Bp]:

$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$

$= 30547 / (2350 \cdot 610)$

$= 2.131 \text{ Kg/cm}^2$

Axial Load [P]:

$= A_p \cdot B_p$

$= 1984 \cdot 2.13$

$= 4228.746 \text{ Kg}$

Area of the Rib and Web [Ar]:

$= \text{Rib Area} + \text{Web Area}$

$= 113 + 65.1$

$= 178.050 \text{ cm}^2$

Compressive Stress [Sc]:

$= P / A_r$

$= 4229 / 178$

$= 2.329 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	585.0	...	117.0	...	0.334E+05

Rib dimension [D]:

$= \text{Saddle Width} - \text{Web Thickness}$

$= 585 - 20$

$= 565.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:

$= AY / A$

$= 0 / 178$

$= 0.000 \text{ mm}$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 150 of 262

Saddle Calcs: Road Transport:

Step: 26 10:16am

Sep 9, 2026

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 585 / 2$$

$$= 292.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(33367 / 178)}$$

$$= 136.895 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1981 - \cos(140 / 2) (1350 + 14.5 + 20) - 50$$

$$= 1457.673 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(140 / 2) (1350 + 14.5 + 20)$$

$$= 2602.009 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2350 - 50.8$$

$$= 2298.700 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (2602 - 2299) / 2$$

$$= 151.654 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{L_v^2 + D_d^2}$$

$$= \sqrt{1458^2 + 152^2}$$

$$= 1465.541 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1466 / 137$$

$$= 10.706$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{\text{plen}}) \cdot e \cdot L / 2$$

$$= 9398 / (2 \cdot 2350) \cdot 651 \cdot 1466 / 2$$

$$= 953.331 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($10.7 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (10.7)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (10.7) / (8 \cdot 123) - (10.7^3) / (8 \cdot 123^3))$$

$$= 152.4 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc / Sca + (Rm \cdot C1 / I) / Sba$$

$$= 2.33 / 152 + (953 \cdot 292 / 333669344) / 173$$

$$= 0.063$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

B	D	Y	A	AY	Io
---	---	---	---	----	----



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 151 of 262

Saddle Calcs: Road Transport:

Step: 26 10:16am

Sep 9, 2026

Rib	20.0	565.0	0.0	113.0	0.0	0.334E+05
Web	650.5	20.0	0.0	130.1	0.0	43.4
Totals	...	...	...	243.1	...	0.334E+05

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/243

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 585/2

= 292.500 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 1981 - cos(1385/2)(50 + 0 + 0) - 0

= 546.700 mm

Radius of Gyration [r]:

= $\sqrt{(\text{Total Inertia} / \text{Total Area})}$

= $\sqrt{(33409/243)}$

= 117.230 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 547/117

= 4.663

Unit Force [Force,u]:

= $FI / (2 \cdot \text{Baseplate Length})$

= 9398/(2 • 2350)

= 2.000 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 2 • 651 • 547

= 711.254 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 711/1142186

= 6.107 N./mm²

Compressive Allowable, $KL/r < C_c$ (4.66 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$

= $(1 - (4.66)^2 / (2 \cdot 123^2)) 260 /$

$(5/3 + 3 \cdot (4.66) / (8 \cdot 123) - (4.66^3) / (8 \cdot 123^3))$

= 154.6 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$

= 3.41/155 + (711 • 292/33409)/173

= 0.057

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 152 of 262

Saddle Calcs: Operating Case:

Step: 27 10:16am

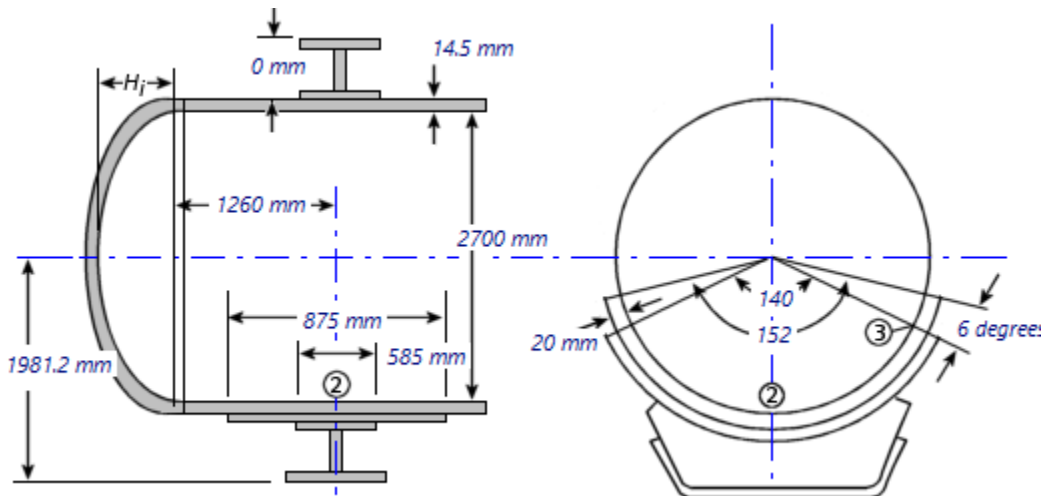
Sep 9, 2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1357.65	mm
Shell Thickness used in this Case	t	14.500	mm
Stiffened Vessel Length per 4.15.6	L	7227.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1260.00	mm
Saddle Width	b1 or b	585.00	mm
Saddle Bearing Angle	delta or theta	140.00	degrees
Wear Plate Width	b2 or b1	875.00	mm
Wear Plate Bearing Angle	delta2 or theta1	152.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		136.40	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1981.20	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	115644.57	Kgf
Pressure used in Analysis	P 1013.233	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	40.57	138.00
Long. Stress at Bottom of Midspan	59.85	138.00
Long. Stress at Top of Saddles	59.54	138.00
Long. Stress at Bottom of Saddles	44.91	138.00
Tangential Shear in Shell	35.75	110.40
Circ. Stress at Horn of Saddle	86.62	172.50
Circ. Compressive Stress in Shell	2.94	138.00



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 153 of 262

Saddle Calcs: Operating Case: Step: 27 10:16am Sep 9, 2026

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (1437/2 + 0) \cdot 1981/2350$$

$$= 1818.0 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(724, 0) \cdot 1981/3780$$

$$= 379.4 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(8818, 0) \cdot 1981/3780$$

$$= 4621.7 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (8818/2 + 0) \cdot 1981/2350$$

$$= 11153.6 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 104491 + \max(379, 1818, 4622, 11154)$$

$$= 115644.6 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (2.73 \cdot 1.2)^2)$$

$$= 7097.894 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	117747.80	Kgf
Transverse Shear Load	4409.00	Kgf
Longitudinal Shear Load	8817.99	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 2349.500 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1412	K2 = 0.9003	K3 = 0.5922	K4 = 0.3269
K5 = 0.6971	K6 = 0.0379	K7 = 0.0338	K8 = 0.3167
K9 = 0.2372	K10 = 0.0432	K1* = 0.2486	K6p = 0.0305
K7p = 0.0272			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -115645 \cdot 1260 [1 - (1 - 1260/7227 + (1358^2 - 0^2)/(2 \cdot 1260 \cdot 7227))/(1 + (4 \cdot 0)/(3 \cdot 7227))]$$

$$= -10657.3 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 [(1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L]$$

$$= 115645 \cdot 7227/4 [(1 + 2(1358^2 - 0^2)/(7227^2))/(1 + (4 \cdot 0)/(3 \cdot 7227)) - 4 \cdot 1260/7227]$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 154 of 262

Saddle Calcs: Operating Case: Step: 27 10:16am Sep 9, 2026

= 77977.6 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 1013 \cdot 1358 / (2 \cdot 13.7) - 77978 / (\pi \cdot 1358^2 \cdot 13.7)$$

$$= 40.57 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 1013 \cdot 1358 / (2 \cdot 13.7) + 77978 / (\pi \cdot 1358^2 \cdot 13.7)$$

$$= 59.85 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 1013 \cdot 1358 / (2 \cdot 13.7) - 10657 / (0.14 \cdot \pi \cdot 1358^2 \cdot 13.7)$$

$$= 59.54 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1013 \cdot 1358 / (2 \cdot 13.7) + 10657 / (0.25 \cdot \pi \cdot 1358^2 \cdot 13.7)$$

$$= 44.91 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 115645 (7227 - 2 \cdot 1260) / (7227 + (4 \cdot 0 / 3))$$

$$= 75320.2 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 0.9 \cdot 75320 / (1358 \cdot 13.7)$$

$$= 35.75 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1358 \cdot 13.7)}$$

$$= 106.377 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(585 + 1.56 \cdot \sqrt{(1358 \cdot 13.7)}, 2 \cdot 1260)$$

$$= 797.75 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(138 / 138, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1 (t + \eta \cdot tr))$$

$$= -0.7 \cdot 115645 \cdot 0.1 / (798 (13.7 + 1 \cdot 20))$$

$$= -2.94 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r_{*}]:

$$= -Q / (4(t + \eta \cdot tr) b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -115645 / (4(13.7 + 1 \cdot 20) 798) -$$

$$12 \cdot 0.034 \cdot 115645 \cdot 1358 / (7227(13.7 + 1 \cdot 20)^2)$$

$$= -86.62 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 3780.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.13009 \text{E-}04 \cdot 3780 (240 - 21.1)$$

$$= 10.764 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 155 of 262

Saddle Calcs: Operating Case:

Step: 27 10:16am

Sep 9, 2026

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2349.5000	mm
Baseplate Thickness	Bpthk	50.0000	mm
Baseplate Width	Bpwid	610.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	218.0	N./mm ²
Height of Web at Center	Hw, c	650.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1351 \cdot 13.7) + 875})^3) / 12$$

$$= 23.297 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{shell}} (C1 - Y)^2$$

$$= 23.3 + 149 (303 - 6.85)^2$$

$$= 130283.680 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	1087.2	13.7	6.9	148.9	102029.9	0.130E+06
Wearplate	875.0	20.0	23.7	175.0	414749.9	0.136E+06
Web	20.0	546.7	307.1	109.3	3357284.2	0.273E+05
BasePlate	610.0	50.0	605.4	305.0	18464700.0	0.280E+06
Totals	...	...	...	738.3	22338764.0	0.574E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 22338764 / 738$$

$$= 302.575 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 140 / 2$$

$$= 110.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(110) - 0.5 \cdot \sin(110)^2) / (\pi - 1.92 + \sin(110) \cos(110))$$

$$= 0.2404$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.24 \cdot 115645$$

$$= 27804.5234 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 4.6268 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 130.8000 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1981 - 1351 \cdot \sin(140 / 2) / 1.22$$

$$= 942.234 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 26198.8770 \text{ Kg-m.}$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 156 of 262

Saddle Calcs: Operating Case: Step: 27 10:16am Sep 9,2026

Bending Stress, $S_b = (M * C_1 / I) = 13.5427 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 * \text{Yield Str} = 145.3333 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{ (3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress}))}$
 $= \sqrt{ (3(115645 + 2103) 610 / (4 * 2350 * 145))}$
 $= 39.334 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 140/2) (1350 + 14.5 + 20)$
 $= 2602.009 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 2602 / (5 - 1)$
 $= 650.502 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e * \text{Bpwid} / 2$
 $= 651 * 610 / 2$
 $= 1984.032 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 117748 / (2350 * 610)$
 $= 8.216 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p * B_p$
 $= 1984 * 8.22$
 $= 16300.321 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 113 + 65.1$
 $= 178.050 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 16300 / 178$
 $= 8.978 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	585.0	...	117.0	...	0.334E+05

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 585 - 20$
 $= 565.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= A_Y / A$
 $= 0 / 178$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 157 of 262

Saddle Calcs: Operating Case: Step: 27 10:16am Sep 9, 2026

$$= 585/2$$

$$= 292.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{33367/178}$$

$$= 136.895 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1981 - \cos(140/2) (1350 + 14.5 + 20) - 50$$

$$= 1457.673 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(140/2) (1350 + 14.5 + 20)$$

$$= 2602.009 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2350 - 50.8$$

$$= 2298.700 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (\text{Lw} - \text{Bd}) / 2$$

$$= (2602 - 2299) / 2$$

$$= 151.654 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{\text{Lv}^2 + \text{Dd}^2}$$

$$= \sqrt{1458^2 + 152^2}$$

$$= 1465.541 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot \text{Elastic Modulus}}{\text{Yield Stress}}}$$

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot 199943008}{218}}$$

$$= 134.556$$

Slenderness ratio [KL/r]:

$$= \text{KL} / r$$

$$= 1 \cdot 1466 / 137$$

$$= 10.706$$

Bending Moment [Rm]:

$$= \text{Fl} / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 8818 / (2 \cdot 2350) \cdot 651 \cdot 1466 / 2$$

$$= 894.519 \text{ Kg-m.}$$

Compressive Allowable, $\text{KL}/r < \text{Cc}$ ($10.7 < 135$) per AISC E2-1 [Sca]:

$$= (1 - (\text{KL}/r)^2 / (2 \cdot \text{Cc}^2)) \text{Fy} / (5/3 + 3 \cdot (\text{KL}/r) / (8 \cdot \text{Cc}) - (\text{KL}/r)^3 / (8 \cdot \text{Cc}^3))$$

$$= (1 - (10.7)^2 / (2 \cdot 135^2)) 218 / (5/3 + 3 \cdot (10.7) / (8 \cdot 135) - (10.7^3) / (8 \cdot 135^3))$$

$$= 128.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= \text{Sc}/\text{Sca} + (\text{Rm} \cdot \text{C1} / \text{I}) / \text{Sba}$$

$$= 8.98 / 128 + (895 \cdot 292 / 333669344) / 145$$

$$= 0.123$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	565.0	0.0	113.0	0.0	0.334E+05



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 158 of 262

Saddle Calcs: Operating Case: Step: 27 10:16am Sep 9,2026

Web	650.5	20.0	0.0	130.1	0.0	43.4
Totals	...	...	...	243.1	...	0.334E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/243$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 585/2$$

$$= 292.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1981 - \cos(1385/2)(50 + 0 + 0) - 0$$

$$= 546.700 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{33409/243}$$

$$= 117.230 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 547/117$$

$$= 4.663$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 8818 / (2 \cdot 2350)$$

$$= 1.877 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.88 \cdot 651 \cdot 547$$

$$= 667.376 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 667/1142186$$

$$= 5.730 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($4.66 < 135$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (4.66)^2 / (2 \cdot 135^2)) 218 / (5/3 + 3 \cdot (4.66) / (8 \cdot 135) - (4.66^3) / (8 \cdot 135^3))$$

$$= 129.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 13.2/130 + (667 \cdot 292/33409) / 145$$

$$= 0.141$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	16.1400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 159 of 262

Saddle Calcs: Operating Case: Step: 27 10:16am Sep 9, 2026

Area Available in a Single Bolt	BltArea	14.0743 cm ²
Saddle Load QO (Weight)	QO	106594.2 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	4621.7 Kgf
Maximum Transverse Force	Ft	4409.0 Kgf
Maximum Longitudinal Force	Fl	8818.0 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 8818 / (14.1 \cdot 8)$$

$$= 7.7 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 4409 / (14.1 \cdot 8)$$

$$= 3.8 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 8818 / (82.8 \cdot 8)$$

$$= 1.3055 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1981 \cdot 4409 + 0$$

$$= 8735.28 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 8735 / 106594$$

$$= 81.95 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0, 1.31, 0]$$

$$= 1.3055 \text{ cm}^2$$

MDMT Calculations for: Left Saddle flo, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 14.5, tr = 13.4, c = 0.8 mm, E* = 0.85

Thickness Ratio = tr * E*/(tg - c) = 0.83, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-17 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1357.65	mm
Shell Thickness used in this Case	t	14.500	mm
Stiffened Vessel Length per 4.15.6	L	7227.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1260.00	mm



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 160 of 262

Saddle Calcs: Operating Case: Step: 27 10:16am Sep 9,2026

Saddle Width	b1 or b	585.00	mm
Saddle Bearing Angle	delta or theta	140.00	degrees
Wear Plate Width	b2 or b1	875.00	mm
Wear Plate Bearing Angle	delta2 or theta1	152.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		136.40	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1981.20	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		61331.39	Kgf
Pressure used in Analysis	P	1013.233	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	45.10	138.00
Long. Stress at Bottom of Midspan	55.32	138.00
Long. Stress at Top of Saddles	55.16	138.00
Long. Stress at Bottom of Saddles	47.40	138.00
Tangential Shear in Shell	18.96	110.40
Circ. Stress at Horn of Saddle	45.94	172.50
Circ. Compressive Stress in Shell	1.56	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (1437/2 + 0) \cdot 1981/2350$$

$$= 1818.0 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(724, 0) \cdot 1981/3780$$

$$= 379.4 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(8818, 0) \cdot 1981/3780$$

$$= 4621.7 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (8818/2 + 0) \cdot 1981/2350$$

$$= 11153.6 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 50178 + \max(379, 1818, 4622, 11154)$$

$$= 61331.4 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (2.73 \cdot 1.2)^2)$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 161 of 262

Saddle Calcs: Operating Case: Step: 27 10:16am Sep 9, 2026

= 7097.894 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	63434.61	Kgf
Transverse Shear Load	4409.00	Kgf
Longitudinal Shear Load	8817.99	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= Baseplate Length

= 2349.500 mm

The Computed K values from Table 4.15.1:

K1 = 0.1412	K2 = 0.9003	K3 = 0.5922	K4 = 0.3269
K5 = 0.6971	K6 = 0.0379	K7 = 0.0338	K8 = 0.3167
K9 = 0.2372	K10 = 0.0432	K1* = 0.2486	K6p = 0.0305
K7p = 0.0272			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(R_m^2 - h m^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4hm)}{3L} \right) \right]$$

$$= -61331 \cdot 1260 \left[1 - \left(1 - \frac{1260}{7227} + \frac{(1358^2 - 0^2)}{(2 \cdot 1260 \cdot 7227)} \right) / \left(1 + \frac{(4 \cdot 0)}{(3 \cdot 7227)} \right) \right]$$

$$= -5652.0 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{(1 + 2(R_m^2 - h m^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L}{(3 \cdot 7227)} - 4 \cdot 1260 / 7227 \right)$$

$$= 61331 \cdot 7227 / 4 \left(\frac{(1 + 2(1358^2 - 0^2)/(7227^2)) / (1 + (4 \cdot 0)/(3 \cdot 7227)) - 4 \cdot 1260 / 7227}{(3 \cdot 7227)} \right)$$

$$= 41354.9 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 t)$$

$$= 1013 \cdot 1358 / (2 \cdot 13.7) - 41355 / (\pi \cdot 1358^2 \cdot 13.7)$$

$$= 45.10 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 1013 \cdot 1358 / (2 \cdot 13.7) + 41355 / (\pi \cdot 1358^2 \cdot 13.7)$$

$$= 55.32 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 1013 \cdot 1358 / (2 \cdot 13.7) - 5652 / (0.14 \cdot \pi \cdot 1358^2 \cdot 13.7)$$

$$= 55.16 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1013 \cdot 1358 / (2 \cdot 13.7) + 5652 / (0.25 \cdot \pi \cdot 1358^2 \cdot 13.7)$$

$$= 47.40 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h m / 3))$$

$$= 61331 (7227 - 2 \cdot 1260) / (7227 + (4 \cdot 0 / 3))$$

$$= 39945.6 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 0.9 \cdot 39946 / (1358 \cdot 13.7)$$

$$= 18.96 \text{ N./mm}^2$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 162 of 262

Saddle Calcs: Operating Case: Step: 27 10:16am Sep 9,2026

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1358 \cdot 13.7)}$$

$$= 106.377 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(585 + 1.56 \cdot \sqrt{(1358 \cdot 13.7)}, 2 \cdot 1260)$$

$$= 797.75 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(S_r/S, 1)$$

$$= \min(138/138, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ6,r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7 \cdot 61331 \cdot 0.1 / (798(13.7 + 1 \cdot 20))$$

$$= -1.56 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.29) [σ7,r*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -61331 / (4(13.7 + 1 \cdot 20)798) -$$

$$12 \cdot 0.034 \cdot 61331 \cdot 1358 / (7227(13.7 + 1 \cdot 20)^2)$$

$$= -45.94 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2349.5000	mm
Baseplate Thickness	Bpthk	50.0000	mm
Baseplate Width	Bpwid	610.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	218.0	N./mm ²
Height of Web at Center	Hw,c	650.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1351 \cdot 13.7)} + 875)13.7^3) / 12$$

$$= 23.297 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 23.3 + 149(303 - 6.85)^2$$

$$= 130283.680 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	1087.2	13.7	6.9	148.9	102029.9	0.130E+06
Wearplate	875.0	20.0	23.7	175.0	414749.9	0.136E+06
Web	20.0	546.7	307.1	109.3	3357284.2	0.273E+05
BasePlate	610.0	50.0	605.4	305.0	18464700.0	0.280E+06
Totals	...	...	...	738.3	22338764.0	0.574E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 22338764 / 738$$

$$= 302.575 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 163 of 262

Saddle Calcs: Operating Case: Step: 27 10:16am Sep 9, 2026

Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 140/2 \\
 &= 110.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(110) - 0.5 \cdot \sin(110)^2) / (\pi - 1.92 + \sin(110) \cos(110)) \\
 &= 0.2404
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.24 \cdot 61331 \\
 &= 14745.9590 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = 2.4538 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 130.8000 \text{ N./mm}^2
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 1981 - 1351 \cdot \sin(140/2) / 1.22 \\
 &= 942.234 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 13894.4131 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 7.1823 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 145.3333 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{3(61331 + 2103) \cdot 610 / (4 \cdot 2350 \cdot 145)} \\
 &= 28.870 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 140/2) (1350 + 14.5 + 20) \\
 &= 2602.009 \text{ mm}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (N_{\text{ribs}} - 1) \\
 &= 2602 / (5 - 1) \\
 &= 650.502 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e \cdot \text{Bpwid} / 2 \\
 &= 651 \cdot 610 / 2 \\
 &= 1984.032 \text{ cm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 63435 / (2350 \cdot 610) \\
 &= 4.426 \text{ Kgf/cm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= Ap \cdot Bp \\
 &= 1984 \cdot 4.43 \\
 &= 8781.519 \text{ Kgf}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 113 + 65.1 \\
 &= 178.050 \text{ cm}^2
 \end{aligned}$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 164 of 262

Saddle Calcs: Operating Case: Step: 27 10:16am Sep 9, 2026

Compressive Stress [Sc]:

= P/A_r

= 8782/178

= 4.837 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	585.0	...	117.0	...	0.334E+05

Rib dimension [D]:

= Saddle Width - Web Thickness

= 585 - 20

= 565.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/178

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 585/2

= 292.500 mm

Radius of Gyration [r]:

= $\sqrt{(Total\ Inertia / Total\ Area)}$

= $\sqrt{(33367/178)}$

= 136.895 mm

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - $\cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$

= 1981 - $\cos(140/2) (1350 + 14.5 + 20) - 50$

= 1457.673 mm

Length of Outer Rib, step 2, [Lw]:

= $2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$

= $2 \cdot \sin(140/2) (1350 + 14.5 + 20)$

= 2602.009 mm

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance

= 2350 - 50.8

= 2298.700 mm

Length of Outer Rib, step 4, [Dd]:

= $(Lw - Bd) / 2$

= $(2602 - 2299) / 2$

= 151.654 mm

Length of Outer Rib [L]:

= $\sqrt{Lv^2 + Dd^2}$

= $\sqrt{1458^2 + 152^2}$

= 1465.541 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$

= $\sqrt{(2 \cdot \pi^2 \cdot 199943008/218)}$

= 134.556

Slenderness ratio [KL/r]:



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 165 of 262

Saddle Calcs: Operating Case: Step: 27 10:16am Sep 9,2026

$$= KL/r$$

$$= 1 \cdot 1466/137$$

$$= 10.706$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 8818 / (2 \cdot 2350) \cdot 651 \cdot 1466/2$$

$$= 894.519 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($10.7 < 135$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (10.7)^2 / (2 \cdot 135^2)) 218 / (5/3 + 3 \cdot (10.7) / (8 \cdot 135) - (10.7^3) / (8 \cdot 135^3))$$

$$= 128.1 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 4.84/128 + (895 \cdot 292/333669344) / 145$$

$$= 0.091$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	565.0	0.0	113.0	0.0	0.334E+05
Web	650.5	20.0	0.0	130.1	0.0	43.4
Totals	...	...	...	243.1	...	0.334E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/243$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 585/2$$

$$= 292.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1981 - \cos(1385/2) (50 + 0 + 0) - 0$$

$$= 546.700 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 33409/243 }$$

$$= 117.230 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 547/117$$

$$= 4.663$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 8818 / (2 \cdot 2350)$$

$$= 1.877 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.88 \cdot 651 \cdot 547$$

$$= 667.376 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 166 of 262

Saddle Calcs: Operating Case: Step: 27 10:16am Sep 9,2026

= Bending Moment / Section Modulus

= 667/1142186

= 5.730 N./mm²

Compressive Allowable, $KL/r < C_c$ (4.66 < 135) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2• C_c^2)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c^3)

= (1-(4.66)²/(2 • 135²))218/

(5/3+3•(4.66)/(8• 135)-(4.66³)/(8•135³)

= 129.7 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$

= 7.09/130 + (667 • 292/33409)/145

= 0.094

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	16.1400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	14.0743 cm ²
Saddle Load QO (Weight)	QO	52281.0 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	4621.7 Kgf
Maximum Transverse Force	Ft	4409.0 Kgf
Maximum Longitudinal Force	Fl	8818.0 Kgf
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

= 31978.3 Kgf

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b, l]:

= $\max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= 31978/(14.1 • 8)

= 27.9 N./mm². Must be less than 82.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b, t]:

= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= 4409/(14.1 • 8)

= 3.8 N./mm². Must be less than 82.8 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$

= 8818/(82.8 • 8)

= 1.3055 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot F_t + \text{Sum of X Moments}$

= 1981 • 4409 + 0

= 8735.28 Kg-m.

Eccentricity (e):

= R_{mom} / QO



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 167 of 262

Saddle Calcs: Operating Case: Step: 27 10:16am Sep 9, 2026

= 8735/52281

= 167.08 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0, 1.31, 0]

= 1.3055 cm²

MDMT Calculations for: RH fixed saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 14.5, tr = 13.4, c = 0.8 mm, E* = 0.85

Thickness Ratio = tr * E*/(tg - c) = 0.83, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -17 °C

Min Metal Temp. at Required thickness (UCS 66.1) -27 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 168 of 262

Saddle Calcs: Test Case:

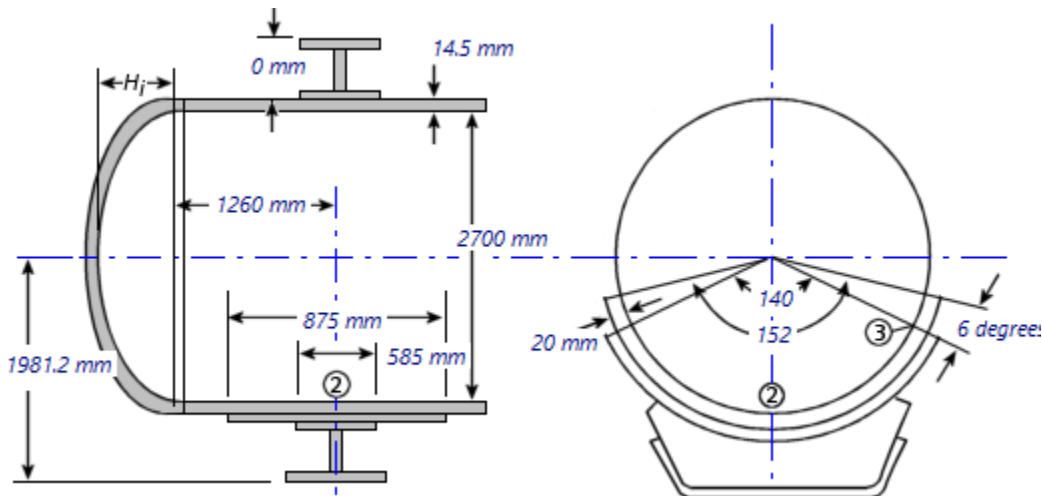
Step: 28 10:16am Sep 9,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1357.65	mm
Shell Thickness used in this Case	t	14.500	mm
Stiffened Vessel Length per 4.15.6	L	7227.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1260.00	mm
Saddle Width	b1 or b	585.00	mm
Saddle Bearing Angle	delta or theta	140.00	degrees
Wear Plate Width	b2 or b1	875.00	mm
Wear Plate Bearing Angle	delta2 or theta1	152.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1981.20	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	101913.61	Kgf
Pressure used in Analysis	P	1313.307 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	56.58	234.00
Long. Stress at Bottom of Midspan	73.57	234.00
Long. Stress at Top of Saddles	73.30	234.00
Long. Stress at Bottom of Saddles	60.41	234.00
Tangential Shear in Shell	31.51	138.58
Circ. Stress at Horn of Saddle	76.34	350.99
Circ. Compressive Stress in Shell	2.59	234.00



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 169 of 262

Saddle Calcs: Test Case:

Step: 28 10:16am

Sep 9, 2026

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3 (719/2 + 0) \cdot 1981/2350$$

$$= 909.0 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(362, 0) \cdot 1981/3780$$

$$= 189.7 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 101005 + \max(190, 909, 0, 0)$$

$$= 101913.6 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4(2.73 \cdot 1.2)^2)$$

$$= 7097.894 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	104016.84	Kgf
Transverse Shear Load	359.34	Kgf
Longitudinal Shear Load	361.90	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 2349.500 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1412	K2 = 0.9003	K3 = 0.5922	K4 = 0.3269
K5 = 0.6971	K6 = 0.0379	K7 = 0.0338	K8 = 0.3167
K9 = 0.2372	K10 = 0.0432	K1* = 0.2486	K6p = 0.0305
K7p = 0.0272			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -101914 \cdot 1260 [1 - (1 - 1260/7227 + (1358^2 - 0^2)/(2 \cdot 1260 \cdot 7227))/(1 + (4 \cdot 0)/(3 \cdot 7227))]$$

$$= -9391.9 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 101914 \cdot 7227/4((1 + 2(1358^2 - 0^2)/(7227^2))/(1 + (4 \cdot 0)/(3 \cdot 7227)) - 4 \cdot 1260/7227)$$

$$= 68719.0 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 1313 \cdot 1358/(2 \cdot 13.7) - 68719/(\pi \cdot 1358^2 \cdot 13.7)$$

$$= 56.58 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 1313 \cdot 1358/(2 \cdot 13.7) + 68719/(\pi \cdot 1358^2 \cdot 13.7)$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 170 of 262

Saddle Calcs: Test Case:

Step: 28 10:16am

Sep 9, 2026

$$= 73.57 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 1313 \cdot 1358 / (2 \cdot 13.7) - 9392 / (0.14 \cdot \pi \cdot 1358^2 \cdot 13.7)$$

$$= 73.30 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1313 \cdot 1358 / (2 \cdot 13.7) + 9392 / (0.25 \cdot \pi \cdot 1358^2 \cdot 13.7)$$

$$= 60.41 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 101914 (7227 - 2 \cdot 1260) / (7227 + (4 \cdot 0 / 3))$$

$$= 66377.1 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 0.9 \cdot 66377 / (1358 \cdot 13.7)$$

$$= 31.51 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1358 \cdot 13.7)}$$

$$= 106.377 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(585 + 1.56 \cdot \sqrt{(1358 \cdot 13.7)}, 2 \cdot 1260)$$

$$= 797.75 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7 \cdot 101914 \cdot 0.1 / (798 (13.7 + 1 \cdot 20))$$

$$= -2.59 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -101914 / (4(13.7 + 1 \cdot 20)798) -$$

$$12 \cdot 0.034 \cdot 101914 \cdot 1358 / (7227(13.7 + 1 \cdot 20)^2)$$

$$= -76.34 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2349.5000	mm
Baseplate Thickness	Bpthk	50.0000	mm
Baseplate Width	Bpwid	610.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	650.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1351 \cdot 13.7)} + 875)^3) / 12$$

$$= 23.297 \text{ cm}^4$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 171 of 262

Saddle Calcs: Test Case:

Step: 28 10:16am Sep 9,2026

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 23.3 + 149 (303 - 6.85)^2$$

$$= 130283.680 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	1087.2	13.7	6.9	148.9	102029.9	0.130E+06
Wearplate	875.0	20.0	23.7	175.0	414749.9	0.136E+06
Web	20.0	546.7	307.1	109.3	3357284.2	0.273E+05
BasePlate	610.0	50.0	605.4	305.0	18464700.0	0.280E+06
Totals	...	...	...	738.3	22338764.0	0.574E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 22338764/738$$

$$= 302.575 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 140/2$$

$$= 110.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(110) - 0.5 \cdot \sin(110)^2) / (\pi - 1.92 + \sin(110) \cos(110))$$

$$= 0.2404$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.24 \cdot 101914$$

$$= 24503.1777 \text{ Kg}$$

$$\text{Tension Stress, } S_t = (F_h / A_s) = 4.0774 \text{ N./mm}^2$$

$$\text{Allowed Stress, } S_a = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1981 - 1351 \cdot \sin(140/2) / 1.22$$

$$= 942.234 \text{ mm}$$

$$\text{Bending Moment, } M = F_h \cdot d = 23088.1738 \text{ Kg-m.}$$

$$\text{Bending Stress, } S_b = (M \cdot C1 / I) = 11.9347 \text{ N./mm}^2$$

$$\text{Allowed Stress, } S_a = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(101914 + 2103) \cdot 610) / (4 \cdot 2350 \cdot 173)}$$

$$= 33.852 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 140/2) (1350 + 14.5 + 20)$$

$$= 2602.009 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 2602 / (5 - 1)$$

$$= 650.502 \text{ mm}$$

Baseplate Pressure Area [Ap]:



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 172 of 262

Saddle Calcs: Test Case:

Step: 28 10:16am

Sep 9, 2026

$$= e \cdot Bpwid / 2$$

$$= 651 \cdot 610 / 2$$

$$= 1984.032 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 104017 / (2350 \cdot 610)$$

$$= 7.258 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1984 \cdot 7.26$$

$$= 14399.485 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 113 + 65.1$$

$$= 178.050 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 14399 / 178$$

$$= 7.931 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	585.0	...	117.0	...	0.334E+05

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 585 - 20$$

$$= 565.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 178$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 585 / 2$$

$$= 292.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (33367 / 178) }$$

$$= 136.895 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= Saddle \text{ Height} - \cos(saddle \text{ bearing angle} / 2)(radius + shlthk + wpdthk) - bpthk$$

$$= 1981 - \cos(140 / 2)(1350 + 14.5 + 20) - 50$$

$$= 1457.673 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(saddle \text{ bearing angle} / 2)(radius + shlthk + wpdthk)$$

$$= 2 \cdot \sin(140 / 2)(1350 + 14.5 + 20)$$

$$= 2602.009 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= Baseplate \text{ length} - clearance$$

$$= 2350 - 50.8$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 173 of 262

Saddle Calcs: Test Case:

Step: 28 10:16am Sep 9,2026

= 2298.700 mm

Length of Outer Rib, step 4, [Dd]:

= (Lw - Bd)/2

= (2602 - 2299)/2

= 151.654 mm

Length of Outer Rib [L]:

= $\sqrt{(Lv^2 + Dd^2)}$

= $\sqrt{(1458^2 + 152^2)}$

= 1465.541 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$

= $\sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)}$

= 123.211

Slenderness ratio [KL/r]:

= KL/r

= 1 • 1466/137

= 10.706

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 362 / (2 • 2350) • 651 • 1466/2

= 36.712 Kg-m.

Compressive Allowable, KL/r < Cc (10.7 < 123) per AISC E2-1 [Sca]:

= (1 - (KL/r)² / (2 • Cc²)) Fy / (5/3 + 3 • (KL/r) / (8 • Cc) - (KL/r)³ / (8 • Cc³))

= (1 - (10.7)² / (2 • 123²)) 260 /

(5/3 + 3 • (10.7) / (8 • 123) - (10.7³) / (8 • 123³))

= 152.4 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba

= 7.93/152 + (36.7 • 292/333669344) / 173

= 0.054

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	565.0	0.0	113.0	0.0	0.334E+05
Web	650.5	20.0	0.0	130.1	0.0	43.4
Totals	...	...	...	243.1	...	0.334E+05

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/243

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 585/2

= 292.500 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 1981 - cos(1385/2) (50 + 0 + 0) - 0

= 546.700 mm

Radius of Gyration [r]:

= $\sqrt{(\text{Total Inertia} / \text{Total Area})}$

= $\sqrt{(33409 / 243)}$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 174 of 262

Saddle Calcs: Test Case:

Step: 28 10:16am

Sep 9, 2026

= 117.230 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 547/117

= 4.663

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 362/(2 • 2350)

= 0.077 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 0.077 • 651 • 547

= 27.390 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 27.4/1142186

= 0.235 N./mm²

Compressive Allowable, KL/r < Cc (4.66 < 123) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2•Cc²))Fy/(5/3+3•(KL/r)/(8•Cc)-(KL/r)³)/(8•Cc³)

= (1-(4.66)²/(2 • 123²))260/

(5/3+3•(4.66)/(8• 123)-(4.66³)/(8•123³)

= 154.6 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 11.6/155 + (27.4 • 292/33409)/173

= 0.077

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	16.1400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	14.0743 cm ²
Saddle Load Q0 (Weight)	Q0	103107.8 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	189.7 Kg
Maximum Transverse Force	Ft	359.3 Kg
Maximum Longitudinal Force	Fl	361.9 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

= FI / (Bolt Area • Number of Bolts)

= 362/(14.1 • 8)

= 0.3 N./mm². Must be less than 29.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

= Ft / (Bolt Area • Number of Bolts)

= 359/(14.1 • 8)

= 0.3 N./mm². Must be less than 29.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)



PV Elite 28 Licensee: SL User
 FileName : 197-C_Channel MOC= CS Page 175 of 262
 Saddle Calcs: Test Case: Step: 28 10:16am Sep 9,2026

Bolt Area due to Shear Load [Bltarears]:
 $= F / (\text{BoltShearAllowable} \cdot N_{\text{bolts}})$
 $= 362 / (29 \cdot 8)$
 $= 0.1531 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot F_t + \text{Sum of X Moments}$
 $= 1981 \cdot 359 + 0$
 $= 711.93 \text{ Kg-m.}$

Eccentricity (e):
 $= R_{\text{mom}} / QO$
 $= 712 / 103108$
 $= 6.90 \text{ mm} < B_{\text{plen}}/6 \rightarrow \text{No Uplift in Transverse direction}$

Bolt Area due to Transverse Load [Bltareart]:
 $= 0 \text{ (No Uplift)}$

Required Area of a Single Bolt [Bltarear]:
 $= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
 $= \max[0, 0.15, 0]$
 $= 0.1531 \text{ cm}^2$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1357.65	mm
Shell Thickness used in this Case	t	14.500	mm
Stiffened Vessel Length per 4.15.6	L	7227.00	mm
Distance from Saddle to Vessel tangent	a1 or a	1260.00	mm
Saddle Width	b1 or b	585.00	mm
Saddle Bearing Angle	delta or theta	140.00	degrees
Wear Plate Width	b2 or b1	875.00	mm
Wear Plate Bearing Angle	delta2 or theta1	152.00	degrees
Wear Plate Thickness	e2 or tr	20.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1981.20	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		47775.52	Kgf
Pressure used in Analysis	P	1313.307	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		61.09	234.00
Long. Stress at Bottom of Midspan		69.06	234.00
Long. Stress at Top of Saddles		68.93	234.00
Long. Stress at Bottom of Saddles		62.89	234.00
Tangential Shear in Shell		14.77	138.58



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 176 of 262

Saddle Calcs: Test Case:

Step: 28 10:16am Sep 9, 2026

Circ. Stress at Horn of Saddle	35.79	350.99
Circ. Compressive Stress in Shell	1.21	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3 (719/2 + 0) \cdot 1981/2350$$

$$= 909.0 \text{ Kg}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(362, 0) \cdot 1981/3780$$

$$= 189.7 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 46866 + \max(190, 909, 0, 0)$$

$$= 47775.5 \text{ Kg}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766 (0 + \pi/4 (2.73 \cdot 1.2)^2)$$

$$= 7097.894 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	49878.74	Kgf
Transverse Shear Load	359.34	Kgf
Longitudinal Shear Load	361.90	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 2349.500 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1412	K2 = 0.9003	K3 = 0.5922	K4 = 0.3269
K5 = 0.6971	K6 = 0.0379	K7 = 0.0338	K8 = 0.3167
K9 = 0.2372	K10 = 0.0432	K1* = 0.2486	K6p = 0.0305
K7p = 0.0272			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -47776 \cdot 1260 [1 - (1 - 1260/7227 + (1358^2 - 0^2)/(2 \cdot 1260 \cdot 7227)) / (1 + (4 \cdot 0)/(3 \cdot 7227))]$$

$$= -4402.8 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 47776 \cdot 7227/4 ((1 + 2(1358^2 - 0^2)/(7227^2)) / (1 + (4 \cdot 0)/(3 \cdot 7227)) - 4 \cdot 1260/7227)$$

$$= 32214.4 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$$

$$= 1313 \cdot 1358/(2 \cdot 13.7) - 32214/(\pi \cdot 1358^2 \cdot 13.7)$$

$$= 61.09 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 177 of 262

Saddle Calcs: Test Case:

Step: 28 10:16am

Sep 9, 2026

$$\begin{aligned}
 &= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\
 &= 1313 \cdot 1358 / (2 \cdot 13.7) + 32214 / (\pi \cdot 1358^2 \cdot 13.7) \\
 &= 69.06 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 1313 \cdot 1358 / (2 \cdot 13.7) - 4403 / (0.14 \cdot \pi \cdot 1358^2 \cdot 13.7) \\
 &= 68.93 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 1313 \cdot 1358 / (2 \cdot 13.7) + 4403 / (0.25 \cdot \pi \cdot 1358^2 \cdot 13.7) \\
 &= 62.89 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L-2a) / (L + (4 \cdot h_m / 3)) \\
 &= 47776 (7227 - 2 \cdot 1260) / (7227 + (4 \cdot 0 / 3)) \\
 &= 31116.6 \text{ Kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K_2 \cdot T / (R_m \cdot t) \\
 &= 0.9 \cdot 31117 / (1358 \cdot 13.7) \\
 &= 14.77 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(R_m \cdot t)} \\
 &= 0.78 \cdot \sqrt{(1358 \cdot 13.7)} \\
 &= 106.377 \text{ mm}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a) \\
 &= \min(585 + 1.56 \cdot \sqrt{(1358 \cdot 13.7)}, 2 \cdot 1260) \\
 &= 797.75 \text{ mm}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$\begin{aligned}
 &= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot t_r)) \\
 &= -0.7 \cdot 47776 \cdot 0.1 / (798(13.7 + 1 \cdot 20)) \\
 &= -1.21 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$\begin{aligned}
 &= -Q / (4(t + \eta \cdot t_r)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot t_r)^2) \\
 &= -47776 / (4(13.7 + 1 \cdot 20)798) - \\
 &\quad 12 \cdot 0.034 \cdot 47776 \cdot 1358 / (7227(13.7 + 1 \cdot 20)^2) \\
 &= -35.79 \text{ N./mm}^2
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2349.5000	mm
Baseplate Thickness	Bpthk	50.0000	mm
Baseplate Width	Bpwid	610.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	20.0000	mm
Web Thickness	Webtk	20.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	650.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 178 of 262

Saddle Calcs: Test Case:

Step: 28 10:16am

Sep 9, 2026

$$= ((1.56 \cdot \sqrt{(1351 \cdot 13.7)} + 875) 13.7^3) / 12$$

$$= 23.297 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 23.3 + 149 (303 - 6.85)^2$$

$$= 130283.680 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	1087.2	13.7	6.9	148.9	102029.9	0.130E+06
Wearplate	875.0	20.0	23.7	175.0	414749.9	0.136E+06
Web	20.0	546.7	307.1	109.3	3357284.2	0.273E+05
BasePlate	610.0	50.0	605.4	305.0	18464700.0	0.280E+06
Totals	...	...	...	738.3	22338764.0	0.574E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 22338764 / 738$$

$$= 302.575 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 140 / 2$$

$$= 110.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(110) - 0.5 \cdot \sin(110)^2) / (\pi - 1.92 + \sin(110) \cos(110))$$

$$= 0.2404$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.24 \cdot 47776$$

$$= 11486.7090 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.9114 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1981 - 1351 \cdot \sin(140/2) / 1.22$$

$$= 942.234 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 10823.3779 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 5.5948 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(47776 + 2103) 610) / (4 \cdot 2350 \cdot 173)}$$

$$= 23.442 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 140/2) (1350 + 14.5 + 20)$$

$$= 2602.009 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 2602 / (5 - 1)$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 179 of 262

Saddle Calcs: Test Case:

Step: 28 10:16am

Sep 9, 2026

= 650.502 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2

= 651 • 610/2

= 1984.032 cm²

Bearing Pressure [Bp]:

= (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)

= 49879/(2350 • 610)

= 3.480 Kg/cm²

Axial Load [P]:

= Ap • Bp

= 1984 • 3.48

= 6904.923 Kg

Area of the Rib and Web [Ar]:

= Rib Area + Web Area

= 113 + 65.1

= 178.050 cm²

Compressive Stress [Sc]:

= P/Ar

= 6905/178

= 3.803 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	20.0	585.0	...	117.0	...	0.334E+05

Rib dimension [D]:

= Saddle Width - Web Thickness

= 585 - 20

= 565.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0/178

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 585/2

= 292.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{33367/178}$

= 136.895 mm

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk

= 1981 - cos(140/2)(1350 + 14.5 + 20) - 50

= 1457.673 mm

Length of Outer Rib, step 2, [Lw]:

= 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)

= 2 • sin(140/2)(1350 + 14.5 + 20)

= 2602.009 mm



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 180 of 262

Saddle Calcs: Test Case:

Step: 28 10:16am

Sep 9, 2026

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2350 - 50.8$$

$$= 2298.700 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (L_w - B_d) / 2$$

$$= (2602 - 2299) / 2$$

$$= 151.654 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{(L_v^2 + D_d^2)}$$

$$= \sqrt{(1458^2 + 152^2)}$$

$$= 1465.541 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008 / 260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1466 / 137$$

$$= 10.706$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 362 / (2 \cdot 2350) \cdot 651 \cdot 1466 / 2$$

$$= 36.712 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($10.7 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (10.7)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (10.7) / (8 \cdot 123) - (10.7^3) / (8 \cdot 123^3))$$

$$= 152.4 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 3.8 / 152 + (36.7 \cdot 292 / 333669344) / 173$$

$$= 0.027$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	20.0	565.0	0.0	113.0	0.0	0.334E+05
Web	650.5	20.0	0.0	130.1	0.0	43.4
Totals	...	...	...	243.1	...	0.334E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 243$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 585 / 2$$

$$= 292.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1981 - \cos(1385/2) (50 + 0 + 0) - 0$$

$$= 546.700 \text{ mm}$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 181 of 262

Saddle Calcs: Test Case:

Step: 28 10:16am Sep 9, 2026

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{33409/243}$$

$$= 117.230 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 547/117$$

$$= 4.663$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 362 / (2 \cdot 2350)$$

$$= 0.077 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.077 \cdot 651 \cdot 547$$

$$= 27.390 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 27.4/1142186$$

$$= 0.235 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($4.66 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (4.66)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (4.66) / (8 \cdot 123) - (4.66^3) / (8 \cdot 123^3))$$

$$= 154.6 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_b$$

$$= 5.57/155 + (27.4 \cdot 292/33409) / 173$$

$$= 0.037$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification	SA-307 A	
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	16.1400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	14.0743 cm ²
Saddle Load Q0 (Weight)	Q0	48969.7 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	189.7 Kg
Maximum Transverse Force	Ft	359.3 Kg
Maximum Longitudinal Force	F1	361.9 Kg
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 362 / (14.1 \cdot 8)$$

$$= 0.3 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 359 / (14.1 \cdot 8)$$

$$= 0.3 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 182 of 262

Saddle Calcs: Test Case:

Step: 28 10:16am

Sep 9, 2026

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (BoltShearAllowable \cdot Nbolts)$
= $362 / (29 \cdot 8)$
= 0.1531 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot Ft + \text{Sum of X Moments}$
= $1981 \cdot 359 + 0$
= 711.93 Kg-m.

Eccentricity (e):
= $Rmom / QO$
= $712 / 48970$
= 14.54 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0, 0.15, 0]$
= 0.1531 cm²

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 183 of 262

Nozzle Calcs.: N1-Tube Inlet

Nozl: 13 10:16am Sep 9,2026

Input, Nozzle Desc: N1-Tube Inlet

From: 30

Pressure for Reinforcement Calculations	P	1000.000	KPa.
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2700.00	mm
Design Length of Section	L	1200.0001	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1106.35	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	906.4000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		250.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



PV Elite 28 Licensee: SL User

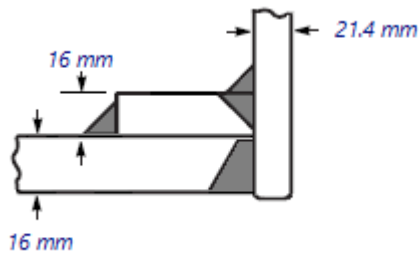
FileName : 197-C_Channel MOC= CS

Page 184 of 262

Nozzle Calcs.: N1-Tube Inlet

Nozl: 13 10:16am

Sep 9, 2026



Reinforcement CALCULATION, Description: N1-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	14.314 in.
Actual Thickness Used in Calculation	0.843 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1000 \cdot 1353 / (138 \cdot 1 - 0.6 \cdot 1000)$
 $= 9.8479 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1000 \cdot 185 / (118 \cdot 1 - 0.6 \cdot 1000)$
 $= 1.5741 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.2211 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	739.1512 mm
Parallel to Vessel Wall, opening length	d	369.5756 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		32.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 185 of 262

Nozzle Calcs.: N1-Tube Inlet

Nozl: 13 10:16am Sep 9,2026

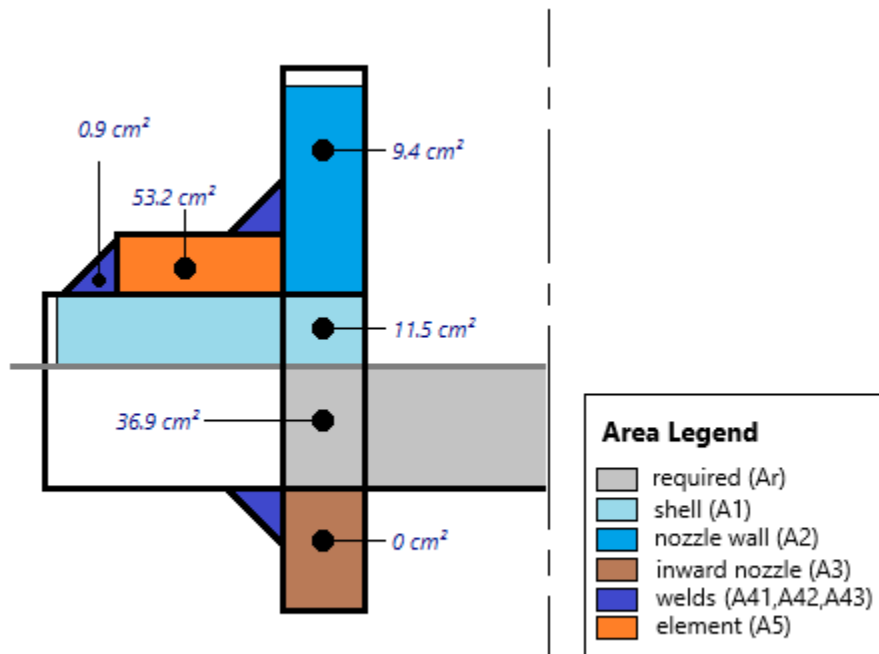


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	36.921	11.660	NA
Area in Shell	A1	11.481	24.696	NA
Area in Nozzle Wall	A2	9.359	9.555	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.855	0.855	NA
Area in Element	A5	53.240	53.240	NA
TOTAL AREA AVAILABLE	Atot	74.935	88.346	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 501.5645 16.0000 mm
Based on given Pad Diameter: 906.4000 4.5759 mm
Based on Shell or Nozzle Thickness: 501.5645 16.0000 mm

Area Required [A]:
= (d • tr • F + 2 • tn • tr • F • (1-fr1)) UG-37(c)
= (370•9.85•1+2•18.4•9.85•1•(1-0.86))
= 36.921 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
= d(E1•t - F•tr) - 2 • tn (E1•t - F•tr) • (1 - fr1)
= 370 (1 • 13 - 1 • 9.85) - 2 • 18.4
(1 • 13 - 1 • 9.85) • (1 - 0.86)



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 186 of 262

Nozzle Calcs.: N1-Tube Inlet

Nozl: 13 10:16am

Sep 9, 2026

$$= 11.481 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 32.5) \cdot (18.4 - 1.57) \cdot 0.86$$

$$= 9.359 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi-can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 0^2 \cdot 1$$

$$= 0.855 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (739 - 406) \cdot 16 \cdot 1$$

$$= 53.240 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 4.5741 \text{ mm}$

Wall Thickness per UG16(b), $tr16b = 4.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb1 = 12.8479 \text{ mm}$

Wall Thickness $tb1 = \max(trb1, tr16b) = 12.8479 \text{ mm}$

Wall Thickness, shell/head, external pressure $trb2 = 4.0145 \text{ mm}$

Wall Thickness $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(12.8, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.57, 11.3)$$

$$= 11.3312 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 21.412 = 18.736 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	18.6,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	276.3 N./mm ²	Passed
Occasional	:	4.8,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	10.7,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $tg = 18.7$, $tr = 1.57$, $c = 3 \text{ mm}$, $E* = 1$

Thickness Ratio = $tr \cdot E*/(tg - c) = 0.1$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A $10 \text{ }^\circ\text{C}$

Min Metal Temp. at Required thickness (UCS 66.1) $-104 \text{ }^\circ\text{C}$

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $tg = 16$, $c = 3 \text{ mm}$, $E* = 1$

Thickness Ratio = $tr \cdot E*/(tg - c) = 0.76$, Temp. Reduction = $13 \text{ }^\circ\text{C}$

Attachment governing, Attachment stress = Shell stress per Code



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 187 of 262

Nozzle Calcs.: N1-Tube Inlet

Nozl: 13 10:16am

Sep 9, 2026

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-28 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 16$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $tr * E^* / (t_g - c) = 0.76$, Temp. Reduction = 13 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-28 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 16$, $tr = 9.85$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $tr * E^* / (t_g - c) = 0.76$, Temp. Reduction = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-28 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 16$, $tr = 9.85$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $tr * E^* / (t_g - c) = 0.76$, Temp. Reduction = 13 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-28 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	:	-29 °C
Gov. MDMT of the Reinforcement Pad	:	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1000.00 / 1960.01 = 0.510$

Weld Size Calculations, Description: N1-Tube Inlet

Intermediate Calc. for nozzle/shell Welds	T_{min}	16.0000 mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$
Pad Weld	$6.5000 = 0.5 * T_{minPad}$	$8.4840 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 * t_n * f_r1 * (E1 * t - tr)) S_v)$
 $= \max(0, (36.9 - 11.5 + 2 * 18.4 * 0.86 * (1 * 13 - 9.85)) 138)$
 $= 37194.63 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 188 of 262

Nozzle Calcs.: N1-Tube Inlet

Nozl: 13 10:16am

Sep 9, 2026

$$\begin{aligned}
 &= (A2 + A5 + A4 - (W_i - C_{an} / .707)^2 \cdot fr2) \cdot Sv \\
 &= (9.36 + 53.2 + 0.86 - 0 \cdot 0.86) \cdot 138 \\
 &= 89289.48 \text{ Kgf}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv \\
 &= (9.36 + 0 + 0.86 + (4.09)) \cdot 138 \\
 &= 20132.20 \text{ Kgf}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S \\
 &= (9.36 + 0 + 0.86 + 53.2 + (4.09)) \cdot 138 \\
 &= 95049.51 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 406 \cdot 10 \cdot 0.49 \cdot 118 \\
 &= 37637. \text{ Kgf}
 \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned}
 &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\
 &= (3.14/2.0) \cdot 906 \cdot 12 \cdot 0.49 \cdot 138 \\
 &= 117804. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - C_{an}) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 194) \cdot (21.4 - 3) \cdot 0.7 \cdot 118 \\
 &= 94512. \text{ Kgf}
 \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot Seg \\
 &= (3.14/2) \cdot 406 \cdot 16 \cdot 0.74 \cdot 138 \\
 &= 106357. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 406 \cdot (16 - 3) \cdot 0.74 \cdot 138 \\
 &= 86415. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (117804 + 94512) = 212316 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (37637 + 106357 + 86415 + 0) = 230410 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Spew + Tngw + Sinw) \\
 &= (117804 + 86415 + 0) = 204219 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 212315 Kgf, must exceed W = 37194 Kgf	or W1 = 89289 Kgf
Path 2-2 = 230409 Kgf, must exceed W = 37194 Kgf	or W2 = 20132 Kgf
Path 3-3 = 204219 Kgf, must exceed W = 37194 Kgf	or W3 = 95049 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1094 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.



PV Elite 28 Licensee: SL User
 FileName : 197-C_Channel MOC= CS Page 189 of 262
 Nozzle Calcs.: N1-Tube Inlet Nozl: 13 10:16am Sep 9,2026

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 15.3803 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 381.3802 mm

Input Echo, WRC297 Item 1, Description: N1-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2700.000	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	363.576	mm
Nozzle Thickness	21.4122	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	906.400	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	1000.00	KPa.
Radial Load	P	-3059.35	Kgf
Circumferential Shear	Vc	2039.57	Kgf
Longitudinal Shear	Vl	2039.57	Kgf
Circumferential Moment	Mc	2283.43	Kg-m.
Longitudinal Moment	Ml	2283.43	Kg-m.
Torsional Moment	Mt	3262.05	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)} \\
 &= 406.4 + 2 \cdot 1.65 \cdot \sqrt{1367.5(16.0 - 3.0)} \\
 &= 846.397 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2735.000 mm
Nozzle Outside Diameter	(d) =	406.400 mm
Vessel Thickness used	(T) =	29.000 mm



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 190 of 262

Nozzle Calcs.: N1-Tube Inlet

Nozl: 13 10:16am

Sep 9, 2026

Nozzle Thickness used (t) = 18.412 mm
 T / t = 1.575
 d / t = 22.072

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.443$$

Nr/P = 0.078
 Mr/P = 0.122
 M0/P = 0.056
 N0/P = 0.142
 MrD/Mc = 0.231
 NrDL/Mc = 0.121
 M0d/Mc = 0.123
 N0DL/Mc = 0.119
 MrD/Ml = 0.150
 NrDL/Ml = 0.086
 M0D/Ml = 0.077
 N0DL/Ml = 0.223

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-14	-14	14	14
Outplane Bending (ML)	-30	30	30	-30
Normal Pressure Stress	46	47	46	47
Outplane Stress Summary	18	55	108	24

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	26	-26	26	-26
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-58	58	58	-58
Inplane Pressure Stress	23	23	23	23
Inplane Stress Summary	-12	52	116	-53

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 191 of 262

Nozzle Calcs.: N1-Tube Inlet

Nozl: 13 10:16am Sep 9,2026

Two * Max Shear Stress	32	59	117	78
------------------------	----	----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	-7	-7	7	7
Outplane Bending (Mc)	-48	48	48	-48
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	23	23	23	23
Outplane Stress Summary	-16	56	96	-24

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	26	-26	26	-26
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-90	90	90	-90
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	46	47	46	47
Inplane Stress Summary	-23	106	173	-59

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	3	3	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	24	106	174	60

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-14	-14	14	14
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	12	11	12
Hoop Stress Summary	1	3	31	32



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 192 of 262

Nozzle Calcs.: N1-Tube Inlet

Nozl: 13 10:16am

Sep 9,2026

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	44	-44	44	-44
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-10	-10	10	10
Axial Bending (ML)	-104	104	104	-104
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-63	56	166	-130

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6
Shear Stress Summary	8	8	4	4

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	67	57	166	163

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-7	-7	7	7
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	12	11	12
Hoop Stress Summary	8	9	24	25

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	44	-44	44	-44
Axial Membrane (Mc)	-10	-10	10	10
Axial Bending (Mc)	-166	166	166	-166
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-126	118	228	-193

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 193 of 262

Nozzle Calcs.: N1-Tube Inlet Nozl: 13 10:16am Sep 9,2026

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	6	6	6	6

Shear Stress Summary	4	4	8	8
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	135	118	229	219

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 406.4 + 2 \cdot 1.65 \cdot \sqrt{1359.5(16.0 - 3.0)}$$

$$= 845.108 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2719.000 mm
Nozzle Outside Diameter	(d) =	906.400 mm
Vessel Thickness used	(T) =	13.000 mm
Nozzle Thickness used	(t) =	268.412 mm
T / t	=	0.048
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.821$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.047
Mr/P	=	0.071
M0/P	=	0.019
N0/P	=	0.022
MrD/Mc	=	0.241
NrDL/Mc	=	0.104
M0d/Mc	=	0.072
N0DL/Mc	=	0.045
MrD/Ml	=	0.049
NrDL/Ml	=	0.053
M0D/Ml	=	0.015
N0DL/Ml	=	0.023

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-13	13	13	-13
Normal Pressure Stress	103	104	103	104
Outplane Stress Summary	111	97	144	78

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 194 of 262

Nozzle Calcs.: N1-Tube Inlet

Nozl: 13 10:16am

Sep 9,2026

Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	75	-75	75	-75
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-42	42	42	-42
Inplane Pressure Stress	51	51	51	51

Inplane Stress Summary	85	19	186	-50
------------------------	----	----	-----	-----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	2	2	0	0
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Two * Max Shear Stress	111	97	186	129
------------------------	-----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-63	63	63	-63
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	51	51	51	51

Outplane Stress Summary	6	91	145	-21
-------------------------	---	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	75	-75	75	-75
Inplane Membrane (Mc)	-15	-15	15	15
Inplane Bending (Mc)	-211	211	211	-211
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	103	104	103	104

Inplane Stress Summary	-38	233	414	-158
------------------------	-----	-----	-----	------

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Shear (Vc)	0	0	0	0
---------------------	---	---	---	---



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 195 of 262

Nozzle Calcs.: N1-Tube Inlet

Nozl: 13 10:16am

Sep 9,2026

Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	45	233	414	158

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	46	47	46	47	46	47	46	47
Circ. Pl (SUS)	-9	-9	19	19	-5	-5	9	9
Circ. Q (SUS)	-18	18	42	-42	-64	64	116	-116
Long. Pm (SUS)	23	23	23	23	23	23	23	23
Long. Pl (SUS)	-3	-3	7	7	-2	-2	12	12
Long. Q (SUS)	-32	32	84	-84	-36	36	60	-60
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	46.0	47.0	46.0	47.0	46.0	47.0	46.0	47.0
Pm+Pl (SUS)	37.1	38.1	65.0	66.0	41.0	42.0	55.0	56.0
Pm+Pl+Q (Total)	32.6	59.4	115.1	78.2	24.0	106.2	171.3	60.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	47.00	138.00	Passed
Pm+Pl (SUS)	66.03	207.00	Passed
Pm+Pl+Q (TOTAL)	171.33	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	103	104	103	104	103	104	103	104
Circ. Pl (SUS)	0	0	6	6	-7	-7	23	23
Circ. Q (SUS)	7	-7	33	-33	-136	136	286	-286
Long. Pm (SUS)	51	51	51	51	51	51	51	51
Long. Pl (SUS)	1	1	15	15	-3	-3	9	9
Long. Q (SUS)	33	-33	117	-117	-43	43	83	-83



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 196 of 262

Nozzle Calcs.: N1-Tube Inlet

Nozl: 13 10:16am

Sep 9, 2026

Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	103.0	104.0	103.0	104.0	103.0	104.0	103.0	104.0
Pm+P1 (SUS)	103.0	104.0	109.0	110.0	96.0	97.0	126.0	127.0
Pm+P1+Q (Total)	110.2	97.1	183.0	128.0	45.0	233.0	412.0	159.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	104.00	138.00	Passed
Pm+P1 (SUS)	127.01	207.00	Passed
Pm+P1+Q (TOTAL)	412.01	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	12	11	12	11	12	11	12
Circ. P1 (SUS)	-9	-9	19	19	-2	-2	12	12
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. P1 (SUS)	-9	-9	11	11	-9	-9	11	11
Long. Q (SUS)	-60	60	148	-148	-122	122	210	-210
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	11.0	12.0	11.0	12.0	11.0	12.0	11.0	12.0
Pm+P1 (SUS)	6.3	7.3	30.1	31.1	13.2	14.1	23.1	24.1
Pm+P1+Q (Total)	67.5	56.9	164.2	163.3	135.4	118.2	226.2	218.4

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	12.00	118.00	Passed
Pm+P1 (SUS)	31.07	177.00	Passed
Pm+P1+Q (TOTAL)	226.24	353.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 197 of 262

Nozzle Calcs.: N2-Tube outlet

Nozl: 14 10:16am

Sep 9,2026

Input, Nozzle Desc: N2-Tube outlet

From: 30

Pressure for Reinforcement Calculations	P	1026.465	KPa.
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2700.00	mm
Design Length of Section	L	1200.0001	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1106.35	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		16.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	926.4000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		260.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



PV Elite 28 Licensee: SL User

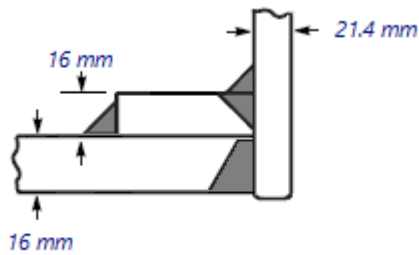
FileName : 197-C_Channel MOC= CS

Page 198 of 262

Nozzle Calcs.: N2-Tube outlet

Nozl: 14 10:16am

Sep 9,2026



Reinforcement CALCULATION, Description: N2-Tube outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	14.314 in.
Actual Thickness Used in Calculation	0.843 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1026 \cdot 1353 / (138 \cdot 1 - 0.6 \cdot 1026)$
 $= 10.1097 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1026 \cdot 185 / (118 \cdot 1 - 0.6 \cdot 1026)$
 $= 1.6160 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.2211 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	739.1512 mm
Parallel to Vessel Wall, opening length	d	369.5756 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		32.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 199 of 262

Nozzle Calcs.: N2-Tube outlet

Nozl: 14 10:16am Sep 9,2026

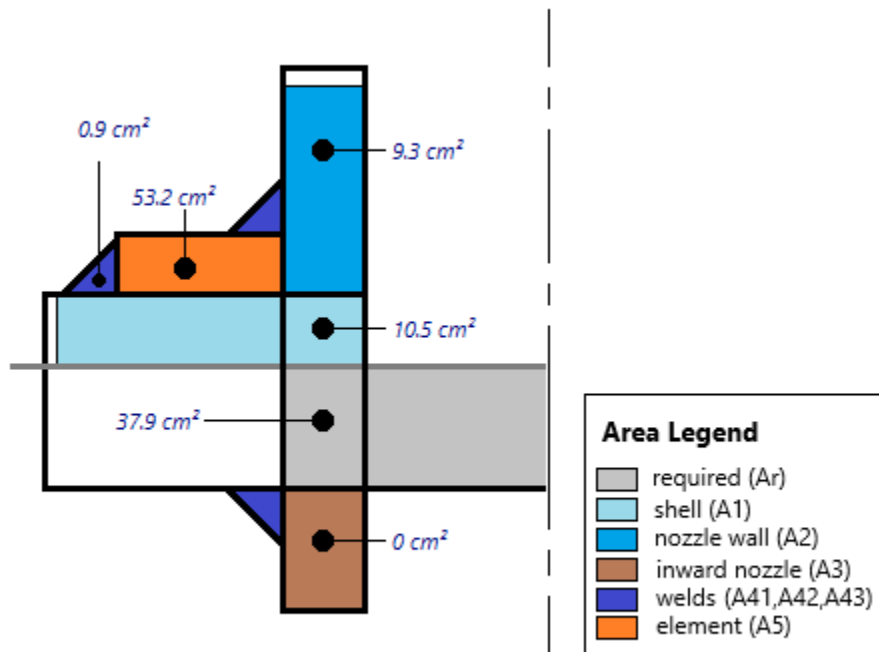


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	37.903	11.660	NA
Area in Shell	A1	10.528	24.696	NA
Area in Nozzle Wall	A2	9.335	9.555	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.855	0.855	NA
Area in Element	A5	53.240	53.240	NA
TOTAL AREA AVAILABLE	Atot	73.958	88.346	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 513.8043 16.0000 mm
Based on given Pad Diameter: 926.4000 5.1644 mm
Based on Shell or Nozzle Thickness: 513.8043 16.0000 mm

Area Required [A]:
= (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)
= (370•10.1•1+2•18.4•10.1•1•(1-0.86))
= 37.903 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
= d(E1•t - F•tr) - 2 • tn (E1•t - F•tr) • (1 - fr1)
= 370 (1 • 13 - 1 • 10.1) - 2 • 18.4
(1 • 13 - 1 • 10.1) • (1 - 0.86)



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 200 of 262

Nozzle Calcs.: N2-Tube outlet

Nozl: 14 10:16am

Sep 9, 2026

$$= 10.528 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot T_{lwp}) \cdot (t_n - t_{rn}) \cdot f_{r2}$$

$$= (2 \cdot 32.5) \cdot (18.4 - 1.62) \cdot 0.86$$

$$= 9.335 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot f_{r3} + (W_i - c_{an}/0.707)^2 \cdot f_{r2} + W_p^2 \cdot f_{r4}$$

$$= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 0^2 \cdot 1$$

$$= 0.855 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot f_{r4}$$

$$= (739 - 406) \cdot 16 \cdot 1$$

$$= 53.240 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 4.6160 \text{ mm}$

Wall Thickness per UG16(b), $t_{r16b} = 4.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $t_{rb1} = 13.1097 \text{ mm}$

Wall Thickness $t_{b1} = \max(t_{rb1}, t_{r16b}) = 13.1097 \text{ mm}$

Wall Thickness, shell/head, external pressure $t_{rb2} = 4.0145 \text{ mm}$

Wall Thickness $t_{b2} = \max(t_{rb2}, t_{r16b}) = 4.5000 \text{ mm}$

Wall Thickness per table UG-45 $t_{b3} = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[t_{b3}, \max(t_{b1}, t_{b2})]$$

$$= \min[11.3, \max(13.1, 4.5)]$$

$$= 11.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(4.62, 11.3)$$

$$= 11.3312 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 21.412 = 18.736 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	18.8,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	276.2 N./mm ²	Passed
Occasional	:	4.9,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	10.7,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 18.7$, $t_r = 1.62$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.1$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A $10 \text{ }^\circ\text{C}$

Min Metal Temp. at Required thickness (UCS 66.1) $-104 \text{ }^\circ\text{C}$

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 16$, $c = 3 \text{ mm}$, $E^* = 1$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.78$, Temp. Reduction = $12 \text{ }^\circ\text{C}$

Attachment governing, Attachment stress = Shell stress per Code



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 201 of 262

Nozzle Calcs.: N2-Tube outlet Nozl: 14 10:16am Sep 9, 2026

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-26 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 16$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.78$, Temp. Reduction = 12 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-26 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 16$, $t_r = 10.1$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.78$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-26 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 16$, $t_r = 10.1$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.78$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-26 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	:	-29 °C
Gov. MDMT of the Reinforcement Pad	:	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-47 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1026.47 / 1960.01 = 0.524$

Weld Size Calculations, Description: N2-Tube outlet

Intermediate Calc. for nozzle/shell Welds	T_{min}	16.0000 mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	13.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$
Pad Weld	$6.5000 = 0.5 * T_{minPad}$	$8.4840 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 * t_n * f_r1 * (E1 * t - t_r)) S_v)$
 $= \max(0, (37.9 - 10.5 + 2 * 18.4 * 0.86 * (1 * 13 - 10.1)) 138)$
 $= 39801.59 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 202 of 262

Nozzle Calcs.: N2-Tube outlet

Nozl: 14 10:16am

Sep 9, 2026

$$\begin{aligned}
 &= (A2 + A5 + A4 - (W_i - C_{an} / .707)^2 \cdot fr2) \cdot Sv \\
 &= (9.34 + 53.2 + 0.86 - 0 \cdot 0.86) \cdot 138 \\
 &= 89256.73 \text{ Kg}
 \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv \\
 &= (9.34 + 0 + 0.86 + (4.09)) \cdot 138 \\
 &= 20099.45 \text{ Kg}
 \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned}
 &= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S \\
 &= (9.34 + 0 + 0.86 + 53.2 + (4.09)) \cdot 138 \\
 &= 95016.75 \text{ Kg}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.14/2.0) \cdot 406 \cdot 10 \cdot 0.49 \cdot 118 \\
 &= 37637. \text{ Kg}
 \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned}
 &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\
 &= (3.14/2.0) \cdot 926 \cdot 12 \cdot 0.49 \cdot 138 \\
 &= 120403. \text{ Kg}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - C_{an}) \cdot 0.7 \cdot S_n \\
 &= (3.14 \cdot 194) \cdot (21.4 - 3) \cdot 0.7 \cdot 118 \\
 &= 94512. \text{ Kg}
 \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot Seg \\
 &= (3.14/2) \cdot 406 \cdot 16 \cdot 0.74 \cdot 138 \\
 &= 106357. \text{ Kg}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\
 &= (3.14/2.0) \cdot 406 \cdot (16 - 3) \cdot 0.74 \cdot 138 \\
 &= 86415. \text{ Kg}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (120403 + 94512) = 214915 \text{ Kg}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (Sonw + Tpgw + Tngw + Sinw) \\
 &= (37637 + 106357 + 86415 + 0) = 230410 \text{ Kg}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (Spew + Tngw + Sinw) \\
 &= (120403 + 86415 + 0) = 206819 \text{ Kg}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 214915 Kg, must exceed W = 39801 Kg	or W1 = 89256 Kg
Path 2-2 = 230409 Kg, must exceed W = 39801 Kg	or W2 = 20099 Kg
Path 3-3 = 206818 Kg, must exceed W = 39801 Kg	or W3 = 95016 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1094 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 203 of 262

Nozzle Calcs.: N2-Tube outlet

Nozl: 14 10:16am

Sep 9, 2026

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 15.3803 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 381.3802 mm

Input Echo, WRC297 Item 1, Description: N2-Tube outlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2700.000	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	363.576	mm
Nozzle Thickness	21.4122	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	926.400	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	1026.47	KPa.
Radial Load	P	-3059.35	Kgf
Circumferential Shear	Vc	2039.57	Kgf
Longitudinal Shear	Vl	2039.57	Kgf
Circumferential Moment	Mc	2283.43	Kg-m.
Longitudinal Moment	Ml	2283.43	Kg-m.
Torsional Moment	Mt	3262.05	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)} \\
 &= 406.4 + 2 \cdot 1.65 \cdot \sqrt{1367.5(16.0 - 3.0)} \\
 &= 846.397 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2735.000	mm
Nozzle Outside Diameter	(d) =	406.400	mm
Vessel Thickness used	(T) =	29.000	mm



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 204 of 262

Nozzle Calcs.: N2-Tube outlet

Nozl: 14 10:16am

Sep 9, 2026

Nozzle Thickness used (t) = 18.412 mm
 T / t = 1.575
 d / t = 22.072

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.443$$

Nr/P = 0.078
 Mr/P = 0.122
 M0/P = 0.056
 N0/P = 0.142
 MrD/Mc = 0.231
 NrDL/Mc = 0.121
 M0d/Mc = 0.123
 N0DL/Mc = 0.119
 MrD/Ml = 0.150
 NrDL/Ml = 0.086
 M0D/Ml = 0.077
 N0DL/Ml = 0.223

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-14	-14	14	14
Outplane Bending (ML)	-30	30	30	-30
Normal Pressure Stress	47	48	47	48
Outplane Stress Summary	19	57	109	25

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	26	-26	26	-26
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-58	58	58	-58
Inplane Pressure Stress	23	23	23	23
Inplane Stress Summary	-11	53	117	-52

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 205 of 262

Nozzle Calcs.: N2-Tube outlet

Nozl: 14 10:16am Sep 9,2026

Two * Max Shear Stress	33	60	118	78
------------------------	----	----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	-7	-7	7	7
Outplane Bending (Mc)	-48	48	48	-48
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	23	23	23	23
Outplane Stress Summary	-15	57	97	-24

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	26	-26	26	-26
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-90	90	90	-90
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	47	48	47	48
Inplane Stress Summary	-22	108	175	-57

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	3	3	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	23	108	175	58

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-14	-14	14	14
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	13	11	13
Hoop Stress Summary	2	3	31	32



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 206 of 262

Nozzle Calcs.: N2-Tube outlet

Nozl: 14 10:16am

Sep 9, 2026

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	44	-44	44	-44
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-10	-10	10	10
Axial Bending (ML)	-104	104	104	-104
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-63	56	166	-130

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6
Shear Stress Summary	8	8	4	4

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	67	57	166	163

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-7	-7	7	7
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	13	11	13
Hoop Stress Summary	9	10	24	25

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	44	-44	44	-44
Axial Membrane (Mc)	-10	-10	10	10
Axial Bending (Mc)	-166	166	166	-166
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-125	118	228	-192

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 207 of 262

Nozzle Calcs.: N2-Tube outlet Nozl: 14 10:16am Sep 9,2026

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	6	6	6	6

Shear Stress Summary	4	4	8	8
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside
Two * Max Shear Stress	135	119	229	219

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 406.4 + 2 \cdot 1.65 \cdot \sqrt{1359.5(16.0 - 3.0)}$$

$$= 845.108 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2719.000 mm
Nozzle Outside Diameter	(d) =	926.400 mm
Vessel Thickness used	(T) =	13.000 mm
Nozzle Thickness used	(t) =	278.412 mm
T / t	=	0.047
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.927$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.047
Mr/P	=	0.071
MØ/P	=	0.019
NØ/P	=	0.022
MrD/Mc	=	0.239
NrDL/Mc	=	0.104
MØd/Mc	=	0.071
NØDL/Mc	=	0.045
MrD/Ml	=	0.048
NrDL/Ml	=	0.052
MØD/Ml	=	0.015
NØDL/Ml	=	0.022

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-12	12	12	-12
Normal Pressure Stress	106	107	106	107
Outplane Stress Summary	114	100	146	81

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 208 of 262

Nozzle Calcs.: N2-Tube outlet

Nozl: 14 10:16am

Sep 9,2026

Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	75	-75	75	-75
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-41	41	41	-41
Inplane Pressure Stress	53	53	53	53

Inplane Stress Summary	88	19	185	-48
------------------------	----	----	-----	-----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	2	2	0	0
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Two * Max Shear Stress	114	100	185	129
------------------------	-----	-----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-61	61	61	-61
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	53	53	53	53

Outplane Stress Summary	9	91	144	-18
-------------------------	---	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	75	-75	75	-75
Inplane Membrane (Mc)	-14	-14	14	14
Inplane Bending (Mc)	-205	205	205	-205
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	106	107	106	107

Inplane Stress Summary	-29	230	410	-150
------------------------	-----	-----	-----	------

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Shear (Vc)	0	0	0	0
---------------------	---	---	---	---



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 209 of 262

Nozzle Calcs.: N2-Tube outlet

Nozl: 14 10:16am

Sep 9,2026

Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	39	230	410	150

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	47	48	47	48	47	48	47	48
Circ. Pl (SUS)	-9	-9	19	19	-5	-5	9	9
Circ. Q (SUS)	-18	18	42	-42	-64	64	116	-116
Long. Pm (SUS)	23	23	23	23	23	23	23	23
Long. Pl (SUS)	-3	-3	7	7	-2	-2	12	12
Long. Q (SUS)	-32	32	84	-84	-36	36	60	-60
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	47.0	48.0	47.0	48.0	47.0	48.0	47.0	48.0
Pm+Pl (SUS)	38.1	39.1	66.0	67.0	42.0	43.0	56.0	57.0
Pm+Pl+Q (Total)	33.5	60.1	115.2	79.2	23.1	107.2	172.3	59.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	48.00	138.00	Passed
Pm+Pl (SUS)	67.03	207.00	Passed
Pm+Pl+Q (TOTAL)	172.32	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	106	107	106	107	106	107	106	107
Circ. Pl (SUS)	0	0	6	6	-6	-6	22	22
Circ. Q (SUS)	8	-8	32	-32	-130	130	280	-280
Long. Pm (SUS)	53	53	53	53	53	53	53	53
Long. Pl (SUS)	1	1	15	15	-3	-3	9	9
Long. Q (SUS)	34	-34	116	-116	-41	41	81	-81



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 210 of 262

Nozzle Calcs.: N2-Tube outlet

Nozl: 14 10:16am

Sep 9, 2026

Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	106.0	107.0	106.0	107.0	106.0	107.0	106.0	107.0
Pm+P1 (SUS)	106.0	107.0	112.0	113.0	100.0	101.0	128.0	129.0
Pm+P1+Q (Total)	114.2	99.1	184.0	129.0	39.0	231.0	408.0	151.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	107.00	138.00	Passed
Pm+P1 (SUS)	129.01	207.00	Passed
Pm+P1+Q (TOTAL)	408.02	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	13	11	13	11	13	11	13
Circ. P1 (SUS)	-9	-9	19	19	-2	-2	12	12
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. P1 (SUS)	-9	-9	11	11	-9	-9	11	11
Long. Q (SUS)	-60	60	148	-148	-122	122	210	-210
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	11.0	13.0	11.0	13.0	11.0	13.0	11.0	13.0
Pm+P1 (SUS)	6.3	8.2	30.1	32.1	13.2	15.1	23.1	25.1
Pm+P1+Q (Total)	67.5	56.9	164.2	164.3	135.4	118.2	226.2	219.4

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	13.00	118.00	Passed
Pm+P1 (SUS)	32.06	177.00	Passed
Pm+P1+Q (TOTAL)	226.24	353.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 211 of 262

Nozzle Calcs.: N3-Shell inlet

Nozl: 15 10:16am Sep 9,2026

Input, Nozzle Desc: N3-Shell inlet

From: 70

Pressure for Reinforcement Calculations	P	1000.000	KPa.
Temperature for Internal Pressure	Temp	240	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2700.00	mm
Shell Finished (Minimum) Thickness	t	22.5000	mm
Shell Internal Corrosion Allowance	c	0.8000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5869.70	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.40	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm
Flange Material		SA-266 2	
Hub Height of Integral Nozzle	h	124.5000	mm
Height of Beveled Transition	L'	104.5000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		210.0000	mm
Corrosion Allowance	can	0.8000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was Design Pressure + static head.



PV Elite 28 Licensee: SL User

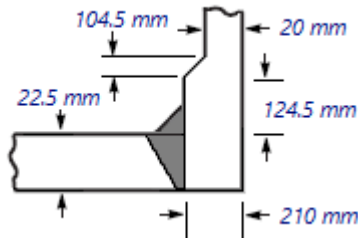
FileName : 197-C_Channel MOC= CS

Page 212 of 262

Nozzle Calcs.: N3-Shell inlet

Nozl: 15 10:16am

Sep 9,2026



Reinforcement CALCULATION, Description: N3-Shell inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	36.000 in.
Actual Thickness Used in Calculation	0.787 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1000 \cdot 1351 / (138 \cdot 1 - 0.6 \cdot 1000)$
 $= 9.8319 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1000 \cdot 458 / (136 \cdot 1 - 0.6 \cdot 1000)$
 $= 3.3728 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 190 / 0.5773$
 $= 329.0898 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1832.0000 mm
Parallel to Vessel Wall, opening length	d	916.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	54.2500 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 136 / 138)$
 $= 0.988$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 136 / 138)$
 $= 0.988$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.99, 1)$
 $= 0.988$



PV Elite 28 Licensee: SL User

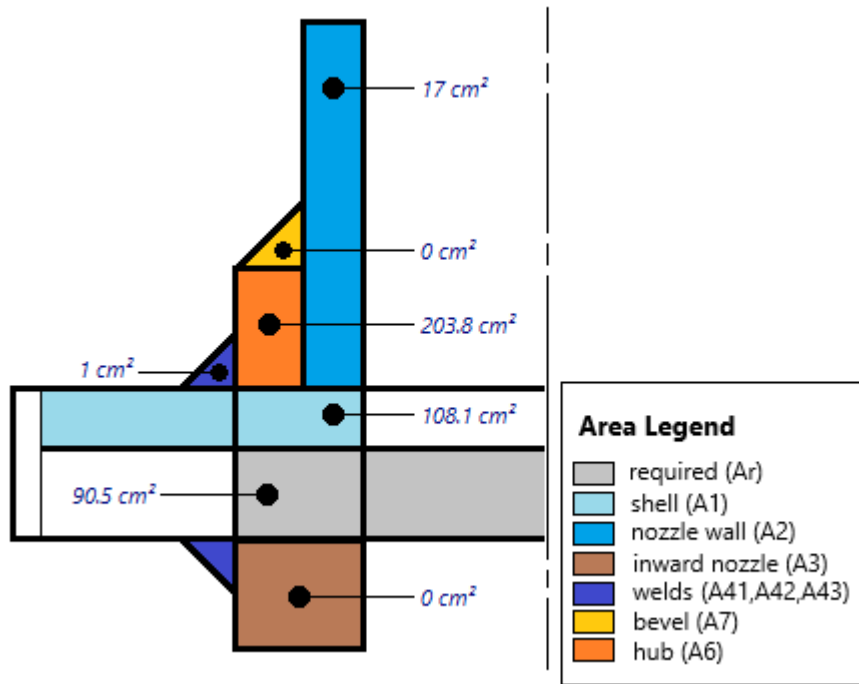
FileName : 197-C_Channel MOC= CS

Page 213 of 262

Nozzle Calcs.: N3-Shell inlet

Nozl: 15 10:16am

Sep 9, 2026



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	90.537	NA	NA
Area in Shell	A1	108.136	NA	NA
Area in Nozzle Wall	A2	16.973	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.988	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	203.761	NA	NA
TOTAL AREA AVAILABLE	Atot	329.859	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (916 \cdot 9.83 \cdot 1 + 2 \cdot 209 \cdot 9.83 \cdot 1 \cdot (1 - 0.99))$$

$$= 90.537 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 916(1 \cdot 21.7 - 1 \cdot 9.83) - 2 \cdot 209$$

$$(1 \cdot 21.7 - 1 \cdot 9.83) \cdot (1 - 0.99)$$

$$= 108.136 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 54.2)(19.2 - 3.37)0.99$$

$$= 16.973 \text{ cm}^2$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 214 of 262

Nozzle Calcs.: N3-Shell inlet

Nozl: 15 10:16am

Sep 9, 2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 0.99 + (0)^2 \cdot 0.99$$

$$= 0.988 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(54.2, 400, 125)) \cdot (210 - 20) \cdot 0.99$$

$$= 203.761 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.1728 mm
Wall Thickness per UG16(b),	tr16b = 2.3000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 10.6319 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 10.6319 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.3000 mm
Wall Thickness per table UG-45	tb3 = 9.1312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[9.13, \max(10.6, 2.3)]$$

$$= 9.1312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.17, 9.13)$$

$$= 9.1312 \text{ mm}$$

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 20, tr = 3.33, c = 0.8 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.17, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 22.5, tr = 9.83, c = 0.8 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.45, Temp. Reduction = 39 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -43 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

$$\text{Design Pressure/Ambient Rating} = 1000.00/5100.00 = 0.196$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:



PV Elite 28 Licensee: SL User
 FileName : 197-C_Channel MOC= CS Page 215 of 262
 Nozzle Calcs.: N3-Shell inlet Nozl: 15 10:16am Sep 9, 2026

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 190 / 0.5773$$

$$= 329.0898 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1374.0000 mm
 Parallel to Vessel Wall, opening length d 687.0000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 54.2500 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	60.361	60.361
Area Available in Shell	A1	53.780	53.780
Area Available in Nozzle Wall	A2	16.973	16.973
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	0.988	0.988
Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	203.761	203.761
TOTAL AREA AVAILABLE	Atot	275.503	275.503

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: N3-Shell inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (90.5 - 108 + 2 \cdot 209 \cdot 0.99 \cdot (1 \cdot 21.7 - 9.83)) \cdot 138)$$

$$= 44299.09 \text{ Kgf}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A4 - (W_i - \text{Can} / .707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (221 + 0.99 - 0 \cdot 0.99) \cdot 138$$

$$= 311999.00 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (221 + 0 + 0.99 + (89.7)) \cdot 138$$

$$= 438278.22 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (221 + 0 + 0.99 + 0 + (89.7)) \cdot 138$$

$$= 438278.22 \text{ Kgf}$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 216 of 262

Nozzle Calcs.: N3-Shell inlet

Nozl: 15 10:16am

Sep 9, 2026

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 1334 \cdot 10 \cdot 0.49 \cdot 136$$

$$= 142851. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 563) \cdot (210 - 0.8) \cdot 0.7 \cdot 136$$

$$= 3599894. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 1334 \cdot (14 - 0) \cdot 0.74 \cdot 138$$

$$= 305568. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (142851 + 3599894) = 3742744 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (142851 + 0 + 305568 + 0) = 448419 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (142851 + 305568 + 0) = 448419 \text{ Kg}$$

Summary of Failure Path Calculations:

$$\text{Path 1-1} = 3742744 \text{ Kg}, \text{ must exceed } W = 44299 \text{ Kg} \quad \text{or } W1 = 311999 \text{ Kg}$$

$$\text{Path 2-2} = 448418 \text{ Kg}, \text{ must exceed } W = 44299 \text{ Kg} \quad \text{or } W2 = 438278 \text{ Kg}$$

$$\text{Path 3-3} = 448418 \text{ Kg}, \text{ must exceed } W = 44299 \text{ Kg} \quad \text{or } W3 = 438278 \text{ Kg}$$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1840 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 176.3969 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 598.8969 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 217 of 262

Nozzle Calcs.: N4-Shell outlet

Nozl: 16 10:16am

Sep 9,2026

Input, Nozzle Desc: N4-Shell outlet

From: 70

Pressure for Reinforcement Calculations	P	1026.465	KPa.
Temperature for Internal Pressure	Temp	240	°C
Parent Material		SA-516	70
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2700.00	mm
Shell Finished (Minimum) Thickness	t	22.5000	mm
Shell Internal Corrosion Allowance	c	0.8000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5869.70	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266	2
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.40	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm
Flange Material		SA-266	2
Hub Height of Integral Nozzle	h	120.0000	mm
Height of Beveled Transition	L'	109.6970	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		210.0000	mm
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was Design Pressure + static head.



PV Elite 28 Licensee: SL User

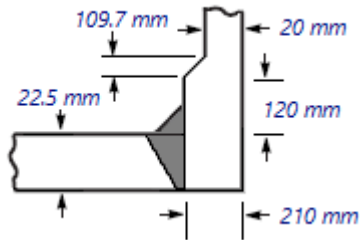
FileName : 197-C_Channel MOC= CS

Page 218 of 262

Nozzle Calcs.: N4-Shell outlet

Nozl: 16 10:16am

Sep 9,2026



Reinforcement CALCULATION, Description: N4-Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	36.000 in.
Actual Thickness Used in Calculation	0.787 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1026 \cdot 1351 / (138 \cdot 1 - 0.6 \cdot 1026)$
 $= 10.0933 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1026 \cdot 460 / (136 \cdot 1 - 0.6 \cdot 1026)$
 $= 3.4791 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 190 / 0.5773$
 $= 329.0898 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1840.8000 mm
Parallel to Vessel Wall, opening length	d	920.4000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	54.2500 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 136 / 138)$
 $= 0.988$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 136 / 138)$
 $= 0.988$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.99, 1)$
 $= 0.988$



PV Elite 28 Licensee: SL User

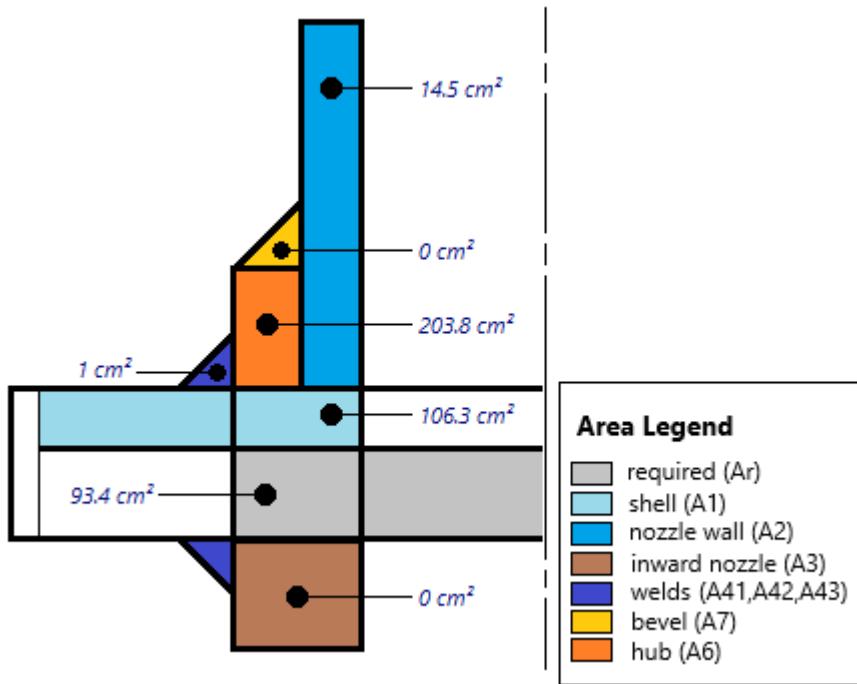
FileName : 197-C_Channel MOC= CS

Page 219 of 262

Nozzle Calcs.: N4-Shell outlet

Nozl: 16 10:16am

Sep 9, 2026



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	93.383	NA	NA
Area in Shell	A1	106.271	NA	NA
Area in Nozzle Wall	A2	14.500	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.988	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	203.761	NA	NA
TOTAL AREA AVAILABLE	Atot	325.521	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \cdot UG-37(c)$$

$$= (920 \cdot 10.1 \cdot 1 + 2 \cdot 207 \cdot 10.1 \cdot 1 \cdot (1 - 0.99))$$

$$= 93.383 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 920(1 \cdot 21.7 - 1 \cdot 10.1) - 2 \cdot 207$$

$$(1 \cdot 21.7 - 1 \cdot 10.1) \cdot (1 - 0.99)$$

$$= 106.271 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 54.2)(17 - 3.48)0.99$$

$$= 14.500 \text{ cm}^2$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 220 of 262

Nozzle Calcs.: N4-Shell outlet

Nozl: 16 10:16am

Sep 9, 2026

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10^2 \cdot 0.99 + (0)^2 \cdot 0.99$$

$$= 0.988 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(54.2, 400, 120)) \cdot (210 - 20) \cdot 0.99$$

$$= 203.761 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 6.4791 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 10.8933 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 10.8933 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.3, \max(10.9, 4.5)]$$

$$= 10.8933 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(6.48, 10.9)$$

$$= 10.8933 \text{ mm}$$

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 26.2,	Allowable	: 136.4 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 316.8 N./mm ²	Passed
Occasional	: 13.6,	Allowable	: 181.4 N./mm ²	Passed
Shear	: 9.7,	Allowable	: 95.5 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 20, tr = 3.44, c = 3 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.2, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 22.5, tr = 10.1, c = 0.8 mm, E* = 1
Thickness Ratio = tr * E*/(tg - c) = 0.47, Temp. Reduction = 37 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-41 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -41 °C

Flange MDMT including Temperature reduction per UCS-66.1:



PV Elite 28 Licensee: SL User
 FileName : 197-C_Channel MOC= CS Page 221 of 262
 Nozzle Calcs.: N4-Shell outlet Nozl: 16 10:16am Sep 9,2026

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 1026.47/5100.00 = 0.201

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 = (Hub Thickness - Neck Thickness) / tan(30)
 = 190/0.5773
 = 329.0898 mm

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1380.6000 mm
 Parallel to Vessel Wall, opening length d 690.3000 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 54.2500 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure] AR		62.258	62.258
Area Available in Shell A1		52.857	52.857
Area Available in Nozzle Wall A2		14.500	14.500
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		0.988	0.988
Area Available in Pad A5		0.000	0.000
Area Available in Hub A6		203.761	203.761
TOTAL AREA AVAILABLE Atot		272.107	272.107

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: N4-Shell outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 = max(0, (A-A1+2*tn*fr1*(E1*t-tr))Sv)
 = max(0, (93.4 - 106 + 2 * 207 * 0.99 *
 (1 * 21.7 - 10.1))138)
 = 48696.45 Kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 = (A2+A4-(Wi-Can/.707)*fr2)*Sv
 = (218 + 0.99 - 0 * 0.99) * 138
 = 308518.62 Kgf



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 222 of 262

Nozzle Calcs.: N4-Shell outlet

Nozl: 16 10:16am

Sep 9, 2026

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (218 + 0 + 0.99 + (88.8)) \cdot 138$$

$$= 433469.84 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (218 + 0 + 0.99 + 0 + (88.8)) \cdot 138$$

$$= 433469.84 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.14/2.0) \cdot 1334 \cdot 10 \cdot 0.49 \cdot 136$$

$$= 142851. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.14 \cdot 564) \cdot (210 - 3) \cdot 0.7 \cdot 136$$

$$= 3569001. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.14/2.0) \cdot 1334 \cdot (14 - 0) \cdot 0.74 \cdot 138$$

$$= 305568. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (142851 + 3569001) = 3711851 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (142851 + 0 + 305568 + 0) = 448419 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (142851 + 305568 + 0) = 448419 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 3711851 Kgf, must exceed W = 48696 Kgf or W1 = 308518 Kgf
 Path 2-2 = 448418 Kgf, must exceed W = 48696 Kgf or W2 = 433469 Kgf
 Path 3-3 = 448418 Kgf, must exceed W = 48696 Kgf or W3 = 433469 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1840 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 176.3969 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 598.8969 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N4-Shell outle:

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	0.8000 mm
Shell External Corrosion Allowance	0.0000 mm



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 223 of 262

Nozzle Calcs.: N4-Shell outlet Nozl: 16 10:16am Sep 9,2026

Shell Diameter	2700.000	mm
Shell Thickness	22.5000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	914.400	mm
Nozzle Thickness	210.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 136.40	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1026.47	KPa.
Radial Load	P	-6883.56	Kgf
Circumferential Shear	Vc	4589.03	Kgf
Longitudinal Shear	Vl	4589.03	Kgf
Circumferential Moment	Mc	11559.91	Kg-m.
Longitudinal Moment	Ml	11559.91	Kg-m.
Torsional Moment	Mt	16514.19	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_n/T_n (6.354) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 1334.4 + 2 \cdot 1.65 \cdot \sqrt{1361.65(22.5 - 0.8)}$$

$$= 1901.652 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2723.300	mm
Nozzle Outside Diameter	(d) =	1334.400	mm
Vessel Thickness used	(T) =	21.700	mm
Nozzle Thickness used	(t) =	207.000	mm
T / t	=	0.105	
d / t	=	10.000	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 5.489$$

Nr/P	=	0.044
Mr/P	=	0.070
M0/P	=	0.020
N0/P	=	0.031
MrD/Mc	=	0.240
NrDL/Mc	=	0.094



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 224 of 262

Nozzle Calcs.: N4-Shell outlet

Nozl: 16 10:16am Sep 9,2026

MØd/Mc = 0.071
 NØDL/Mc = 0.057
 MrD/Ml = 0.048
 NrDL/Ml = 0.042
 MØD/Ml = 0.015
 NØDL/Ml = 0.028

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-16	16	16	-16
Normal Pressure Stress	63	64	63	64
Outplane Stress Summary	63	62	106	40

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	60	-60	60	-60
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-51	51	51	-51
Inplane Pressure Stress	31	31	31	31
Inplane Stress Summary	39	21	157	-66

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	64	63	157	107

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	-10	-10	10	10
Outplane Bending (Mc)	-76	76	76	-76
Outplane Membrane (ML)	0	0	0	0



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 225 of 262

Nozzle Calcs.: N4-Shell outlet

Nozl: 16 10:16am

Sep 9, 2026

Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	31	31	31	31

Outplane Stress Summary	-33	85	140	-47
-------------------------	-----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	60	-60	60	-60
Inplane Membrane (Mc)	-16	-16	16	16
Inplane Bending (Mc)	-260	260	260	-260
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	63	64	63	64

Inplane Stress Summary	-147	253	407	-232
------------------------	------	-----	-----	------

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	147	253	407	232
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-5	-5	5	5
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	3	2	3

Hoop Stress Summary	1	2	11	12
---------------------	---	---	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 226 of 262

Nozzle Calcs.: N4-Shell outlet

Nozl: 16 10:16am

Sep 9,2026

Axial Stress Summary	0	0	2	1
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	1	2	11	12
------------------------	---	---	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-10	-10	10	10
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	2	3	2	3

Hoop Stress Summary	-3	-2	17	18
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-2	2	2	-2
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-1	2	4	0
----------------------	----	---	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
----------------------	---	---	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	3	4	17	19
------------------------	---	---	----	----



WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	63	64	63	64	63	64	63	64
Circ. Pl (SUS)	-1	-1	9	9	-10	-10	22	22
Circ. Q (SUS)	1	-1	33	-33	-200	200	320	-320
Long. Pm (SUS)	31	31	31	31	31	31	31	31
Long. Pl (SUS)	-1	-1	13	13	-6	-6	14	14
Long. Q (SUS)	9	-9	111	-111	-59	59	93	-93
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	63.0	64.0	63.0	64.0	63.0	64.0	63.0	64.0
Pm+Pl (SUS)	62.0	63.0	72.0	73.0	53.0	54.0	85.0	86.0
Pm+Pl+Q (Total)	63.2	62.1	155.1	107.1	147.0	254.0	405.0	234.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	64.00	138.00	Passed
Pm+Pl (SUS)	86.00	207.00	Passed
Pm+Pl+Q (TOTAL)	405.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	2	3	2	3	2	3	2	3
Circ. Pl (SUS)	-1	-1	9	9	-6	-6	14	14
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	0	0	0	0	-2	2	2	-2
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	2.0	3.0	2.0	3.0	2.0	3.0	2.0	3.0
Pm+Pl (SUS)	1.0	2.0	11.0	12.0	5.0	4.0	16.0	17.0
Pm+Pl+Q (Total)	1.0	2.0	11.0	12.0	4.0	6.0	16.0	18.0

Nozzle Stress Summation Comparison (N./mm²):



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 228 of 262

Nozzle Calcs.: N4-Shell outlet

Nozl: 16 10:16am

Sep 9, 2026

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	136.40	Passed
Pm+Pl (SUS)	17.00	204.60	Passed
Pm+Pl+Q (TOTAL)	18.00	409.20	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 229 of 262

Nozzle Calcs.: Drain

Nozl: 17 10:16am Sep 9,2026

Input, Nozzle Desc: Drain

From: 70

Pressure for Reinforcement Calculations	P	1026.465	KPa.
Temperature for Internal Pressure	Temp	240	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2700.00	mm
Shell Finished (Minimum) Thickness	t	22.5000	mm
Shell Internal Corrosion Allowance	c	0.8000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4869.70	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.40	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		XXS	
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.5000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	162.0000	mm
Thickness of Pad	te	24.0000	mm
Weld leg size between Pad and Shell	Wp	18.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	24.0000	mm
Reinforcing Pad Width		50.8375	mm
Flange Class		2500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



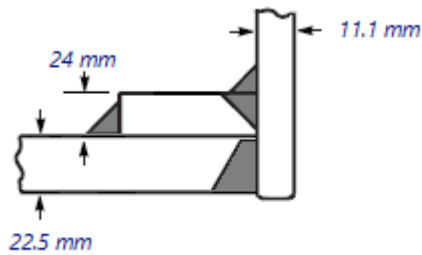
PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 230 of 262

Nozzle Calcs.: Drain

Nozl: 17 10:16am Sep 9,2026



Reinforcement CALCULATION, Description: Drain

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.503 in.
Actual Thickness Used in Calculation	0.436 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1026 \cdot 1351 / (138 \cdot 1 - 0.6 \cdot 1026)$
 $= 10.0933 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1026 \cdot 22.1 / (136 \cdot 1 - 0.6 \cdot 1026)$
 $= 0.1670 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	103.7250 mm
Parallel to Vessel Wall	Rn+tn+t	51.8625 mm
Normal to Vessel Wall (Thickness Limit), pad side TLwp		44.1860 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

*Taking a UG-36(c)(3)(a) exemption for nozzle: Drain.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.1670 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 10.8933 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 10.8933 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(10.9, 4.5)]$
 $= 6.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.17, 6.42)$
 $= 6.4200 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 11.074 = 9.690 \text{ mm} \rightarrow \text{OK}$



PV Elite 28 Licensee: SL User
 FileName : 197-C_Channel MOC= CS Page 231 of 262
 Nozzle Calcs.: Drain Nozl: 17 10:16am Sep 9,2026

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Pad Weld for the Nozzle, Curve: A

Govrn. thk, tg = 9.69, tr = 0.17, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.69, tr = 0.17, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 22.5, tr = 10.1, c = 0.8 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.47$, Temp. Reduction = 37 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-41 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 9.69, tr = 0.17, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 0.025$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -41 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -41 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1026.47 / 42509.99 = 0.024$

Weld Size Calculations, Description: Drain

Intermediate Calc. for nozzle/shell Welds	Tmin	8.0744 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.6521 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$9.5000 = 0.5 * T_{minPad}$	$12.7260 = 0.7 * W_p$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.3370 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 422.8370 mm



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 232 of 262

Nozzle Calcs.: Drain

Nozl: 17 10:16am Sep 9, 2026

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 233 of 262

Nozzle Calcs.: Vent

Nozl: 18 10:16am Sep 9,2026

Input, Nozzle Desc: Vent

From: 70

Pressure for Reinforcement Calculations	P	1000.000	KPa.
Temperature for Internal Pressure	Temp	240	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2700.00	mm
Shell Finished (Minimum) Thickness	t	22.5000	mm
Shell Internal Corrosion Allowance	c	0.8000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4869.70	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	136.40	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		XXS	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	22.5000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	162.0000	mm
Thickness of Pad	te	24.0000	mm
Weld leg size between Pad and Shell	Wp	18.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	24.0000	mm
Reinforcing Pad Width		50.8375	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



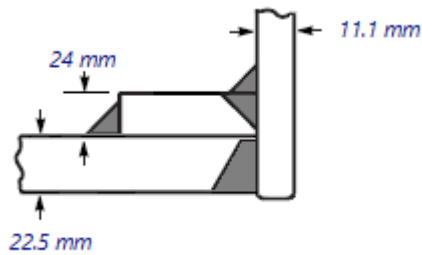
PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 234 of 262

Nozzle Calcs.: Vent

Nozl: 18 10:16am Sep 9,2026



Reinforcement CALCULATION, Description: Vent

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.503 in.
Actual Thickness Used in Calculation	0.436 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1000 \cdot 1351 / (138 \cdot 1 - 0.6 \cdot 1000)$
 $= 9.8319 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1000 \cdot 22.1 / (136 \cdot 1 - 0.6 \cdot 1000)$
 $= 0.1627 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	103.7250 mm
Parallel to Vessel Wall	Rn+tn+t	51.8625 mm
Normal to Vessel Wall (Thickness Limit), pad side TLwp		44.1860 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

*Taking a UG-36(c)(3)(a) exemption for nozzle: Vent.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.1627 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 10.6319 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 10.6319 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 6.4200 mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[6.42, \max(10.6, 4.5)]$
 $= 6.4200 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.16, 6.42)$
 $= 6.4200 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 \cdot 11.074 = 9.690 \text{ mm} \rightarrow \text{OK}$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 235 of 262

Nozzle Calcs.: Vent

Nozl: 18 10:16am Sep 9, 2026

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 9.69, tr = 0.16, c = 3 mm, E* = 1

Thickness Ratio = $tr * E* / (tg - c) = 0.024$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: A

Govrn. thk, tg = 9.69, tr = 0.16, c = 3 mm, E* = 1

Thickness Ratio = $tr * E* / (tg - c) = 0.024$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.69, tr = 0.16, c = 3 mm, E* = 1

Thickness Ratio = $tr * E* / (tg - c) = 0.024$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 22.5, tr = 9.83, c = 0.8 mm, E* = 1

Thickness Ratio = $tr * E* / (tg - c) = 0.45$, Temp. Reduction = 39 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 9.69, tr = 0.16, c = 3 mm, E* = 1

Thickness Ratio = $tr * E* / (tg - c) = 0.024$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -43 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -43 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $1000.00 / 1960.01 = 0.510$

Weld Size Calculations, Description: Vent

Intermediate Calc. for nozzle/shell Welds	Tmin	8.0744 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

Required Thickness	Actual Thickness
--------------------	------------------



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 236 of 262

Nozzle Calcs.: Vent

Nozl: 18 10:16am Sep 9, 2026

Nozzle Weld $5.6521 = 0.7 * t_{min}$. $7.0700 = 0.7 * W_o$ mm

Pad Weld $9.5000 = 0.5 * T_{minPad}$ $12.7260 = 0.7 * W_p$ mm

[Skipping the nozzle attachment weld strength calculations. Per UW-15\(b\)\(2\) the nozzles exempted by UG-36\(c\)\(3\)\(a\) \(small nozzles\) do not require a weld strength check.](#)

The Drop for this Nozzle is : 0.3370 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 422.8370 mm

[PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.](#)



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 237 of 262

ASME TS Calc:

Case: 1 10:16a Sep 9,2026

Input Echo, Tubesheet Number 1, Description: Fixed Tubeheet

Shell Data:

Main Shell Description: Shell-1

Shell Maximum Design Pressure	Psd,max	1000.00 KPa.
Shell Maximum Operating Pressure	Psox,max	1000.00 KPa.
Shell Minimum Operating Pressure	Psox,min	103.42 KPa.
Shell Thickness	ts	16.0000 mm
Shell Internal Corrosion Allowance	cas	0.8000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2700.000 mm
Shell Temperature for Internal Pressure	Ts	240.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	1000.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	1000.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	16.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2700.000 mm
Channel Design Temperature	TEMPC	82.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	4924
Tube Wall Thickness	et	1.6510 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	6550.00 mm
Straight Tube Length (bet. inner tubsht faces) L		6270.00 mm
Design Temperature of the Tubes		240.00 °C
Tube Material		SA-789
Tube Material UNS Number		S32205
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		170.80 N./mm ²
Tube Allowable Stress At Ambient		187.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	372.40 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.6510 mm
Groove Weld Leg	ag	1.6510 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	1207.750 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 238 of 262

ASME TS Calc:

Case: 1 10:16a Sep 9,2026

Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	42498.0 cm ²
Perimeter of Tube Layout	Cp	7703.8 mm

Tubesheet Data:

Tubesheet TYPE: Stationary, Gasketed both Sides, Conf. d
Exchanger TYPE: With an Immersed Floating Head, Conf. a

Tubesheet Design Metal Temperature	T	240.00 °C
Tubesheet Material	SA-182 F60	
Tubesheet Material UNS Number	S32205	
Tubesheet Allowable Stress at Temperature	S	170.80 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	187.00 N./mm ²
Thickness of Tubesheet	h	140.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	2806.000 mm
Dimension G for the Channel Side	Gc	2763.285 mm
Area of the Untubed Lanes	AL	428.6 cm ²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	1068.500 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Channel Side Min. Design Pressure	Ptd,min	-103.0000 KPa.
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side	Both	
Flange Outside Diameter	A	3000.000 mm
Flange Inside Diameter	B	2710.000 mm
Flange Face Outside Diameter	Fod	2809.000 mm
Flange Face Inside Diameter	Fid	2710.000 mm
Gasket Outside Diameter	Go	2780.000 mm
Gasket Inside Diameter	Gi	2736.000 mm
Small end Hub thk.	g0	14.5000 mm
Large end Hub thk.	g1	21.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch	Code Sketch 1a	
Gasket Thickness	tg	6.3500 mm
Full face Gasket Flange Option	Program Selects	

Bolting Information:

Diameter of Bolt Circle	C	2900.000 mm
Nominal Bolt Diameter	dB	47.6250 mm
Thread Series	TEMA Thread Series	
Number of Bolts	n	56
Bolt Material	SA-193 B7	
Bolt Material UNS Number	G41400	
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm
Alternate Flange Operating Bolt Load, Wm1		673310.31 KgF
Alternate Flange Seating Bolt Load, Wm2		510110.72 KgF
Alternate Flange Design Bolt Load, W		1459922.62 KgF

Note: The full bolt load Sa*Ab, was used for all load cases.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 239 of 262

ASME TS Calc:

Case: 1 10:16a Sep 9,2026

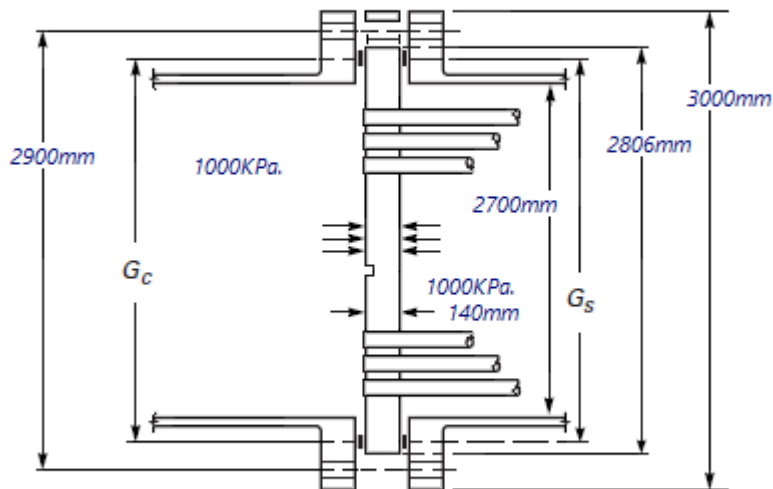
Tubesheet Integral with Tubesheet Extended as Flange

None
No

Is Exchanger in Creep range (skip EP, Use 3S for Sps)

No

Tubesheet Gasketed w/Shell and Channel



Configuration d:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	240.0 °C	0.18971E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	82.0 °C	0.19921E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Low Alloy Steels, Group 2 Group 1
Tubes - TM-1 Material Group H

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000125990 /°C
Elastic Mod. at Design Temperature	240.0 °C	0.18351E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20179E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	240.0 °C	0.18351E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19994E+09 KPa.

TubeSheet - TE-1 Low Alloy Steels, Group 2 Group 1
TubeSheet - TM-1 Material Group H



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 240 of 262

ASME TS Calc:

Case: 1 10:16a Sep 9,2026

Elastic Mod. at Design Temperature	240.0 °C	0.18351E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19994E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (1000 \cdot (25.4/2 - 0)) / (171 \cdot 1 + 0.4 \cdot 1000)$
 $= 0.0742 + 0.0000 = 0.0742 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	HA-5	at	240.00 °C
Elastic Modulus for Material			180060896.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6510	25.40	6270.00	15.38	50.0000	0.0046475	136.46

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 136) / (3 \cdot 15.4) = 11826 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.5154	25.40	6270.00	49.28	50.0000	0.0004529	40.77

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 40.8) / (3 \cdot 49.3) = 1103 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.5154 mm
Allowable Internal Pressure at Corroded thickness	23420.38 KPa.
Required Internal Design Pressure	1000.00 KPa.
Allowable External Pressure at Corroded thickness	11825.91 KPa.
Required External Design Pressure	1103.00 KPa.
Required Thickness due to Shell Side pressure	0.5154 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Gasketed Tubesheets:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	22.000 mm
Basic Gasket Width,	$b_0 = N / 2$	11.000 mm
Effective Gasket Width,	$b = \text{sqrt}(b_0) \cdot 2.5$	8.355 mm
Gasket Reaction Diameter,	$G = G_o - 2.0 \cdot b$	2763.291 mm

Results for ASME Stationary Tubesheet Calculations for Configuration d:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.9.4 Step 1:

Tube expansion depth ratio $[\rho]$:

$= l_{tx} / h$ (modified for corrosion if present)
 $= 0/140$
 $= 0.0000$ (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter $[d^*]$:



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 241 of 262

ASME TS Calc: Case: 1 10:16a Sep 9, 2026

$$\begin{aligned}
 &= \max(dt - 2 \cdot tt(Et/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt) \\
 &= \max(25.4 - 2 \cdot 1.65 (183508208/183508208) \cdot \\
 &\quad (171/171) \cdot (0); 25.4 - 2 \cdot 1.65) \\
 &= 25.4000 \text{ mm}
 \end{aligned}$$

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned}
 &= 2 \cdot ro + dt \\
 &= 2 \cdot 1208 + 25.4 \\
 &= 2440.900 \text{ mm}
 \end{aligned}$$

Determine the basic ligament efficiency for shear [μ]:

$$\begin{aligned}
 &= (p - dt) / p \\
 &= (31.8 - 25.4) / 31.8 \\
 &= 0.200
 \end{aligned}$$

Equivalent outer tube limit radius [ao]:

$$\begin{aligned}
 &= Do / 2 \\
 &= 2441/2 \\
 &= 1220.4500 \text{ mm}
 \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned}
 &= p / \sqrt{ 1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2) } \\
 &= 31.8 / \sqrt{ 1 - 4 \cdot \min(429 \cdot 100 ; 4 \cdot 2441 \cdot 31.8) / (\pi \cdot 2441^2) } \\
 &= 31.8964 \text{ mm}
 \end{aligned}$$

Effective ligament efficiency for bending [μ *]:

$$\begin{aligned}
 &= (p* - d*) / p* \\
 &= (31.9 - 25.4) / 31.9 \\
 &= 0.2037
 \end{aligned}$$

Ratio [ps]:

$$\begin{aligned}
 &= as / ao \\
 &= 1382/1220 \\
 &= 1.1321
 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned}
 &= ac / ao \\
 &= 1382/1220 \\
 &= 1.1321
 \end{aligned}$$

Compute parameter [xt]:

$$\begin{aligned}
 &= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2 \\
 &= 1 - 4924 ((25.4 - 2 \cdot 1.65) / (2 \cdot 1220))^2 \\
 &= 0.5964
 \end{aligned}$$

Determine parameter [xs]:

$$\begin{aligned}
 &= 1 - Nt(dt / (2 \cdot ao))^2 \\
 &= 1 - 4924 (25.4 / (2 \cdot 1220))^2 \\
 &= 0.4668
 \end{aligned}$$

Determine the value [h'g]:

$$\begin{aligned}
 &= \max((hg - CATC); 0) \text{ (For pressure only cases)} \\
 &= \max((8 - 0); 0) \\
 &= 8.000 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.9.4 Step 2:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

β_s , k_s , λ_s , δ_s are all = 0

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

β_C , k_C , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.9.4 Step 3:



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 242 of 262

ASME TS Calc:

Case: 1 10:16a Sep 9, 2026

$h/p = 4.409449$; $\mu^* = 0.203672$; $E^* = 29304318$ KPa.
 $E^*/E = 0.159689$; $\nu^* = 0.453250$

Note: As h/p (4.409) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [Xa]:

$$= \left(\left(24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2 \right) / (E^* \cdot L \cdot H^3) \right)^{1/4}$$

$$= \left(\left(24(1 - 0.45^2)4924 \cdot 183508208 \cdot 1.65 \right. \right. \\ \left. \left. (25.4 - 1.65)1220^2 \right) / (29304318 \cdot 6270 \cdot 140^3) \right)^{1/4}$$

$$= 6.6841$$

Values from Table 4.18.3

$Z_d = 0.004960$; $Z_v = 0.022184$; $Z_m = 0.211940$
 $Z_a = 0.198049E+03$; $Z_w = 0.022184$

ASME Section VIII-2 Part 4.18.9.4 Step 4:

Diameter ratio [K]:

$$= A / Do$$

$$= 2806/2441$$

$$= 1.1496$$

Compute coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.45) / 29304318 \cdot (0 + 0 + 183508208 \cdot \ln(1.15))$$

$$= 0.4773$$

Compute parameter [ϕ]:

$$= (1 + \nu^*)^F$$

$$= (1 + 0.45)^{0.48}$$

$$= 0.6936$$

Compute parameter [Q1]:

$$= (ps - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.13 - 1 - 0.69 \cdot 0.022) / (1 + 0.69 \cdot 0.21)$$

$$= 0.101735$$

ASME Section VIII-2 Part 4.18.9.4 Step 5:

Determine factor [γ_b]:

$$= (G_c - G_s) / Do \text{ (config d)}$$

$$= (2763 - 2763) / 2441 = 0$$

Calculate parameter [ω_s]:

$$= ps \cdot ks \cdot \beta_s \cdot \delta_s(1 + h \cdot \beta_s)$$

$$= 1.13 \cdot 0 \cdot 0 \cdot 0.000(1 + 140 \cdot 0)$$

$$= 0.0000 \text{ mm}^2$$

Calculate parameter [ω^*s]:

$$= ao^2(ps^2 - 1) \cdot (ps - 1) / 4 - \omega_s$$

$$= 1220^2(1.13^2 - 1) \cdot (1.13 - 1) / 4 - 0$$

$$= 13849.4170 \text{ mm}^2$$

Parameter [ω_c]:

$$= pc \cdot kc[\delta C(CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) - \\ \zeta c(CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2)]$$

$$= 1.13 \cdot 0 [0.000(0 + 0 \cdot 0 \cdot 140) - 0.000(0/140 + 0 \cdot 0/2)]$$

$$= 0.0000 \text{ mm}^2$$

Parameter [ω^*c]:

$$= ao^2[(pc^2 + 1) \cdot (pc - 1) / 4 - (ps - 1) / 2] - \omega_c$$

$$= 1220^2[(1.13^2 + 1) \cdot (1.13 - 1) / 4 - (1.13 - 1) / 2] - 0$$

$$= 13849.4150 \text{ mm}^2$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 243 of 262

ASME TS Calc:

Case: 1 10:16a Sep 9, 2026

Pressure [P*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.9.4 Step 6:

$P_e = P_s - P_t$

= 1000 - 1000 = 0 KPa.

ASME Section VIII-2 Part 4.18.9.4 Step 7:

Determine factor [Q2]:

$= [((\omega * s * P_s - \omega * c * P_t) - (\omega_s * P * s - \omega_c * P * c)) +$

$W * \gamma_b / (2 * \pi)] / (1 + \phi * Z_m)$

$= [((13849 * 1000 - 13849 * 1000) -$

$(0 * 0 - 0 * 0)) * 0.0001 +$

$1459922 * 0 / (2 * \pi)] / (1 + 0.69 * 0.21)$

= 0.0002 Kgf

Tubesheet Bending Stress - original thickness [σ]:

$= 6 * Q2 / (\mu * (H - h'g)^2)$

$= 6 * 0.00019 / (0.2 * (140 - 8)^2)$

= 0.0000 N./mm²

Allowable Tubesheet Bending Stress [σ allowed]:

= 1.5 * S = 1.5 * 171 = 256 N./mm²

Could not converge on a min. required tubesheet thk. to satisfy the Bending Stress criteria.



Use the best Estimated Required Tubesheet Thickness value as design guidance & Compare the Actual Stresses versus Allowable Stresses.

Tubesheet thickness = 19.0133 mm

Tubesheet Bending Stress = 0.0005 N./mm²

ASME Section VIII-2 Part 4.18.9.4 Step 8:

Shear Stress check [τ limit]:

$= (2 * \mu * h / a_o) * \min(0.8S; 0.533S_y)$

$= (2 * 0.2 * 140 / 1220) * 137$

= 6.2696 N./mm²

Shear Stress is not required to be computed; [P_e] <= T_{τ} limit

Tubesheet Shear Stress - final thickness [τ_f]:

$= (A_p * P_e) / (\mu * h * C_p)$

$= (42498 * 0) / (0.2 * 19 * 7704)$

= 0.0000 N./mm²

Reqd Tubesheet Thickness, for Shear Stress (including CA) [H_{reqS}]:

= H + Cats + Catc

= 19 + 0 + 0

= 19.0133 mm

Reqd Tubesheet Thickness for Given Loadings (including CA) [H_{reqd}]:

= max(H_{reqB} ; H_{reqS})



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 244 of 262

ASME TS Calc:

Case: 1 10:16a Sep 9, 2026

$$= \max(19; 19)$$

$$= 19.0133 \text{ mm}$$

ASME Section VIII-2 Part 4.18.9.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = -6.42, F_{tmax} = 22.1$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - 2 \cdot Q_2 \cdot F_{tmin} / a_o^2) / (X_t - X_s)$$

$$= ((1000 \cdot 0.47 - 1000 \cdot 0.6) - 2 \cdot 0.00043 \cdot -6.42 / 1220^2) / (0.6 - 0.47)$$

$$= -1.0001 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - 2 \cdot Q_2 \cdot F_{tmax} / a_o^2) / (X_t - X_s)$$

$$= ((1000 \cdot 0.47 - 1000 \cdot 0.6) - 2 \cdot 0.00043 \cdot 22.1 / 1220^2) / (0.6 - 0.47)$$

$$= -1.0001 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 1 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= S_{ot} = 170.7986 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= 1.2500 \text{ (as } P_e \text{ is 0.0)}$$

Determine the factor [rt]:

$$= (\sqrt{(d_t^2 + (d_t - 2 \cdot t_t)^2)}) / 4$$

$$= (\sqrt{(25.4^2 + (25.4 - 2 \cdot 1.65)^2)}) / 4$$

$$= 8.4168 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{(2 \cdot \pi^2 \cdot E_t / S_{yt})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 183508208 / 372)}$$

$$= 98.6283$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 1 \cdot 1068 / 8.42$$

$$= 126.9484$$

Buckling Allowable Stress [S_{tb}]:

$$= (1 / F_s) \cdot (\pi^2 \cdot E_t / F_t^2) \cdot (C_t \leq F_t)$$

$$= (1 / 1.25) \cdot (\pi^2 \cdot 183508208 / 127^2)$$

$$= 89.912 \text{ N./mm}^2 \text{ (Never greater than } S_{ot})$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 1 \cdot 1.23$$

$$= 12.56 \text{ Kg}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 1.65) 1.65$$

$$= 1.232 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 245 of 262

ASME TS Calc:

Case: 1 10:16a Sep 9, 2026

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 1.65 (25.4 - 1.65) 171$$

$$= 2145.348 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 171$$

$$= 2035.2529 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 171/171$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 1.65 (25.4 + 0.67 \cdot 1.65) 171$$

$$= 1316.9283 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 1.65 \cdot (25.4 - 1.65) 1 \cdot 1)} - 0.75 \cdot 25.4 \right)$$

$$= 2.142 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0711 mm

Minimum Required Groove Weld Leg [agr]: 1.0711 mm

Tube-Tubesheet Jt allowable, 2145 is >= tube strength 2145 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Computations completed for ASME tubesheet configuration: d

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	0.0 <=	256.2 N./mm ²	Pass
Tubesheet Shear Stress	0.0 <=	136.6 N./mm ²	Pass
Maximum Stress in the Tubes	1.0 <=	170.8 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-1.0 <=	-89.9 N./mm ²	Pass
Maximum Force on any one Tube	12.6 <=	2145.3 Kg	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	19.013	140.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.071	1.651	Ok
Tube-Tubesheet Groove Weld Leg :	1.071	1.651	Ok

Stationary Tubesheet results per ASME Div. 2 Part 4.18.9 2025

Results for 6 Load Cases:

Case#	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd	Case Type	Pass/Fail



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 246 of 262

ASME TS Calc:

Case: 1 10:16a Sep 9,2026

D1uc	126.058	5.006	208	256	20	137	Ps+Pt-Th	D1	Ok
D2uc	132.432	5.522	230	256	22	137	Ps+Pt-Th	D2	Ok
D3uc	19.013	...	...	256	...	137	Ps+Pt-Th	D3	Ok
D1c	126.058	5.006	208	256	20	137	Ps+Pt-Th-c	D1	Ok
D2c	132.432	5.522	230	256	22	137	Ps+Pt-Th-c	D2	Ok
D3c	19.013	...	...	256	...	137	Ps+Pt-Th-c	D3	Ok

Max: 132.4317 5.522 mm 0.896 0.159 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Tube, Shell and Channel Stress Summary:

---- Tube Stresses Tube Loads Shell Stress Channel Stress Pass											
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	54	170	-13	-89	675	2145	...	...	...	...	Ok
D2uc	60	170	-60	-89	757	2145	...	...	...	...	Ok
D3uc	1	170	-1	-89	13	2145	...	...	...	...	Ok
D1c	54	170	-13	-89	675	2145	...	...	...	...	Ok
D2c	60	170	-60	-89	757	2145	...	...	...	...	Ok
D3c	1	170	-1	-89	13	2145	...	...	...	...	Ok
Max RATIO	0.353		0.670		0.353		...		...		

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	132.432	140.000	Ok
Tubesheet thickness flanged extension :	5.522	124.000	Ok
Tube thickness :	0.515	1.651	Ok
Tube-tubesheet fillet weld leg :	1.071	1.651	Ok
Tube-tubesheet groove weld leg :	1.071	1.651	Ok

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
--------------------------	---------------	-------------------------	----------------	------------------------



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 247 of 262

ASME TS Calc:

Case: 1 10:16a Sep 9, 2026

Tubesheet Bending Stress	1231.25	1.000	1231.25	1.000
Tubesheet Shear Stress	6934.99	1.000	6934.99	1.000
Tube Tensile Stress	3178.45	1.000	3120.37	1.000
Tube Compressive Stress	6872.61	1.000	1642.62	1.000
Tube-Tubesheet Joint load	3178.45	1.000	3120.37	1.000
Tube Pressure Stress	23420.37	1.000	11825.90	1.000
Minimum MAWP	1231.25		1231.25	

Tubesheet MAPnc used to Compute Hydrottest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	1348.02	1.000	1348.02	1.000
Tubesheet Shear Stress	7592.68	1.000	7592.68	1.000
Tube Tensile Stress	3479.88	1.000	3416.30	1.000
Tube Compressive Stress	7488.13	1.000	1789.74	1.000
Tube-Tubesheet Joint load	3479.88	1.000	3416.30	1.000
Tube Pressure Stress	25641.50	1.000	15881.17	1.000
Minimum MAPnc	1348.02		1348.02	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 248 of 262

ASME Fl-TS Calc:

Case: 1 10:16a Sep 9,2026

Input Echo, Tubesheet Number 1, Description: Floatating tube

Shell Data:

Main Shell Description: Shell-1

Shell Maximum Design Pressure	Psd,max	1000.00 KPa.
Shell Maximum Operating Pressure	Psox,max	1000.00 KPa.
Shell Minimum Operating Pressure	Psox,min	103.42 KPa.
Shell Thickness	ts	16.0000 mm
Shell Internal Corrosion Allowance	cas	0.8000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2700.000 mm
Shell Temperature for Internal Pressure	Ts	240.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	1000.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	1000.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	16.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2700.000 mm
Channel Design Temperature	TEMPC	82.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	4924
Tube Wall Thickness	et	1.6510 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	6550.00 mm
Straight Tube Length (bet. inner tubsht faces) L		6270.00 mm
Design Temperature of the Tubes		240.00 °C
Tube Material		SA-789
Tube Material UNS Number		S32205
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		170.80 N./mm ²
Tube Allowable Stress At Ambient		187.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	372.40 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.6510 mm
Groove Weld Leg	ag	1.6510 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	1207.750 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 249 of 262

ASME Fl-TS Calc:

Case: 1 10:16a Sep 9,2026

Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm
Area of Tube Layout	Ap	42498.0 cm ²
Perimeter of Tube Layout	Cp	7703.8 mm

Tubesheet Data:

Tubesheet TYPE: Floating, Tubesheet Gasketed, Conf. B
Exchanger TYPE: With an Immersed Floating Head, Conf. a

Tubesheet Design Metal Temperature	T	240.00 °C
Tubesheet Material		SA-182 F60
Tubesheet Material UNS Number		S32205
Tubesheet Allowable Stress at Temperature	S	170.80 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	187.00 N./mm ²
Thickness of Tubesheet	h	140.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	2650.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped? No

Area of the Untubed Lanes	AL	428.6 cm ²
---------------------------	----	-----------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	1068.500 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Channel Side Min. Design Pressure	Ptd,min	-103.0000 KPa.
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Channel
Flange Outside Diameter	A	2650.000 mm
Flange Inside Diameter	B	2454.000 mm
Flange Face Outside Diameter	Fod	2486.000 mm
Flange Face Inside Diameter	Fid	2454.000 mm
Gasket Outside Diameter	Go	2483.000 mm
Gasket Inside Diameter	Gi	2457.000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	4.5000 mm
Full face Gasket Flange Option		Program Selects
Length of Partition Gasket	lp	2700.000 mm
Width of Partition Gasket	wp	22.0000 mm
Partition Gasket Factor,	mPart	3.0000
Partition Gasket Design Seating Stress	yPart	68.95 N./mm ²
Partition Gasket Facing Sketch		Code Sketch 1a
Partition Gasket Column for Gasket Seating		Code Column I

Bolting Information:

Diameter of Bolt Circle	C	2550.000 mm
Nominal Bolt Diameter	dB	47.6250 mm
Thread Series		UNC Thread Series
Number of Bolts	n	40
Bolt Material		SA-193 B7
Bolt Material UNS Number		G41400
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 250 of 262

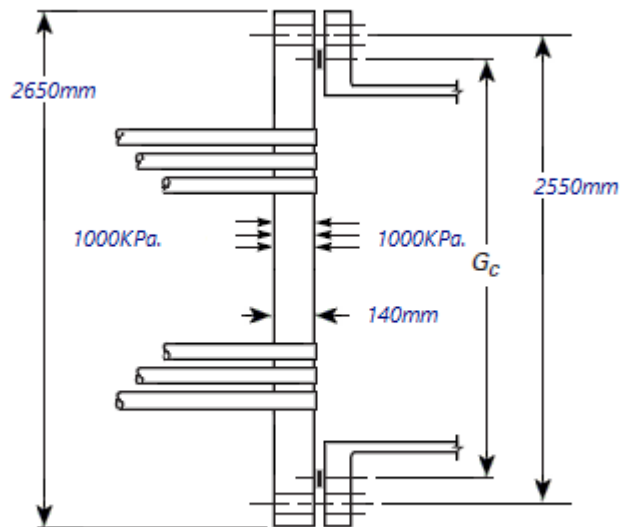
ASME Fl-TS Calc:

Case: 1 10:16a Sep 9,2026

Note: The full bolt load $S_a \cdot A_b$, was used for all load cases.

Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet Tf	114.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes
Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No

Tubesheet Gasketed, Extended as a Flange



Configuration B:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	240.0 °C	0.18971E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	82.0 °C	0.19921E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Low Alloy Steels, Group 2 Group 1
Tubes - TM-1 Material Group H

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000125990 /°C
Elastic Mod. at Design Temperature	240.0 °C	0.18351E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20179E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	240.0 °C	0.18351E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19994E+09 KPa.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 251 of 262

ASME Fl-TS Calc:

Case: 1 10:16a Sep 9,2026

TubeSheet - TE-1 Low Alloy Steels, Group 2 Group 1

TubeSheet - TM-1 Material Group H

Elastic Mod. at Design Temperature	240.0 °C	0.18351E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19994E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)

= $(1000 \cdot (25.4/2 - 0)) / (171 \cdot 1 + 0.4 \cdot 1000)$

= 0.0742 + 0.0000 = 0.0742 mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	HA-5	at	240.00 °C
Elastic Modulus for Material			180060896.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6510	25.40	6270.00	15.38	50.0000	0.0046475	136.46

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 136) / (3 \cdot 15.4) = 11826 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.5154	25.40	6270.00	49.28	50.0000	0.0004529	40.77

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 40.8) / (3 \cdot 49.3) = 1103 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.5154 mm
Allowable Internal Pressure at Corroded thickness	23420.38 KPa.
Required Internal Design Pressure	1000.00 KPa.
Allowable External Pressure at Corroded thickness	11825.91 KPa.
Required External Design Pressure	1103.00 KPa.
Required Thickness due to Shell Side pressure	0.5154 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm
Basic Gasket Width,	$b0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	2470.156 mm

Results for ASME Floating Tubesheet Calculations for Configuration B:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.9.4 Step 1:

Tube expansion depth ratio $[\rho]$:

= lt_x / h (modified for corrosion if present)

= 0/140



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 252 of 262

ASME Fl-TS Calc:

Case: 1 10:16a Sep 9, 2026

$$= 0.0000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole diameter [d*]:

$$= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (p); dt - 2 \cdot tt)$$

$$= \max(25.4 - 2 \cdot 1.65 (183508208/183508208) \cdot (171/171) \cdot (0); 25.4 - 2 \cdot 1.65)$$

$$= 25.4000 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 1208 + 25.4$$

$$= 2440.900 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.8 - 25.4) / 31.8$$

$$= 0.200$$

Equivalent outer tube limit radius [ao]:

$$= Do / 2$$

$$= 2441/2$$

$$= 1220.4500 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$$

$$= 31.8 / \sqrt{ (1 - 4 \cdot \min(429 \cdot 100 ; 4 \cdot 2441 \cdot 31.8) / (\pi \cdot 2441^2)) }$$

$$= 31.8964 \text{ mm}$$

Effective ligament efficiency for bending [μ*]:

$$= (p* - d*) / p*$$

$$= (31.9 - 25.4) / 31.9$$

$$= 0.2037$$

Ratio [ps]:

$$= as / ao$$

$$= 1235/1220$$

$$= 1.0120$$

Ratio [pc]:

$$= ac / ao$$

$$= 1235/1220$$

$$= 1.0120$$

Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 4924 ((25.4 - 2 \cdot 1.65) / (2 \cdot 1220))^2$$

$$= 0.5964$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 4924 (25.4 / (2 \cdot 1220))^2$$

$$= 0.4668$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((8 - 0); 0)$$

$$= 8.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.9.4 Step 2:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

βs, ks, λs, δs are all = 0

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

βC, kc, λC, δC are all = 0



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 253 of 262

ASME Fl-TS Calc:

Case: 1 10:16a Sep 9, 2026

ASME Section VIII-2 Part 4.18.9.4 Step 3:

$h/p = 4.409449$; $\mu^* = 0.203672$; $E^* = 29304318$ KPa.
 $E^*/E = 0.159689$; $\nu^* = 0.453250$

Note: As h/p (4.409) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [Xa]:
 $= ((24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$
 $= ((24(1 - 0.45^2)4924 \cdot 183508208 \cdot 1.65$
 $(25.4 - 1.65)1220^2) / (29304318 \cdot 6270 \cdot 140^3))^{1/4}$
 $= 6.6841$

Values from Table 4.18.3

$Z_d = 0.004960$; $Z_v = 0.022184$; $Z_m = 0.211940$
 $Z_a = 0.198049E+03$; $Z_w = 0.022184$

ASME Section VIII-2 Part 4.18.9.4 Step 4:

Diameter ratio [K]:

$= A / Do$
 $= 2650/2441$
 $= 1.0857$

Compute coefficient [F]:

$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$
 $= (1 - 0.45) / 29304318 \cdot (0 + 0 +$
 $183508208 \cdot \ln(1.09))$
 $= 0.2814$

Compute parameter [ϕ]:

$= (1 + \nu^*) F$
 $= (1 + 0.45) 0.28$
 $= 0.4090$

Compute parameter [Q1]:

$= (p_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$
 $= (1.01 - 1 - 0.41 \cdot 0.022) / (1 + 0.41 \cdot 0.21)$
 $= 0.002681$

ASME Section VIII-2 Part 4.18.9.4 Step 5:

Determine factor [γ_b]:

$= (G_c - C) / Do$ (config b)
 $= (2470 - 2550) / 2441 = -0.033$

Calculate parameter [ω_s]:

$= p_s \cdot k_s \cdot \beta_s \cdot \delta S(1 + h \cdot \beta_s)$
 $= 1.01 \cdot 0 \cdot 0 \cdot 0.000 (1 + 140 \cdot 0)$
 $= 0.0000 \text{ mm}^2$

Calculate parameter [ω^*s]:

$= ao^2(p_s^2 - 1) \cdot (p_s - 1) / 4 - \omega_s$
 $= 1220^2(1.01^2 - 1) \cdot (1.01 - 1) / 4 - 0$
 $= 107.6267 \text{ mm}^2$

Parameter [ω_c]:

$= p_c \cdot k_c [\delta C(C_m \Theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) -$
 $\zeta c(C_m \Theta + h \cdot C_q \Theta + h \cdot \beta C / 2))$
 $= 1.01 \cdot 0 [0.000 (0 + 0 \cdot 0 \cdot 140) -$
 $0.000 (0/140 + 0 \cdot 0/2))$
 $= 0.0000 \text{ mm}^2$

Parameter [ω^*c]:



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 254 of 262

ASME Fl-TS Calc:

Case: 1 10:16a Sep 9, 2026

$$\begin{aligned}
 &= a_o^2 [(p_c^2 + 1) \cdot (p_c - 1)/4 - (p_s - 1)/2] - w_c \\
 &= 1220^2 [(1.01^2 + 1) \cdot (1.01 - 1)/4 - (1.01 - 1)/2] - 0 \\
 &= 107.6262 \text{ mm}^2
 \end{aligned}$$

Pressure [P*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.9.4 Step 6:

$$P_e = P_s - P_t$$

$$= 1000 - 1000 = 0 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.9.4 Step 7:

Determine factor [Q2]:

$$\begin{aligned}
 &= [((w_s \cdot P_s - w_c \cdot P_t) - (w_s \cdot P_s - w_c \cdot P_c)) + \\
 &\quad W \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m) \\
 &= [((108 \cdot 1000 - 108 \cdot 1000) - \\
 &\quad (0 \cdot 0 - 0 \cdot 0)) \cdot 0.0001 + \\
 &\quad 923767 \cdot -0.033 / (2 \cdot \pi)] / (1 + 0.41 \cdot 0.21) \\
 &= -4425.6455 \text{ Kgf}
 \end{aligned}$$

Tubesheet Bending Stress - original thickness [σ]:

$$\begin{aligned}
 &= 6 \cdot Q_2 / (\mu \cdot (H - h'g)^2) \\
 &= 6 \cdot -4426 / (0.2 \cdot (140 - 8)^2) \\
 &= -73.3798 \text{ N./mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 171 = 256 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$\begin{aligned}
 &= 6 \cdot Q_2 / (\mu \cdot (H - h'g)^2) \\
 &= 6 \cdot -4549 / (0.2 \cdot (79.6 - 8)^2) \\
 &= 256.1978 \text{ N./mm}^2
 \end{aligned}$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$\begin{aligned}
 &= h + C_{ats} + C_{atc} \\
 &= 79.6 + 0 + 0 \\
 &= 79.6242 \text{ mm}
 \end{aligned}$$

$$H_{reqd.} = H_{ReqB} = 79.6242 \text{ mm}$$

Computations completed for ASME tubesheet configuration: B

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	73.4 <=	256.2 N./mm ²	Pass

Floating Tubesheet results per ASME Div. 2 Part 4.18.9 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail	
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd			
D1uc	103.603	47.717	148	256	...	...	Ps+Pt-Th	D1	Ok
D2uc	77.794	...	104	256	...	...	Ps+Pt-Th	D2	Ok
D3uc	79.624	...	73	256	...	...	Ps+Pt-Th	D3	Ok



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 255 of 262

ASME Fl-TS Calc:

Case: 1 10:16a Sep 9, 2026

D1c	103.603	47.717	148	256	...	...	Ps+Pt-Th-c	D1	Ok
D2c	77.794	...	104	256	...	...	Ps+Pt-Th-c	D2	Ok
D3c	79.624	...	73	256	...	...	Ps+Pt-Th-c	D3	Ok

Max: 103.6031 47.717 mm 0.578 ... (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	103.603	140.000	Ok
Tubesheet thickness flanged extension :	47.717	114.000	Ok
Tube thickness :	0.515	1.651	Ok
Tube-tubesheet fillet weld leg :	0.000	1.651	Ok
Tube-tubesheet groove weld leg :	0.000	1.651	Ok

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	2060.78	1.000	2718.42	1.000
Tube Pressure Stress	23420.37	1.000	11825.90	1.000
Tubesheet Extension Stress	1733.18	...	No Calc	No Calc
Minimum MAWP	1733.18		2718.42	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	0 shellside Stress Rat.	Shellside MAPnc	0 tubeside Stress Rat.
Tubesheet Bending Stress	2256.22	1.000	2976.23	1.000
Tube Pressure Stress	25641.50	1.000	15881.17	1.000
Tubesheet Extension Stress	1733.18	...	No Calc	No Calc
Minimum MAPnc	1733.18		2976.23	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 256 of 262

ASME Fl-TS Calc:

Case: 1 10:16a Sep 9, 2026

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 257 of 262

Floating Head: Operating

Case: 1 10:16a Sep 9,2026

Input Echo, Floating Head Item 1, Description: Operating

Floating Head Type		D	
Tube Side (Internal) Design Pressure	Pts	1000.00	KPa.
Shell Side (External) Design Pressure	Pss	1103.00	KPa.
Design Temperature for Spherical Head	Temp	200.00	°C
Head Material		SA-240	
Head Material UNS Number		S32205	
Head Allowable Stress at Temperature	Sho	174.00	N./mm ²
Head Allowable Stress at Ambient	Sha	187.00	N./mm ²
Crown Radius for Spherical Head	L	2208.600	mm
Head Thickness	Th	30.000	mm
Tube Side (Internal) Corrosion Allowance	Cats	0.0000	mm
Shell Side (External) Corrosion Allowance	Cass	0.0000	mm
Flange Material		SA-182 F60	
Flange Material UNS Number		S32205	
Flange Allowable Stress at Temperature	Sfo	174.00	N./mm ²
Flange Allowable Stress at Ambient	Sfa	187.00	N./mm ²
Flange Outside Diameter	Fod	2650.000	mm
Flange Inside Diameter	Fid	2454.000	mm
Flange Thickness	Tf	150.0000	mm
Bolt Material		SA-193 B7	
Bolt Material UNS Number		G41400	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2550.000	mm
Nominal Bolt Diameter	dB	47.6250	mm
Type of Threads		UNC Thread Series	
Number of Bolts		40	
Full face Gasket Flange Option		Program Selects	
Flange Face Outside Diameter	Ffod	2486.000	mm
Flange Face Inside Diameter	Ffid	2454.000	mm
Gasket Outside Diameter	Go	2483.000	mm
Gasket Inside Diameter	Gi	2457.000	mm
Gasket Factor,	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Flange Facing Sketch		Code Sketch 1a	
Column for Gasket Seating		Code Column I	
Gasket Thickness		4.5000	mm
Width of raised face		0.0000	mm
Length of Partition Gasket		2700.000	mm
Width of Partition Gasket		22.0000	mm
Partition Gasket Factor,	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²
Partition Gasket Facing Sketch		Code Sketch 1a	
Partition Gasket Column for Gasket Seating		Code Column I	
Dist. from Head Centerline to Flange Centroid	hr	27.0400	mm

The Flange is not Slotted.

This unit does not have a Backing Ring.

Use Full Bolt Load in Flange Design Yes



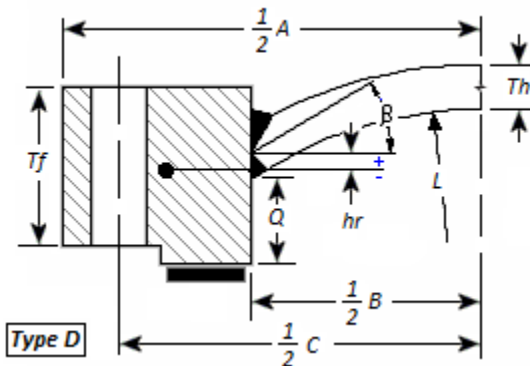
PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 258 of 262

Floating Head: Operating

Case: 1 10:16a Sep 9,2026



Internal Pressure results for Spherical Heads:

ASME Code, Section VIII, Division 1, 2025, App. 1-6

Thickness Due to Internal Pressure[thr]:

$$= 5PL / 6Sho$$

$$= (5 \cdot 1000 \cdot 2209) / (6 \cdot 174)$$

$$= 10.5783 \text{ mm}$$

Maximum Allowable Working Pressure at Given Thickness[Pa]:

$$= 6Sho \cdot (Th - Cass - Cats) / 5L$$

$$= (6 \cdot 174 \cdot 30) / (5 \cdot 2209)$$

$$= 2836.00 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold[Pnc]:

$$= 6Sha / 5L$$

$$= (6 \cdot 187 \cdot 30) / (5 \cdot 2209)$$

$$= 3047.85 \text{ KPa.}$$

Actual Stress at given Pressure and Thickness[Sact]:

$$= 5PL / 6(Th - Cass - Cats)$$

$$= (5 \cdot 1000 \cdot 2209) / (6 \cdot 30)$$

$$= 61. \text{ N./mm}^2$$

Spherical head Percent Elongation $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$: 1.012 %

External Pressure results, Spherical Head

ASME Code, Section VIII Division 1, 2025

External Pressure Chart HA-5 at 200.00 °C
Elastic Modulus for Material 183008944.00 KPa.

Results for Maximum Allowable External Pressure:

Corroded Thickness of Shell	TCA	30.0000	mm
Outside Crown Radius of Shell	OD/2	2238.5999	mm
Crown Radius / Thickness Ratio	$(OD/2)/T$	74.6200	
Geometry Factor, A	f(D/T, LD)	0.0016752	
Materials Factor, B	f(A, Chart)	113.1410	N./mm ²
Maximum Allowable Working Pressure		1516.14	KPa.
EMAWP = $B / ((OD/T)/2) = 113 / (149/2.0) = 1516$			

Results for Req'd Thickness for Ext. Pressure (Tca):

Corroded Thickness of Head	TCA	23.2605	mm
Outside Diameter of Shell	ODCA	4477.200	mm
Diameter / Thickness Ratio	(D/T)	192.4807	
Geometry Factor, A	f(DT, LD)	0.0012988	
Materials Factor, B	f(A, Chart)	106.1707	N./mm ²
Maximum Allowable Working Pressure		1103.12	KPa.
EMAWP = $B / ((D/T)/2) = 106 / (192/2) = 1103$			



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 259 of 262

Floating Head: Operating

Case: 1 10:16a Sep 9, 2026

Intermediate Calculations for Flanged portion:

ASME Code, Section VIII, Div. 1, 2025 App. 2

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	13.000 mm
Basic Gasket Width,	$b_0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b_0) * 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = G_o - 2.0 * b$	2470.156 mm

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsm_{ax}]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 * 47.6 + 6 * 150 / (3 + 0.5)$$

$$= 352.393 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C * \sin(\pi / n)$$

$$= 2550 * \sin(\pi / 40)$$

$$= 200.071 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{200 / (2 * 47.6 + 150)}, 1)$$

$$= 1.0000$$

Bolting Information for UNC Thread Series (Non Mandatory):

Total Area of Bolts 526.709 cm²

	Minimum	Actual	Maximum
Radial Distance between Bolts and Edge:	47.625	50.000	
Circ. Spacing between the Bolts:	101.600	200.071	352.393

Results for the Internal Pressure case:

Basic Flange and Bolt loads:

Hydrostatic End Load due to Pressure	H	488691.8	Kgf
Contact Load on Gasket Surfaces	Hp	44295.3	Kgf
Hydrostatic End Load at Flange ID	Hd	482320.3	Kgf
Pressure Force on Flange Face	Ht	6371.4	Kgf
Radial Component of Head Membrane Force	Hr	727362.9	Kgf
Operating Bolt Load:	Wm1	532987.1	Kgf
Gasket Seating Bolt Load	Wm2	508996.8	Kgf
Required Bolt Area	Am	303.896	cm ²

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calculation]:

$$= Ab * Sb / (y * PI * (Go + Gi))$$

$$= 527 * 172 / (68.9 * 3.14 * (2483 + 2457))$$

$$= 8.466 \text{ mm}$$

Note: Recommended Min. Width for Sheet and Composite Gaskets per table 2-4 :

= 38.100 mm [Note: Exceeds actual gasket width, 13.000]

Flange Design Bolt Load (Seating)	W	923766.8	Kgf
Gasket Seating Force (Operating)	Hg	44295.3	Kgf
Distance to Gasket Load Reaction	hg	39.9222	mm
Distance to Face Pressure Reaction	ht	43.9611	mm
Distance to End Pressure Reaction	hd	48.0000	mm

Summary of Moments for Internal Pressure:

Loading	Force	Distance	Bolt Corr	Moment
End Pressure,	Md	482320.	48.0000	1.0000
				23152. Kg-m.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 260 of 262

Floating Head: Operating

Case: 1 10:16a Sep 9, 2026

Face Pressure,	Mt	6371.	43.9611	1.0000	280. Kg-m.
Gasket Load,	Mg	44295.	39.9222	1.0000	1768. Kg-m.
Floating Hd. Load,	Mh	727363.	27.0400	1.0000	-19668. Kg-m.
Gasket Seating,	Ma	923767.	39.9222	1.0000	36880. Kg-m.

Total Moment for Operation (Internal Pressure) 5532.041 Kg-m.

Total Moment for Gasket Seating (Int. Pressure) 36879.508 Kg-m.

Results for the External Pressure case:

Basic Flange and Bolt loads:

Hydrostatic End Load due to Pressure	H	539027.1	Kgf
Hydrostatic End Load at Flange ID	Hd	531999.3	Kgf
Pressure Force on Flange Face	Ht	7027.7	Kgf
Radial Component of Head Membrane Force	Hr	802281.2	Kgf
Gasket Seating Bolt Load	Wm2	508996.8	Kgf
Required Bolt Area	Am	290.217	cm ²

Flange Design Bolt Load (Seating)	W	923766.8	Kgf
Distance to Gasket Load Reaction	hg	39.9222	mm
Distance to Face Pressure Reaction	ht	43.9611	mm
Distance to End Pressure Reaction	hd	48.0000	mm

Summary of Moments for External Pressure:

Loading	Force	Distance	Bolt Corr	Moment
End Pressure,	Md	531999.	8.0778	1.0000 4297. Kg-m.
Face Pressure,	Mt	7028.	4.0389	1.0000 28. Kg-m.
Floating Hd. Load,	Mh	802281.	27.0400	1.0000 -21694. Kg-m.
Gasket Seating,	Ma	923767.	39.9222	1.0000 36880. Kg-m.

Total Moment for Operation (External Pressure) 17368.260 Kg-m.

Total Moment for Gasket Seating (Ext. Pressure) 36879.508 Kg-m.

Factor for integral type flanges [F]:

$$= P \cdot B \cdot \sqrt{(4 \cdot L \cdot L - B \cdot B) / (8 \cdot S(A - B))}$$

$$= 1000 \cdot 2454 \cdot \sqrt{(4 \cdot 2209 \cdot 2209 - 2454 \cdot 2454) / (8 \cdot 174(2650 - 2454))}$$

$$= 33.0374 \text{ mm}$$

Factor [J]:

$$= Mo(A + B) / (S \cdot B)(A - B)$$

$$= 5532(2650 + 2454) / (174 \cdot 2454)(2650 - 2454)$$

$$= 3308.5576 \text{ mm}^2$$

Required thickness for Main Flange, Internal Operating Condition [T]:

$$= F + \sqrt{(F \cdot F + J)} \text{ per 1-6(g)}$$

$$= 33 + \sqrt{(33 \cdot 33 + 3309)}$$

$$= 99.3701 \text{ mm}$$

Factor [J]:

$$= Mo(A + B) / (S \cdot B)(A - B)$$

$$= 36880(2650 + 2454) / (187 \cdot 2454)(2650 - 2454)$$

$$= 20523.4980 \text{ mm}^2$$

Required thickness for Main Flange, Internal Bolt-up Condition [T]:

$$= F + \sqrt{(F \cdot F + J)} \text{ per 1-6(g)}$$

$$= 0 + \sqrt{(0 \cdot 0 + 20523)}$$

$$= 143.2602 \text{ mm}$$

Factor for integral type flanges [F]:

$$= P \cdot B \cdot \sqrt{(4 \cdot L \cdot L - B \cdot B) / (8 \cdot S(A - B))}$$

$$= 1103 \cdot 2454 \cdot \sqrt{(4 \cdot 2209 \cdot 2209 - 2454 \cdot 2454) / (8 \cdot 174(2650 - 2454))}$$



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 261 of 262

Floating Head: Operating

Case: 1 10:16a Sep 9, 2026

= 36.4402 mm

Factor [J]:

= $Mo(A + B)/(S \cdot B)(A - B)$

= $17368(2650+2454)/(174 \cdot 2454)(2650-2454)$

= 10387.4678 mm²

Required thickness for Main Flange, External Operating Condition [T]:

= $F + \sqrt{(F \cdot F + J)}$ per 1-6(g)

= $36.4 + \sqrt{(36.4 \cdot 36.4 + 10387)}$

= 144.6777 mm

Factor [J]:

= $Mo(A + B)/(S \cdot B)(A - B)$

= $36880(2650+2454)/(187 \cdot 2454)(2650-2454)$

= 20523.4980 mm²

Required thickness for Main Flange, External Bolt-up Condition [T]:

= $F + \sqrt{(F \cdot F + J)}$ per 1-6(g)

= $0 + \sqrt{(0 \cdot 0 + 20523)}$

= 143.2602 mm

Summary of Required Thicknesses(mm):

	Head	Flange
Tubeside (Internal) Pressure	10.5783	99.3701
Shellside (External) Pressure	23.2605	144.6777
Tubeside Gasket Seating Load		143.2602
Shellside Gasket Seating Load		143.2602
Maximum + Corrosion Allowance	23.2605	144.6777
Actual Thickness as Given	30.0000	150.0000

Floating Head MAWP and MAPnc (KPa.):

Stress Condition	Tubeside		Shellside	
	MAWP	MAPnc	MAWP	MAPnc
Flange Stress	1702.34	1828.69	1163.71	1251.02
Floating head Stress	2835.90	3047.75	1516.04	1852.72
Bolt Stress	1733.09	1733.09	...	...
Minimum MAWP & MAPnc	1702.34	1733.09	1163.71	1251.02

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User

FileName : 197-C_Channel MOC= CS

Page 262 of 262

Problems/Failures Summary:

Step: 40 10:16am Sep 9, 2026

Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: Insufficient BasePlate bolting area!**

**** Error: Baseplate Bolt Shear Stress Failure!**

Please review all reports carefully!

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: